$$E_1 = \frac{KQ_1}{(1+x)^2} \text{ and } E_2 = \frac{KQ_2}{x^2}$$

$$\Rightarrow \frac{KQ_1}{(1+x)^2} = \frac{KQ_2}{x^2}$$

$$\frac{4}{(1+x)^2} = \frac{1}{x^2} \Rightarrow 4x^2 = x^2 + 2x + 1$$

$$3x^2 - 2x - 1 = 0 \Rightarrow x = 1\text{m}.$$

Answer: C

122. The electric field required to exert a force on proton equal to its weight is

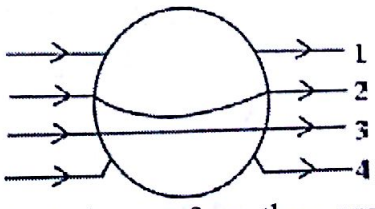
A) 1.67×10^{-27}N/C
B) 1.02×10^{-7}v/m
C) 1.6×10^{-9} v/m
D) 1.04×10^{-27}v/m

Solution:

$$F = EQ = mg \Rightarrow E = \frac{mg}{e} = \frac{1.67 \times 9.81 \times 10^{-27}}{1.6 \times 10^{-19}} = 1.02 \times 10^{-7}\text{v/m}$$

Answer: B

123. A metallic solid sphere is placed in a uniform electric field. The field lines follow the path



A) 1 B) 2 C) 3 D) 4

Explanation: No electric field inside and at the surface they are perpendicular.

Answer: D

124. An electric dipole consisting of two opposite charges of 2μC each, separated by a distance of 0.03m, is placed in an lectric field of 2×10^5N/C. The maximum torque on the dipole will be

A) 12×10^{-1}Nm
B) 12×10^{-3}Nm
C) 24×10^{-1}Nm
D) 24×10^{-3}Nm

Explanation: $\tau = PE = QdE = (2 \times 10^{-6})\ (0.03)\ (2 \times 10^{-5})$

$\tau = 12 \times 10^{-3}$Nm

Answer: B

125. A charge q is placed at the centre of the line joining two equal charges Q. The system of the three charges will be in equilibrium if q is equal to

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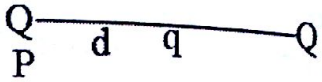
A) $-Q/2$ B) $-Q/4$ C) $Q/4$ D) $Q/2$

Explanation: consider the arrangement at point p,

$$\frac{Qq}{d^2}=\frac{-Q^2}{4d^2}\Rightarrow q=-\frac{Q}{4}$$

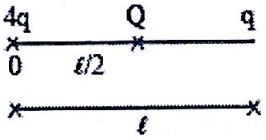


Answer: B

126. Three charges 4q, Q and q are placed in a straight line of length ℓ at point 0, $\ell/2$ and ℓ respectively. What should be Q in order to make the net force on q to be zero?

A) –q B) $-\frac{1}{2}q$ C) –2q D) 4q

Solution:



$$\frac{4Qq}{\ell^2}=\frac{-4Q^2}{\ell^2}\Rightarrow Q=-q$$

Answer: A

127. A hallow metallic sphere of radius 10cm is given a charge of 3.2nc. The electric intensity at a point 4cm from the centre is

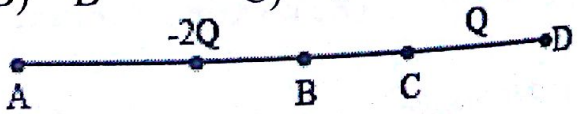
A) 2.88V B) 288V C) 9×10^{-9}V D) zero

Explanation: The electric field is zero inside.

Answer: D

128. Two point charges -2Q and Q are situated as shown. At which point could the resultant field be zero?

A) A B) B C) C D) D



Answer: D

129. A metallic particle having no net charge is placed near a finite metal plate carrying a positve charge. The electric force on the particle will be

A) towards the plate C) parallel to the plate

B) away from the plate D) zero

Explanation: Negative charge will be induced on the particle.

Answer: A

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130. An electric dipole is placed in a uniform electric field. The net electric force on the dipole.
A) is always zero
B) depends on the orientation of the dipole
C) can never be zero
D) depends on the strength of the dipole

Explanation: Charges in a dipole experience forces of equal in magnitude and opposite in direction.

Answer: A

131. A circular metal plate has radius 2m and is given a charge of 40mc. The electric field near the center
A) 0.9×10^8 v/m
B) 1.8×10^8 v/m
C) 3.6×10^8 v/m
D) 7.2×10^8 v/m

Solution: $$E = \frac{\sigma}{\epsilon_o} = \frac{Q}{A\epsilon_o} = \frac{40 \times 10^{-3}}{\left(2 \times \pi \times 2^2\right)\left(8.85 \times 10^{-12}\right)} = 1.8 \times 10^8 \text{ v/m}$$

Answer: B

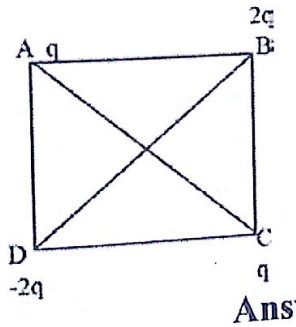
132. A given charge placed at a certain distance from an electric dipole in the end experiences a force F. If the distance of the charge is doubled, the force acting on the charge will be
A) 2F B) F/2 C) F/4 D) F/8

Solution: Force on charge q due to a dipole at the end is,

$$F = K\frac{2pq}{r^3} \Rightarrow F = \frac{1}{r^3}$$

Answer: D

133. Four charges are placed at the corners of a square ABCD. The force on a charge placed at the centre is
A) zero
B) along diagonals AC
C) along diagonal BD
D) perpendicular to AB.



Answer: C

Electric Potential and Gauss's Law

134. A hollow metal sphere of radius 0.1m is charged such that the potential on its surface is 4v. The potential at the centre of the sphere is
A) 0v B) 4v C) 0.1v D) 0.25v

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Explanation: The electric potential at every point inside the hollow metal sphere is the same as on the surface of the sphere.

Answer: B

135. Electric potential of Earth is taken to be zero because Earth is a good.

A) insulator C) semiconductor
B) conductor D) dielectric

Answer: B

136. An electric charge q is placed at the centre of a cube of side ℓ. The electric flux through one of its face will be

A) q/ϵ_o B) $q/6\epsilon_o$ C) $q/\epsilon_o \ell^2$ D) $q/4\pi\epsilon_o \ell^2$

Explanation: The total electric flux in the cube is q/ϵ_o. Flux through one face is one-sixth of this flux.

Answer: B

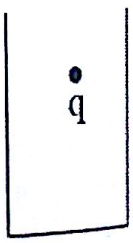
137. There is 40 units of positive charge at the centre of a circle of radius 10m. The work done in moving 4 unit of charge around the circle once is

A) zero B) 40 units C) 400 units D) 4 units

Explanation: The circle around the charge is an equi-potential line.

Answer: A

138. A charge q is placed at the centre of an open end vessel. The flux of the field through the vessel surface is.



A) zero B. q/ϵ_o

C. $2q/\epsilon_o$ D. $q/2\epsilon_o \ell^2$

Explanation: From symmetry.

Answer: D

139. A charge of 10C is moved along an equipotential surface having a potential of 2V. The work done is

A) 10J B) zero C) 2J D) 20J

Explanation: Work done in an equi-potential surface is zero.

Answer: B

140. Which of the following is equivalent to SI unit of potential?

A) NmC^{-1} B) N^2mC^{-1} C) N^3mC^{-2} D) Nm

Explanation: $1V = \frac{1J}{1C} = \frac{Nm}{C} = NmC^{-1}$

Answer: A

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141. A -5C of charge is carried from infinity to a point in a field and perform 100J of work. The potential at the given point is
A) 10v B) -10v C) 20v D) -20v

Solution: $V = \frac{W}{Q} = -\frac{100}{5} = -20V$ **Answer: D**

142. S_1 and S_2 are two equipotential surfaces on which the potential are not equal. Which of the following is incorrect?
A) S_1 and S_2 cannot interesect
B) S_1 and S_2 cannot both be plane surfaces
C) In the region between S_1 and S_2, the field is maximum
D) A line of force from S_1 to S_2 are perpendicular

Explanation: Two plane equipotential surfaces can have different potential energy of the system. **Answer: B**

143. In bringing an electron towards the second electron, the electric potential enregy of the system.
A) increases C) remain same
B) decreases D) become zero **Answer: A**

144. A nucleus has a charge of 50e. If the proton is at a distance of 10^{-12}m from the nucleus, the potential is then
A) 14.4×10^4v C) 7.2×10^8v
B) 7.2×10^4v D) 14.4×10^8v

Solution: $V = \frac{KQ}{r} = \frac{(9\times10^9)(50\times1.6\times10^{-19})}{(10^{-12})} = 7.2\times10^4 v$

Answer: B

145. Two conducting spheres of radii r_1 and r_2 are given the same charge. The ratio of their potentials will be
A) $\frac{r_2}{r_1}$ B) $\frac{r_1}{r_2}$ C) $\frac{r_1}{1+r_2}$ D) $\frac{r_2}{1+r_1}$

Explanation: $V \sim \frac{1}{r}$ **Answer: A**

146. When the separation between two charges is increased, the electric potential energy of the charges.
A) increases C) remain same
B) decreases D) may increase or decrease

Explanation: The increase or decrease of potential energy will be decided by the nature of the charge. **Answer: D**

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147. A cloud is at a potential 10^6v relative to the ground. A charge of 6C is transferred in a lightning stroke between the cloud and the earth. The energy dissipated is

A) 10^6J C) 6×10^6J

B) 3×10^6J D) 12×10^6J

Explanation: $W = QV = (6)(10^6) = 6 \times 10^6$J **Answer: C**

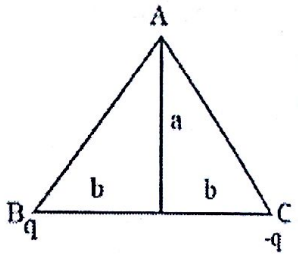
148. Charges q and –q are placed at the vertices B and C of an isosceles triangle. The potential at A is

A) $K\frac{2q}{\sqrt{a^2+b^2}}$

B) Zero

C) $K\frac{q}{\sqrt{a^2+b^2}}$

D) $K\frac{(-q)}{\sqrt{a^2+b^2}}$



Solution: $V = V_1 + V_2 = K\frac{q}{\sqrt{a^2+b^2}} + K\frac{(-q)}{\sqrt{a^2+b^2}} = 0$

Answer: B

149. A charge of Q is placed on a solid piece of metal of irregular shape. The charge will distribute itself

A) uniformly inside C) to have minimum potential energy

B) uniformly outside D) to have maximum potential energy

Explanation: Every system in nature tends to have least potential energy. **Answer: C**

150. The electric potential at any point inside a charged hollow spherical conductor is

A) zero

B) greater than that on its surface

C) the same as that on its surface

D) less than that on the surface **Answer: C**

151. A hollow metal sphere of radius 5cm is charged such that the potential on its surface is 10v. The potential at a distance of 2cm from the centre of the sphere is

A) zero B) 10v C) 4v D) $\frac{10}{3}$v

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Explanation: potential at any point inside a sphere is the same as on the surface of the sphere. **Answer: B**

152. Two charges of 1μC each are kept at a distance of 1m from each other. Their potential energy will be

A) 1J
B) 1.6×10^{-19}V
C) 9×10^{-3}J
D) 9×10^{-9}J

Explanation: $U = \frac{KQ_1Q_2}{r^2} = \frac{(9\times10^9)(10^{-6})(10^{-6})}{1} = 9\times10^{-3}$J

Answer: C

153. The flux through a sphere of radius 0.1m surrounding a charge is 1.5vm. The flux over a concentric sphere of radius 0.15m will be

A) 1.5vm B) 5vm C) 10vm D) 20vm

Explanation: From Gauss, $\phi = \frac{Q}{\epsilon_o}$ = constant **Answer: A**

154. The electric field strength at a distance r from the centre of a charged sphere of radius R is E . If r > R, how much work will be done in bringing a test charge q_o from infinity to that point

A) q_oRE
B) $\frac{1}{2}q_oRE$
C) q_orE
D) $\frac{1}{2}q_orE$

Solution: $E = \frac{V}{r} \Rightarrow V = Er$, and $U = Vq_o = q_oEr$ **Answer: C**

155. The total electric flux in a dipole of radius 1m is

A) zero B. q/ϵ_o C) $8\pi r^2/\epsilon_o$ D) $2q/\epsilon_o$

Explanation: $\phi = \frac{Q_{net}}{\epsilon_o}$, for a dipole, $Q_{net} = 0$. **Answer: A**

156. The electric flux through a hemisphere of radius R, placed in a uniform electric field E is

A) $2\pi RE$ B) $2\pi R^2E$ C) πR^2E D) $\frac{4}{3}R^3E$

Solution: $\phi = EA = E\pi R^2$ **Answer: C**

157. When a positively charged conductor is earth connected

A) electrons flow from the earth to the conductor

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B) protons flow from the conductor to the earth
C) no charge flow occurs
D) electrons flow from the conductor to the earth

Answer: A

158. A hollow metal sphere of radius 5cm is charged such that the potential on its surface is 10v. The electric field at the centre of the sphere will be

A) 50v/m B) 10v/m C) 5v/m D) zero

Explanation: No electric field can present inside a charged conductor:

Answer: D

159. A point charge q is placed at one corner of a cube of edge a. The flux through each of the cube face is

A) q/ϵ_o B) $q/16\epsilon_o$ C) $q/24\epsilon_o$ D) $q/48\epsilon_o$

Explanation: $\phi = \left(\frac{1}{8}\right)\left(\frac{1}{3}\right)\left(q/\epsilon_o\right) = q/24\epsilon_o$

Answer: C

160. An infinite number of charges each numerically equal to q and of the same sign, are placed along the x-axis at x = 1, x = 2, x = 4, x = 8 and so on. Then the potential at x = 0 due to this set of charges is

A) 2q B) $2q/5$ C) $\frac{4}{3}q$ D) $\frac{4}{5}q$

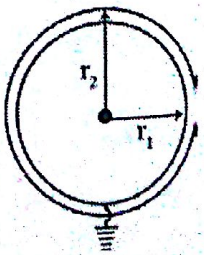
Solution: $v = \frac{q}{1} + \frac{q}{2} + \frac{q}{4} + \frac{q}{8} + \ldots$

$$v = q\left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} \ldots\ldots\right) \Rightarrow v = q\left(\frac{1}{1 - \frac{1}{2}}\right) \Rightarrow v = 2q$$

Answer: A

161. Charge on the outer sphere is q, and the inner sphere is grounded. Then the charge q^1 on the inner sphere is

A) zero C) $q^1 = \frac{-r_1}{r_2}q$



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B) $q^1 = q$ D) $q^1 = \frac{r_1}{r_2} q$

Explanation: potential at the inner surface is zero (grounded)

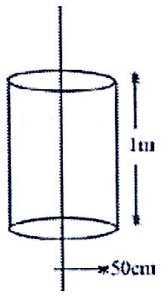
$$\frac{Kq}{r_2} + \frac{Kq^1}{r_1} = 0 \Rightarrow q^1 = -q\frac{r_1}{r_2}$$

Answer: C

162. Electric charge is uniformly distriubted along a long straight wire; if the charge is Q per centimeter length of were and, a cylinder of radius length encloses the wire. What is the flux passing through the cylinderical surface.

A) Q/ϵ_o C) $\frac{10Q}{\pi \epsilon_o}$

B) $100\,Q/\epsilon_o$ D) $\frac{100Q}{\pi \epsilon_o}$



Solution: $\phi = \frac{Q_{enclosed}}{\epsilon_o}$, $Q_{enclosed} = 100Q = \frac{100Q}{\epsilon_o}$ **Answer: B**

163. A ball of mass 1g and charge 10^{-8}C moves from point A whose potential is 600v to point B whose potential is zero. If the velocity of the ball at B is 20cm/s, what is its velocity at point A?

A) 16.7cm/s C) 16.7m/s

B) 2.8cm/s D) 2.8m/s

Explanation: Work energy theorem

$$W = \Delta KE \Rightarrow VQ = \frac{1}{2} m\left(v_B^{\,2} - v_A^{\,2}\right)$$

$$(600)\left(10^{-8}\right) = \frac{1}{2}\left(10^{-3}\right)\left(0.2^2 - v_A^{\,2}\right)$$

$$\Rightarrow 0.04 - v_A^{\,2} = 0.012,\ v_A = 16.7\text{cm/s}$$

Answer: A

164. Two spherical conductors of radii 4m and 5m are charged to the same potential of σ_1 and σ_2 be the respective values of the surface charge densities, then the ratio $\frac{\sigma_1}{\sigma_2}$ is

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A) $\frac{25}{16}$ B) $\frac{16}{25}$ C) $5/4$ D) $4/5$

Solution: $\frac{\sigma_1}{\sigma_2}=\frac{\frac{Q_1}{4\pi R_1^2}}{\frac{Q_2}{4\pi R_2^2}}=\frac{Q_1R_2^2}{Q_2R_1^2}=\frac{C_1V}{C_2V}\frac{R_2^2}{R_1^2}=\frac{R_1R_2^2}{R_2R_1^2}=\frac{R_2}{R_1}=5/4$

Answer: D

165. A hollow metal ball 8cm in diameter is given a charge of -4×10^{-9}C. The potential on the surface of the ball is

A) -9000V B) -900V C) -90V D) zero

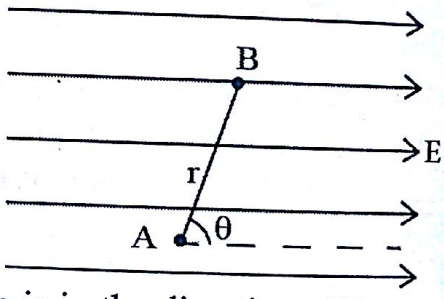
Explanation: $V=\frac{KQ}{r}=\frac{(9\times10^9)(-4\times10^{-8})}{4\times10^{-2}}=-9000V$

Answer: A

166. From the figure shown, what is the potential difference between point A and B

A) $Er\cos\theta$ C) $\frac{Er}{\cos\theta}$

B) $E/r\cos\theta$ D) $\frac{E}{r}\cos\theta$



Explanation: The effective distance is in the direction of E

Answer: A

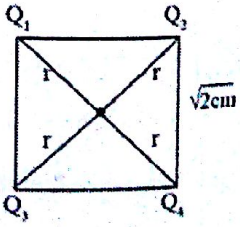
167. Charge of 1μC, 2μC and 3μC are placed at the three coners of a square of side $\sqrt{2}$ cm. What charge should be placed at the fourth corner so that potential at the centre may be zero.

A) 6μC B) −6μC C) 4μC D) −4μC

Solution: $v=v_1+v_2+v_3+v_4$

$=\frac{KQ_1}{r_1}+\frac{KQ_2}{r_2}+\frac{KQ_3}{r_3}+\frac{KQ_4}{r_4}$

$=\frac{Q_1}{r}+\frac{Q_2}{r}+\frac{Q_3}{r}+\frac{Q_4}{r}=0$



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$$=\frac{1}{1}+\frac{2}{1}+\frac{3}{1}+\frac{Q_4}{1}=0,\quad r^2+r^2=\sqrt{2},r=1\text{cm}.$$

$$Q_4=-6\mu C$$

Answer: B

168. Two concentric spheres of radii R and r have similar charges with equal surface charge densities. The electric potential at their common center is

A) $\frac{\sigma}{\epsilon_o}$ C) $\frac{\sigma}{\epsilon_o}(R+r)$

B) $\frac{\sigma}{\epsilon_o}(R-r)$ D) $\frac{\sigma}{4\pi\epsilon_o}(R+r)$

Solution: $v=v_1+v_2=K\frac{Q}{R}+K\frac{Q}{r}$

$$v=\frac{\sigma}{4\pi\epsilon_o}\frac{Q}{R}+\frac{\sigma}{4\pi\epsilon_o}\frac{Q}{r}=\frac{1}{\epsilon_o}\left[\frac{Q}{4\pi R^2}R+\frac{Q}{4\pi r^2}r\right]$$

$$V=\frac{\sigma}{\epsilon_o}(R+r)$$

Answer: C

169. The electric potential in space is given by, $v = 4x^2$. The electric field at (x, y, z) is (1m, 0m, 2m), what is the field.

A) 8v/m along negative x –axis
B) 8v/m along positive x-axis
C) 16v/m along positive x-axis
D) 16v/m along positive x-axis

Solution: $\Delta E=\frac{\Delta v}{\Delta d}\Rightarrow\Delta E=\frac{dv}{dx}=\frac{dv}{dx}=-8x=-8(1)=-8\text{v/m}$

Answer: A

170. Two concentric spheres of radii r_1 and r_2 ($r_1 > r_2$) carry charges q_1 and q_2 respectively. The electric potential at a distance $r(r_2 < r < r_1)$ will be $\frac{1}{4\pi\epsilon_o}$ of

A) $\frac{q_1+q_2}{r}$ C) $\frac{q_1}{r_1}+\frac{q_2}{r_2}$

B) $\frac{q_1}{r_1}+\frac{q_2}{r}$ D) $\frac{q_1}{r_2}+\frac{q_2}{r_1}$

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Explanation: $v = v_1 + v_2 = \frac{1}{4\pi \in_o} \frac{q_1}{r_1} + \frac{1}{4\pi \in_o} \frac{q_2}{r}$

Answer: B

171. If a positive charge is shifted from a low potential region to a high potential region, the electric potential energy.
A) increase C) remain same
B) decreases D) may increase or decrease

Explanation: U = qv, as v increase, U increases. **Answer: A**

172. Twenty seven drops of water of the same size are equally and similarly charged. They are then united to form a bigger drop. By what factor will the electric potential change?
A) 9 times C) 6 times
B) 27 times D) 3 times

Explanation: $27\left(\frac{4}{3}\pi r^3\right) = \frac{4}{3}\pi R^3 = R = 3r$

$$\frac{v^1}{v} = \frac{27q}{3r} \times \frac{r}{q} = 9$$

Answer: A

173. A 5C charge experiences a force of 2000N when moved between two points separated by a distance of 2cm in a uniform electric field. The potential difference between the two points is
A) 8v B) 200v C) 800v D) 20,000v

Explanation: $E = \frac{F}{Q} = \frac{2000}{5} = 400N/C$

$$V = Er = (400)\left(\frac{2}{100}\right) = 8V$$

Answer: A

174. When two uncharged metal balls each of radius 0.09mm collide, two electrons are transferred between them. the potential difference between them would be
A) 16μV B) 16pv C) 32μV D) 64μV

Explanation: Potential of spheres $= \frac{\pm 2e}{4\pi \in_o r}$

Potential difference $= \frac{4e}{4\pi \in_o r}$

$$= \frac{4 \times 1.6 \times 10^{-19} \times 9 \times 10^9}{0.09 \times 10^{-3}} = 64\mu v$$

Answer: D

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175. Eight charged water drops, each with a radius of 1mm and a charge of 10^{-10}C are made to form a single drop. The potential of the big drop is

A) 3.6V B) 36V C) 360V D) 3600v

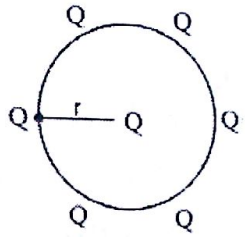
Solution: Volume is constant,

$$\Rightarrow \frac{4}{3}\pi R^3 = 8\left(\frac{4}{3}\pi r^3\right) \Rightarrow R = 2r \Rightarrow V = K\frac{8q}{2r} = 3600V$$

Answer: D

176. A point charge is surrounded symmetrically by six identical charges. How much work is done by the forces of electrostatic repulsion when the point charge at the center is taken to infinity?

A) zero B) $\frac{6Q^2}{4\pi \in_o r}$



C) $\frac{6Q}{4\pi \in_o r}$ D) $\frac{6Q}{4\pi \in_o r^2}$

Explanation: The potential energy due to individual charge is $U = \frac{Q^2}{4\pi \in_o r}$, the potential energy due to the six charges is,

$U = \frac{6Q^2}{4\pi \in_o r}$ = work done

Answer: B

177. The electrostatic potential at a distance of 40cm from the centre of a charged metal sphere of radius 50cm is 18v. The charge carried by the sphere is

A) 1mc B) 1nc C) 1μC D) 1Pc

Solution: $V = \frac{KQ}{r} \Rightarrow 18 = \frac{(9\times10^9)(Q)}{0.5} \Rightarrow Q = 1\times10^{-9}C = 1nc$

Answer: B

178. A point charge q is situated at point x between two parallel plates which have a potential difference v. What is the electric field strength at point x?

A) v/d B) $\frac{vq}{d}$ C) $\frac{Q}{4\pi \in_o a^2}$ D) $\frac{qQ}{4\pi \in_o a^2}$

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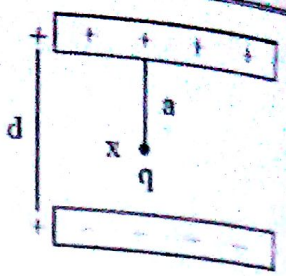
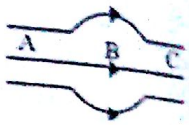
Explanation: The electric field in a parallel plate is constant. And is given by, $E = {}^{V}/_{d}$



Answer: A

179. The figure represents electric field lines. The figure suggests that:



A) $E_A > E_B > E_C$ C. $E_A = E_C > E_B$
B) $E_A = E_B = E_C$ D. $E_A = E_C < E_B$

Explanation: The closer the electric field lines the stronger the electric field strength.

Answer: C

180. On brining 0.4C of charge from infinity to point P the work done is 20J. The potential at point P is?

A) 50v B) 8v C) 1000v d) 160v

Solution: $V = \frac{W}{Q} = \frac{20}{0.4} = 50v$

Answer: A

181. value of potential at a point due to a point charge is
A) invesely proportional to square of the distance
B) directly proportional to square of the distance
C) invesely proportional to the distance
D) directly proportional to the distance

Explanation: $V = \frac{KQ}{r}$

Answer: C

182. If two conducting spheres are separately charged and then brought in contact, then
A) the total energy of the sphere is conserved
B) the total charge on the spheres is conserved
C) both total energy and charge are conserved
D) the final potential is the average of the two sphere intitial potnetials.

Answer: B

183. Work done in moving a positive charge on an equipotnetial surface is

A) positive B) negative C) zero D) infinite

Explanation: $W = Q\Delta V = \Delta V \Rightarrow W = 0$

Answer: C

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184. A hollow metallic sphere of radius 3cm is charged so that potential on its surface becomes 5v. The potential at the center of the sphere will be

A) zero B) 5 C) 3 D) 10

Answer: B

185. The electric potential in a region along x-axis varies with x according to the relation $v(x) = 4 + 5x^2$. What is the electric field at x = 2m.

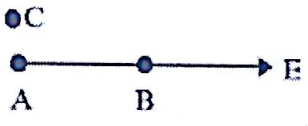
A) 20v/m along the +ve x−axis C) 24v/m along –ve x−axis
B) 20v/m along –ve x−axis D) 24v/vm along the x−axis

Solution: $E = \frac{\Delta V}{\Delta x} = \frac{-dv}{dx} = 4 + 10x$ at $x = 2$

$E = 4 + (10)(2) = 24v/m$

Answer: C

186. A uniform electric field pointing in x direction exists. Let A be the origin, B is at x = 1 and C is at y = 1cm. Then the potential at points A, B and C satisfy.

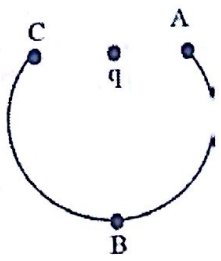


A) $V_A < V_B$ C) $V_A < V_C$
B) $V_A > V_B$ D) $V_A > V_C$

Explanation: Potential decreases in the direction of the field.

Answer: B

187. The work done in taking a point charge Q from P to A is W_A, from P to B is W_B and from P to C is W_C. Then



A) $W_A < W_B < W_C$
B) $W_A > W_B > W_C$
C) $W_A = W_B = W_C$
D) $W_A > W_B < W_C$

Explanation: The semi-circle is an equipotential line So, $V_{PA} = V_{PB} = V_{PC}$. Since $W = QV$, therefore, $W_A = W_B = W_C$

Answer: C

188. An electron is released from rest in a region of space with non zero field. As the electron moves, it experience

A) an increasing electric potential
B) decreasing electric potential
C) the same electric potential
D) decreasing electric potential energy

Explanation: The electron moves in a direction opposite to the electric field, hence its potential increases.

Answer: A

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Capacitor

189. The air between the plates of a parallel plate capacitor is replaced by a medium of dielectric constant K. The capacitance becomes

A) K times
C) $\sqrt{K}$ times
B) $\frac{1}{K}$ times
D) $\frac{1}{\sqrt{K}}$ times

Explanation: $C = \frac{K \in_o A}{d}$ **Answer: A**

190. If we treat the earth as a conducting sphere of radius 6400km, its capacitance would be the order of

A) 1μF B) 1mF C) 1F D) 10^3F

Explanation: $C = \frac{Q}{V} = \frac{Q}{\frac{KQ}{r}} = \frac{6400 \times 1000}{9 \times 10^9} \simeq 1\text{mF}$ **Answer: B**

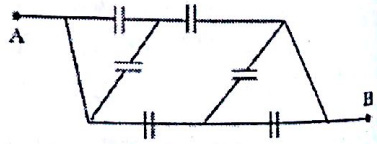
191. The capacitance of a capacitor does not depend on

A) the shape of the plates
B) the size of the plates
C) the charges on the plates
D) the separation between the plates

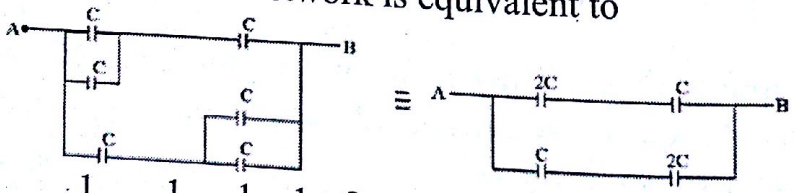
Explanation: $C = K \in_o \frac{A}{d}$ **Answer: C**

192. A network of six identical capacitors, each of value C is made as shown. The equivalent capacitance between points A and B is

A) $\frac{C}{4}$ B) $\frac{3C}{4}$
C) $\frac{4}{3}C$ D) 3C

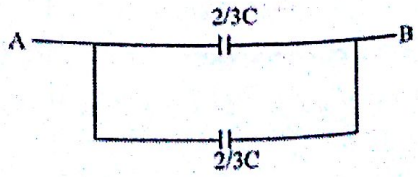


Solution: The network is equivalent to



$$\Rightarrow \frac{1}{CT_1} = \frac{1}{2C} + \frac{1}{C} = \frac{1+2}{2C} = \frac{3}{2C} \Rightarrow CT_1 = \frac{2}{3}C$$

$$C_T = 2CT_1 = \frac{4}{3}C,$$



Answer: C

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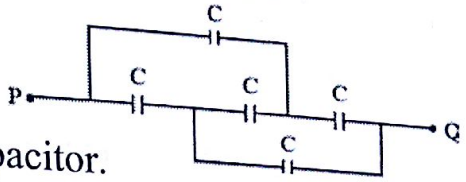
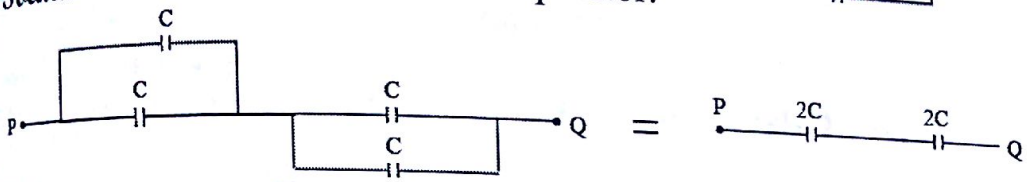
193. What is the equivalent capacitance between points P and Q

A) $C/5$ B) $C/3$

C) $C/2$ D) C



Solution: Take out the common capacitor.



$$\frac{1}{C_T}=\frac{1}{2C}+\frac{1}{2C}\Rightarrow\frac{1}{C_T}=\frac{2}{2C}\Rightarrow C_T=C$$

Answer: D

194. A capacitor which has a capacitance of 1 Farad will

A) be fully charged in 1 sec by a current of 1A

B) store 1C of charge at a potential difference of 1V

C) gain 1J of energy when 1C of charge is stored on it

D) discharge in 1 sec when connected to a 1Ω resistor.

Explanation: $C=\frac{Q}{V}\Rightarrow 1F=\frac{1C}{1V}$ **Answer: B**

195. In a charged capacitor, the energy is stored in

A) the electric field between the plates

B) the edges of the plates

C) positive charges

D) both in positive and negative charges.

Explanation: Field is the store house of energy. **Answer: A**

196. Three identical capacitors are combined differently. For the same voltage to each combination, the one that stores the greatest energy is

A) Two in parallel C) Three in parallel

B) Three in series D) Two in series

Explanation: $U=\frac{1}{2}CV^2\Rightarrow U\sim C$ **Answer: C**

197. Two capacitors of equal capacitance are first connected in parallel and then in series. The ratio of the total capacitance in the two cases is

A) 2:1 B) 1:2 C) 4:1 D) 1:4

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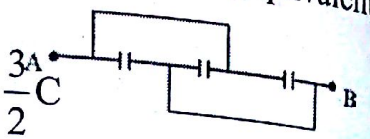
Explanation: $C_P = C + C = 2C,\ C_S = \frac{C^2}{2C} = \frac{C}{2} \Rightarrow \frac{C_P}{C_S} = \frac{2C}{\frac{C}{2}} = \frac{4}{1}$

Answer: C

198. Three equal capacitors are connected as shown. The equivalent capacitance between A and B is

A) C B) 3C C) $\frac{C}{3}$ D) $\frac{3}{2}C$



Explanation: The three capacitors are in parallel. **Answer: B**

199. A parallel plate capacitor has circular plates of 0.08m radius and 1×10^{-3}m separation. If a potential difference of 100V is applied, the charge will be

A) 4.8×10^{-10}C
B) 1.8×10^{-8}C
C) 11.8×10^{-20}C
D) 0C

Solution: $C = \epsilon_o \frac{A}{d} = \frac{\left(8.85 \times 10^{-12}\right)\left(\pi \times 10^{-6}\right)}{0.08} = 1.8 \times 10^{-10}\text{F}$

$Q = CV = 1.8 \times 10^{-8}\text{C}$

Answer: B

200. Two capacitors, each of capacity 2μF, are connected in parallel. This system is connected in series with a third capacitor of 12μF. The equivalent capacitance of the system will be

A) 16μF B) 13μF C) 4μF D) 3μF

Solution: $C_{T1} = 2 + 2 = 4\mu\text{F}$

$\frac{1}{C_T} = \frac{1}{4} + \frac{1}{12} = \frac{4}{12} = \frac{1}{3} \Rightarrow C_T = 3\mu\text{F}$

Answer: D

201. The capacitance of a parallel plate capacitor is 5μF. when a glass plate is introduced, the potential becomes $\frac{1}{8}$ of the original value. The value of dielectric constant is

A) 1.6 B) 5 C) 8 D) 40

Solution: $V = \frac{V_o}{K} \Rightarrow \frac{1}{8}V_o = \frac{V_o}{K} \Rightarrow K = 8.$

Answer: C

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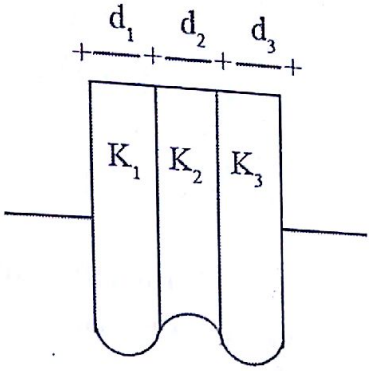
202. Three dielectrics are in a parallel plate capacitor. The capacitance of the capacitor will be



A) $\dfrac{\epsilon_o A}{\frac{d_1}{K_1}+\frac{d_2}{K_2}+\frac{d_3}{K_3}}$

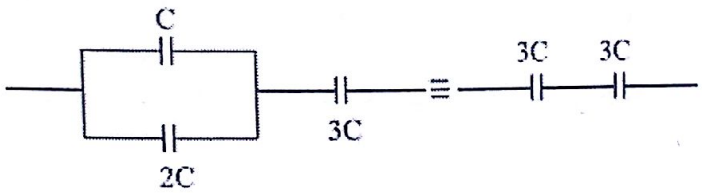
B) $\dfrac{\epsilon_o A}{d_1+d_2+d_3}$

C) $\dfrac{\epsilon_o A(K_1 K_2 K_3)}{d_1 d_2 d_3}$

D) $\epsilon_o A\left(\dfrac{K_1}{d_1}+\dfrac{K_2}{d_2}+\dfrac{K_3}{d_3}\right)$

Explanation: The effective distance is $\dfrac{d}{K}$. **Answer: A**

203. Two capacitors of C and 2C capacity are connected in parallel and then connected in series with a 3C capacitor. If the combination is supplied with V volts, the charge in capacitor C is



A) $\frac{1}{2}CV$

B) CV

C) $2CV$

D) $\frac{3}{2}CV$

Explanation: Given that, the potential is equally divided, $\dfrac{V}{2}$,

hence the charge on C is $Q=\dfrac{CV}{2}$. **Answer: A**

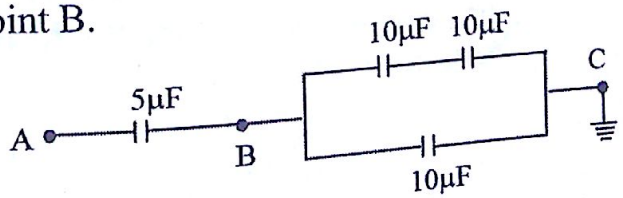
204. If a potential of 2000V is given at A and point C is grounded. What is the potential at point B.

A) 1500V

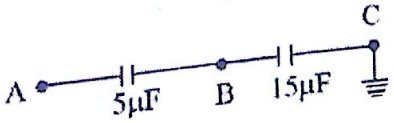
B) 1000V

C) 500V

D) 400V



Solution: The circuit is reduced to



charge is same $\Rightarrow Q_{AB}=Q_{BC}$

$C_{AB}V_{AB}=C_{BC}V_{BC}$

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$$\frac{V_{AB}}{V_{BC}} = \frac{C_{BC}}{C_{AB}} = \frac{15}{5} = 3$$

$\Rightarrow V_{AB} = 3V_{BC}$

But, $V_{AB} + 3V_{BC} = 2000V \Rightarrow 4V_{BC} = 2000 \Rightarrow V_{BC} = 500V$

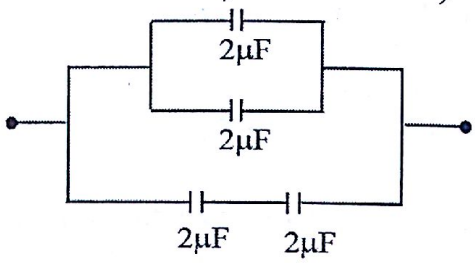
$V_{AB} = 3 \times 500 = 1500V$, but $V_{AB} = V_A - V_B = 500V$

Answer: C

205. The minimum number of capacitors 2μF each required to obtain a capacitance of 5μF will be

A) 3 B) 4 C) 5 D) 6

Explanation:



Answer: B

206. A parallel plate capacitor of 12PF gets its area doubled and its separation halfed. The capacitance will then be

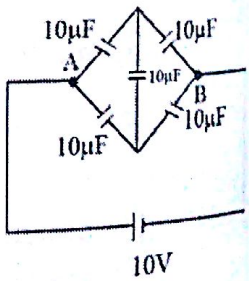
A) 124PF B) 48PF C) 196PF D) 136PF

Solution: $C = \epsilon_o \frac{A}{d} = C_2 = \epsilon_o \frac{2A}{\frac{d}{2}} = 4\left(\epsilon_o \frac{A}{d}\right) = 4(12) = 48PF$

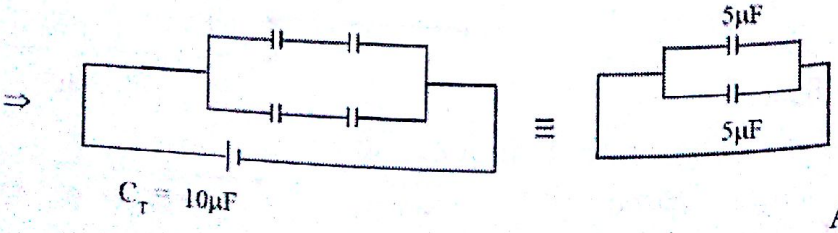
Answer: B

207. For the network shown, find the equivalent capacitance between points A and B.

A) 40μF B) 20μF C) 30μF D) 10μF



Explanation: Think interms bridge arrangement. Hence, ignore the balancing capacitor.



Answer: D

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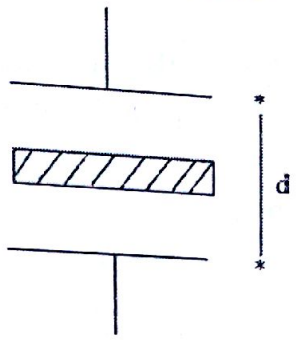
208. Which one of the following methods will reduce the capacitance of a parallel plate capacitor?
A) connecting another capacitor in series with this
B) Reducing the potential difference between the plates
C) Introducing a dielectric between the plates
D) Introducing a metal plate between the plates

Answer: A

209. A slab of copper of thickness $t = \frac{d}{2}$ is inserted between the plates of a capacitor. The ratio of capacitances after and before inserting the slab is



A) $\sqrt{2}:1$ C) 1:1
B) 2:1 D) $1:\sqrt{2}$

Explanation: effective distance is halfed. ⇒Capacitance is doubled.

Answer: B

210. Three capacitors of capacitance 6μF each are available. The minimum and maximum capacitance which may be obtained are
A) 6μF, 19μF C) 26μF, 12μF
B) 3μF, 12μF D) 2μF, 18μF

Explanation: $\frac{1}{C_S}=\frac{1}{6}+\frac{1}{6}+\frac{1}{6} \Rightarrow C_s = 2\mu F$

$C_P = 6 + 6 + 6 = C_P = 18\mu F$

Answer: D

211. While a capaictor remains connected to a battery, a dielectric slab is slipped between the plates.
A) the potential difference between the plates is charged
B) charges flow from the battery to the capacitor
C) the electric field between the plates increases
D) the energy stored in the capacitor decreases

Answer: B

212. A parallel plate capacitor is filled with a uniform dielectric. Maximum charge that can be given to it does not depend on
A) dielectric constant C) Separation between plate
B) dielectric strength D) area of plate

Explanation: $\frac{Q_{max}}{K \in_o A}$ = dielectric strength

Answer: C

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213. In a parallel plate capacitor, a dielectric slab is introduced between the plates. If a battery is connected and the slab is taken out
A) some charge is drawn from the battery
B) some charge is returned to the battery
C) the potential difference is reduced in the capacitor
D) No work is done by an external agent to take the slab out
Explanation: When the slab is taken, C is reduced, so does Q hence some charge will be returned to the battery.

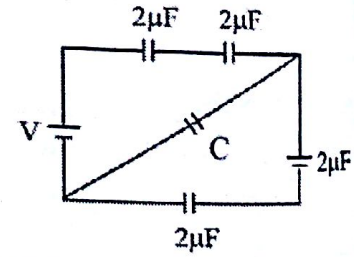
Answer: B

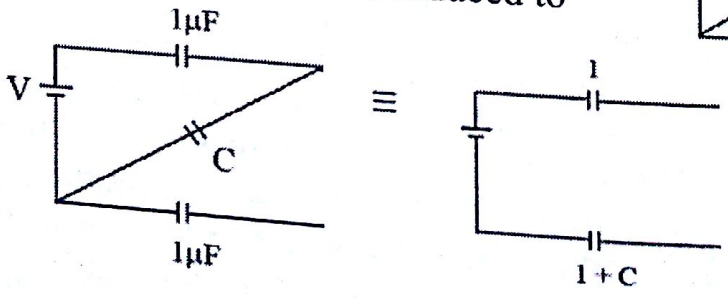
214. A metal slab of negligible thickness is placed between the plates of a parallel plate capacitor. Which of the following is correct?
A) the battery will supply more charge
B) the capacitance will increase
C) the potential difference between plates will increase
D) equal and opposite charges will appear on the face of metal plate.
Explanation: Since the thickness is taken to be zero.

Answer: D

215. When a potential difference of 10^3V is applied, a charge of 0.75mC is stored in the network. The value of C in μF is
A) 3 C) 2.5
B) 2 D) 0.5

Solution: The network is reduced to





$$C_T = \frac{(1+C)1}{(1+C)+1}$$

$$Q = C_T V \Rightarrow C_T = \frac{Q}{V} \Rightarrow \frac{0.75\times10^{-3}}{10^3} = \frac{(1+C)1}{2+C}$$

$$0.75\times10^{-6} = \frac{1+C}{2+C} \Rightarrow C = 2\mu F$$

Answer: B

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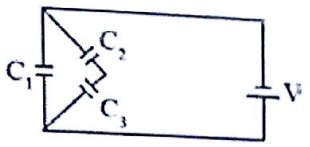
216. A parallel plate capacitor is charged to a potential difference V. After disconnecting the battery, the distance between the plates in increased. The potential difference between the plates will
A) decrease B) does not change
C. become zero D. increase

Explanation: As distance increase, capacity decrease, charge constant, potential increase.

Answer: D

217. The equivalent capacitance of the network shown is



A. $C_1 + C_2 + C_3$

B) $\dfrac{C_1C_2C_3}{C_1+C_2+C_3}$

C) $\dfrac{C_1C_2+C_2C_3+C_3C_1}{C_2+C_3}$

D) $C_1C_2\,C_3$

Explanation: C_2 and C_3 are in series and their equivalent capacitance is parallel to C_1.

$$\Rightarrow C_1+\frac{C_2C_3}{C_2+C_3}=\frac{C_1C_2+C_2C_3+C_3C_1}{C_2C_3}$$

Answer: C

218. If radius of circular plates of a capacitor is halved, its capacitance
A) halved B) doubled C) quadrupled D) one-quarter

Explanation: $C=\dfrac{\epsilon_o A}{d}$, $A=\pi r^2$. When $r^1=r/2 \Rightarrow C^1=\dfrac{1}{4}C$

Answer: D

219. To obtain 3μF capacitance from three capacitors of 2μF each. They should be.
A) All in series C) two in series and the third in parallel
B) All in parallel D) two in parallel and one in series with them

Answer: C

220. A conducting sphere of radius R carrying a charge Q is joined to uncharged conducting sphere of radius 2R. The charge flowing between them will be

A) $Q/4$ B) $\dfrac{Q}{3}$ C) $\dfrac{Q}{2}$ D) $\dfrac{2}{3}Q$

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Explanation: If C_1 and C_2 denote the capacitance of spheres o radii R and 2R. Then, $C_2 = 2C_1$

$$\Rightarrow V = \frac{Q}{C_1 + C_2} \text{ and the charge on } R_2 \text{ is,}$$

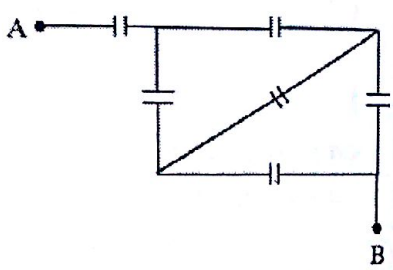
$$Q_2 = (V)\,(C_2) = \frac{Q}{C_1 + C_2} C_2 = \frac{2C_1 Q}{3C_1} = \frac{2}{3} Q$$

Answer: D

221. If the capacitanc of each capacitor is 1μF. The effective capacitance of the network between points A and B is

A) $\frac{2}{3}\mu F$ B) $\frac{3}{2}\mu F$ C) $\frac{1}{6}\mu F$ D) $6\mu F$

Explanation: The given network can be reduced in to a series connection of a 1μF and a 2μF capacitor.



Answer: A

222. 125 identical drops, each charged to a 50v are combined to form a signle drop. The potential of the big drop will be

A) 50V B) 250V C) 500V D) 1250V

Explanation: $\frac{4}{3}\pi R^3 = 125 \times \frac{4}{3}\pi r^3 \Rightarrow R = 5r \Rightarrow C = \frac{5q}{v}$

$$\Rightarrow \frac{125q}{V} = \frac{5q}{v} \Rightarrow V = 25v = 25 \times 50 = 1250V.$$

Answer: D

223. What percentage of the energy drawn from the charging battery is stored in a capacitor?

A) 100% B) 75% C) 50% D) 25%

Explanation: energy stored $\Rightarrow \frac{1}{2}CV$

Answer: C

224. What is the total energy stored in the network shown

A) 0 B) $\frac{1}{2}CV^2$ C) CV^2 D) $2CV^2$

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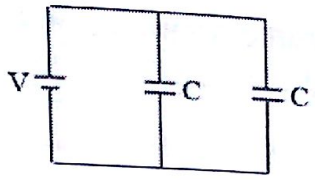
Explanation: $U=\frac{1}{2}C_T V.\ C_T=2C$

$\Rightarrow U=CV^2$



Answer: C

225. The areas of the plates of a capacitor is A. And the distance between the plates is 10mm. If there are two dielectric with dielectric constant 10 and thickness 6mm and the other with dielectric constant 5 and thickness 4mm. The capacitance of the capacitor is

A) $\frac{12}{35}\epsilon_o A$ C) $\frac{5000}{7}\epsilon_o A$

B) $\frac{2}{3}\epsilon_o A$ D) $1500\,\epsilon_o A$

Explanation: $C=\frac{\epsilon_o A}{\frac{t_1}{K_1}+\frac{t_2}{K_2}}=\frac{\epsilon_o A}{\frac{6\times10^{-3}}{10}+\frac{4\times10^{-3}}{5}}=\frac{5000}{7}\epsilon_o A$

Answer: C

226. A 10μF capacitor is charged to a 50V and is connected to uncharged capacitor in parallel and the common potential is found to be 20V. The capacitance of the second capacitor is

A) 10μF B) 20μF C) 30μF D) 15μF

Solution: $V=\frac{Q}{C}=\frac{Q_1+Q_2}{C_1+C_2}=\frac{C_1V_1+C_2V_2}{C_1+C_2}$

$20=\frac{(10)(50)+(0)(C_2)}{10+C_2}\Rightarrow C_2=15\mu F$ **Answer: D**

227. A capacitor of 20μF charged to 500V is connected in parallel with a capacitor of 10μF charged to 200V. The common potential is

A) 100V B) 200V C) 300V D) 400V

Explanation:

$V=\frac{Q}{C}=\frac{Q_1+Q_2}{C_1+C_2}=\frac{C_1V_1+C_2V_2}{C_1+C_2}=\frac{(20)(500)+(10)(200)}{20+10}=400V$

Answer: D

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228. Force of attraction between the plates of a parallel plate capacitor is

A) $\frac{q^2}{2\epsilon_o AK}$ C) $\frac{q}{2\epsilon_o A}$

B) $\frac{q^2}{K\epsilon_o A}$ D) $\frac{q^2}{2\epsilon_o A^2 K}$

Explanation: $E = \frac{\sigma}{\epsilon_o} = \frac{q}{K\epsilon_o A}$ and $F = Eq = \frac{q^2}{K\epsilon_o A}$

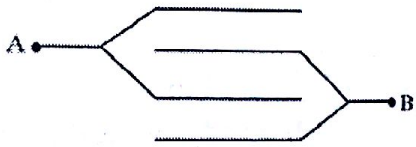
Answer: B

229. Four plates, each of area A and separation d are shown below. The capacitance between A and B is

A) $\frac{\epsilon_o A}{d}$ B) $\frac{2\epsilon_o A}{d}$ C) $\frac{3\epsilon_o A}{d}$ D) $\frac{4\epsilon_o A}{d}$

Explanation: There are three parallel plate capacitors which are connected in parallel.

$$C = C_1 + C_2 + C_3 = \frac{3\epsilon_o A}{d}$$



Answer: C

230. If the potential at A is V_1 and that at B is V_2, then the potential at D will be

A) $\frac{1}{2}(V_1 + V_2)$

B) $\frac{C_1V_2 + C_2V_1}{C_1 + C_2}$

C) $\frac{C_1V_1 + C_2V_2}{C_1 + C_2}$

D) $\frac{C_2V_1 - C_1V_2}{C_1 + C_2}$

Explanation: $\Delta Q_1 = \Delta Q_2 \Rightarrow \Delta V_1 C_1 = \Delta V_2 C_2$

$$\Rightarrow (V_1 - V)(C_1) = (V - V_2)C_2 \Rightarrow V = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$$

Answer: C

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231. A 500μF capacitor is charged at a constant rate of $100\frac{\mu C}{S}$. How long will it take to raise the potential between the plates to 10V?

A) 5 sec B) 10sec C) 50sec D) 100sec

Solution: $Q = CV = (500 \times 10^{-6})(10) = 5 \times 10^{-3} C$, and

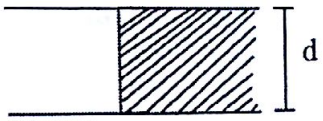
$$Q = qt = t = \frac{Q}{q} = \frac{5 \times 10^{-3}}{100 \times 10^{-6}} = 50sec$$

Answer: C

232. A dielectric of constant K is introduced in half the length of a parallel plate capacitor of plate area A and plates separation d what is the capacitance of this capacitor

A) $\frac{2\epsilon_o A}{d}$ C) $\epsilon_o A\frac{(1+k)}{2d}$

B) $\epsilon_o A\frac{(1+k)}{2d}$ D) $\frac{3\epsilon_o A}{d}$



Solution: C = Air filled, area $A/2$ + Dielectric field area $\frac{A}{2}$

$$C = \frac{\epsilon_o A}{2d} + \frac{K\epsilon_o A}{2d} \Rightarrow C = \frac{\epsilon_o A}{2d}(1+K)$$ **Answer: C**

233. Two capacitors, each of capacitance C and break down voltage V, are joined in sereis. The capacitance and breakdwon voltage of the combination are respectively.

A) 2C and V C) $\frac{C}{2}$ and $\frac{V}{2}$

B) 2C and 2V D) $\frac{C}{2}$ and 2V

Explanation: capacitance $= \frac{C}{2}$

If V^1 is breakdown, $CV = \frac{C}{2}V^1 = V^1 = 2V$ **Answer: D**

234. The distance between the plates of a parallel plate capacitor is 4mm and potential difference is 60V. If the distance between the plates is increased to 12mm, then

A) the potential difference will be unchanged

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B) the potential difference will become 20v
C) the potential difference will become 180v
D) the charge on capacitor will reduce to one-third

Explanation: $Q = CV = \frac{\epsilon_o AV}{d} \Rightarrow V = \frac{Qd}{\epsilon_o A}$, thus $V \sim d$. Since d is increased 3 times, so did the potential difference

$$V = 3 \times 60 = 180V$$

Answer: C

235. A parallel plate capacitor with air has a capacitance of 10μF. The area of the capacitor is divided into two equal parts and filled with dielectrics of constants $K_1 = 2$ and $K_2 = 4$. The capacitance of the system will now be

A) 40μF B) 30μF C) 20μF D) 10μF

Explanation: $C = C_1 + C_2 \Rightarrow 2\left(\frac{10}{2}\right) + 4\left(\frac{10}{2}\right) = 30\mu F$

Answer: B

236. Eight indentical capacitors when connected in series have equivalent capacitance of C . The equivalent capacitance when they are connected in parallel is

A) 8C B) 64C C) $C/8$ D) $C/64$

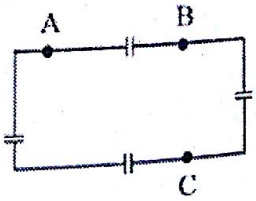
Explanation: $C = C_1 + C_2 + C_3 + C_4 + C_5 + C_6 + C_7 + C_8 = 8C$

Answer: A

237. The ratio of the equivalent capacitance between A and B and between A and C will be (each capacitance = 3μF).

A) 4:3 B) 3:4 C) 2:3 D) 3:2

Explanation: $\frac{C_{AB}}{C_{BC}} = \frac{3+1}{3} = 4:3$



Answer: A

238. The capacitance and the energy stored in parallel plate capacitor when air filled is C_o and U_o. If the capacitor is filled with a dielectric of constant 5, the capacitance and energy will be

A) $5C_o, 5U_o$ C) $\frac{C_o}{5}, 5U_o$

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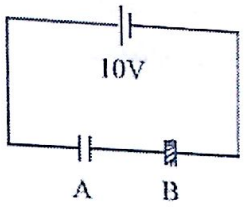
B) $5C_o, \frac{U_o}{5}$ D) $\frac{C_o}{5}, \frac{U_o}{5}$

Explanation: $C = 5C_o$ and $U = \frac{Q}{2C} = \frac{Q}{2(5C_o)} = \frac{U_o}{5}$

Answer: B

239. Capacitors A and B have identical geometry. If there is a dielectric material of constant 3 in capacitor B, the potential difference across A and B is respectively

A) 2.5V, 7.5V C) 7.5V, 2.5V
B) 2V, 8V D) 8V, 2V



Explanation: $C_A = C$, and $C_B = 3C$

$Q_A = Q_B \Rightarrow V_A C = V_B(3C) \Rightarrow V_A = 3V_B$

$V = V_A + V_B \Rightarrow 10 = 3V_B + V_B \Rightarrow 2.5V, V_A = 7.5V$

Answer: C

240. The p.d across the plates of a capacitor is 60v and its capacitance is 12μF. Another capacitor of 24μF is connected with it in seires and its free plate is earthed. The p.d across the 12μF will be

A) 60V B) 40V C) 20V D) 90V

Explanation: first case, $Q = (12)(60) = 720\mu C$

second case, $V = \frac{720}{12} = 60V$

Answer: A

241. A capacitor has plate area A, separatin d, dielectric constant K and charge q. What is the electric field intensity between the plates.

A) $\frac{q}{KA\epsilon_o}$ B) $\frac{Kq}{A\epsilon_o}$ C) $\frac{Kq\epsilon_o}{Ad}$ D) $\frac{\epsilon_o q}{KAd}$

Explanation: $q = CV = CEd \Rightarrow E = \frac{q}{Cd} = \frac{q}{\left(\frac{K\epsilon_o A}{d}\right)d} = \frac{q}{K\epsilon_o A}$

Answer: A

242. A parallel plate capacitor has capacitance C. And, when the plate separation is doubled and a dielectric is introduced between the plates, the capacitance become 2C. The dielectric constant is

A) 2 B) 1 C) 4 D) 8

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Explanation: $C = \frac{\epsilon_o A}{d}$ and $2C = \frac{K \epsilon_o A}{2d} = \frac{KC}{2} \Rightarrow K = 4$

Answer: C

243. A capacitor has separation d and capacitance C. If a matallic foil of thickness $d/3$ is introduced between the plates, the capacitance would become

A) 3C B) 1.5C C) C D) 0.7C

Explanation: Effective separation,

$$d - \frac{d}{3} = \frac{2d}{3} \Rightarrow C^1 = \frac{\epsilon_o A}{2d/3} = \frac{3}{2}C$$

Answer: B

244. A 10μF capacitor is charged by a battery of 100V. The energy drawn from the battery and the energy stored in the capacitor, are respectively

A) 0.1J and 0.05J
B) 0.05J and 0J
C) 1mJ and 0.5mJ
D) 0.05J and 0.05mJ

Explanation:

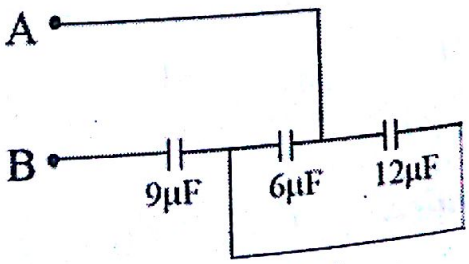
Energy stored in capcitor $= \frac{1}{2}CV^2 = \frac{1}{2}\left(10 \times 10^{-6}\right)(100) = 0.05J$

Energy supplied by the battery $= CV = \left(10 \times 10^{-6}\right)(100) = 0.1J$

Answer: A

245. The equivalent capacitance of the given network is

A) 27μF
B) 15μF
C) 6μF
D) 30μF



Solution: The 6μF and 12μF are in parallel

$\Rightarrow C = 6 + 12 = 18\mu F$

The equivalent of C and the 9μF are in series

$$\frac{1}{C_T} = \frac{1}{C} + \frac{1}{9} \Rightarrow \frac{1}{C_T} = \frac{1}{18} + \frac{1}{9} \Rightarrow C_T = 6\mu F$$

Answer: C

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246. A 4μF capacitor is connected in parallel to another capacitor of 8μF. Both are then connected in series with a 12μF capacitor and charged to 20V. What is the charge on the 4μF capacitor?

A) 3.3μC B) 40μ C) 80μC D) 240μC

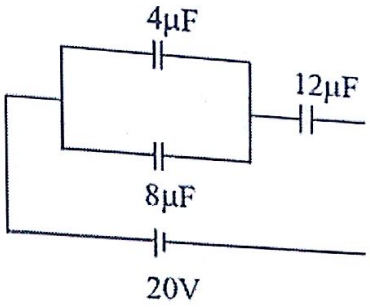
Solution: $C = 4 + 8 = 12\mu F$

$$\frac{1}{C_T} = \frac{1}{12} + \frac{1}{12} \Rightarrow C_T = 6\mu F$$

$$Q = CV = \left(6\times10^{-6}\right)(20) = 120\mu C$$

$Q_4 = C_4V_4$, but $V_4 = V - 10 = 10V$

$$Q_4 = (4\times10)10^{-6} = 40\mu C$$



Answer: B

247. Three capacitors of 2μF, 3μF and 6μF are connected in series to a 10V source. the charge on the 3μF capacitor is

A) 5μC B) 10μC C) 12μC D) 15μC

Solution:
$$\frac{1}{C_T} = \frac{1}{2} + \frac{1}{3} + \frac{1}{6} \Rightarrow C_T = 1\mu F$$

$$Q = CV = \left(1\times10^{-6}\right)(10) = 10\mu C$$

Answer: B

248. Two identical capacitors are joined in parallel, charged to a potential V and then separated and then connected in series. which of the following is correct?

A) The charges on the plates destroyed
B) The charges on the plates increased
C) The energy stored in the system increased
D) The potential difference in the plates is 2V

Explanation: charge is conserved

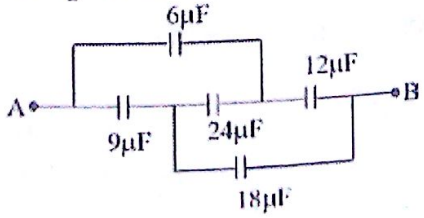
$$CV + CV = CV_1 + CV_2$$

$$\Rightarrow V_1 + V_2 = 2V$$

Answer: D

249. What is the equivalent capacitance between A and B

A) 11μF
B) 69μF
C) 15μF
D) 10μF



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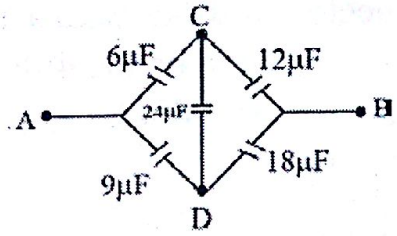
Explanation: The equivalent circuit is



There is no potential difference across C and D.

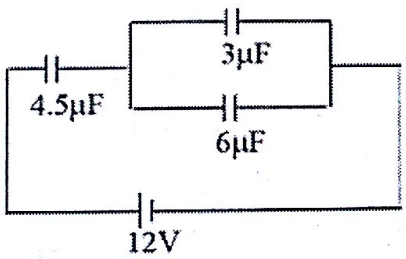
$$\Rightarrow \frac{1}{C_1}=\frac{1}{6}+\frac{1}{12} \Rightarrow C_1 = 4\mu F \text{ and } \frac{1}{C_2}=\frac{1}{9}+\frac{1}{18} \Rightarrow C_2 = 6\mu F$$

$$C = C_1 + C_2 = 4 + 6 = 10\mu F$$

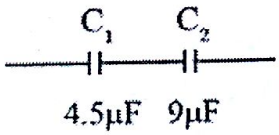
Answer: D

250. The potential difference across the 4.5μF capacitor is

A) 3V
B) 6V
C) 4V
D) 8V



Solution: We can have

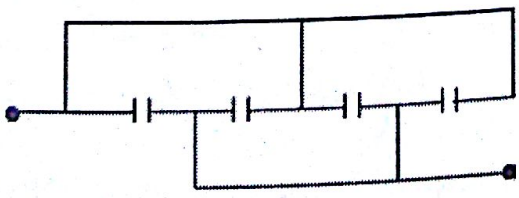


$$Q_1 = Q_2 \Rightarrow C_1V_1 = C_2V_2 \Rightarrow \frac{V_1}{V_2}=\frac{C_1}{C_2}=\frac{9}{4.5}=\frac{2}{1}=\frac{8}{4} \Rightarrow V_1 = 8V$$

Answer: D

251. The eqivalent capacitance of four capacitors of 8μF capacity each is

A) 32μF
B) 2μF
C) 8μF
D) 16μF



Explanation: The connection is in parallel, 4(8), 32μF

Answer: A

252. The capacitance of a capacitor with dielectric constant 5 is C. If the dielectric is replaced by another with dielectric constant 20, the capacitance will become

A) 0.2C B) 0.5C C) 4C D) 2C

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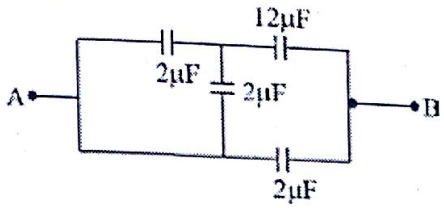
Explanation: $C_1 = 5\frac{\epsilon_o A}{d}$ and $C_2 = 20\frac{\epsilon_o A}{d}$

taking ratio, $\frac{C_1}{C_2} = 4 \Rightarrow C_2 = 4C$

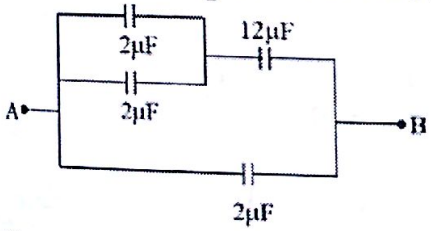
Answer: C

253. The effective capacitance between A and B will be

A) 3μF
B) 4μF
C) 5μF
D) 18μF



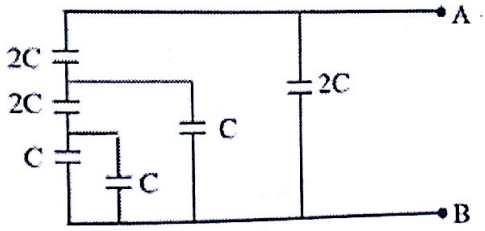
Explanation: The equivalent circuit is,



Answer: C

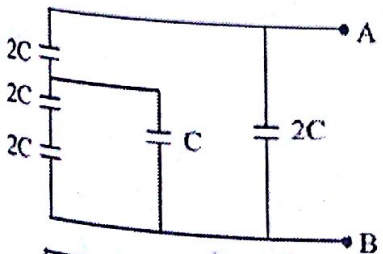
254. The effective capaictance between A and B of the given network is

A) 3C
B) 2C
C) C
D) 9C

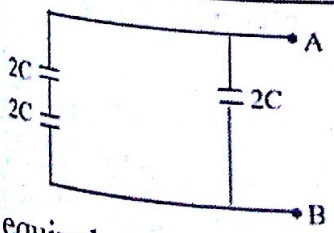


Explanation: The equivalent circuit is,

step 1



step 2



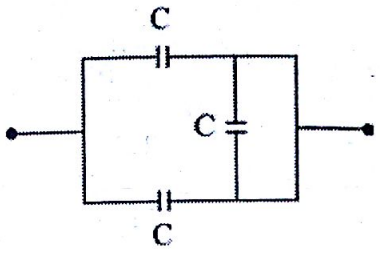
Answer: A

255. The equivalent capacitance of combination shown is

A) C
B) 2C

C) $\frac{1}{2}C$

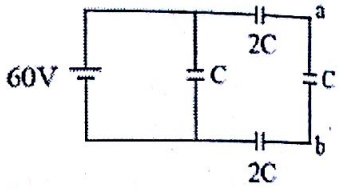
D) $\frac{1}{3}C$



Explanation: The given arrangement is a parallel combination of two capacitors. **Answer: B**

256. The potential difference between point a and b is

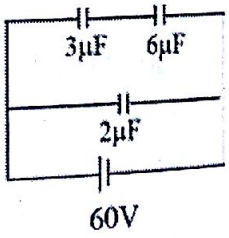
A) 60V
B) 45V
C) 40V
D) 30V



Explanation: $\frac{Q}{2C}+\frac{Q}{C}+\frac{Q}{2C}=60 \Rightarrow \frac{Q}{C}=30V$ **Answer: D**

257. The potential difference across the 3μF capacitor is

A) 20V
B) 40V
C) 45V
D) 60V



Explanation: The equivalent capacitance of the 3μF and 6μF is 2μF and the charge is 120μC. Hence the voltage across the 3μF is 40V. **Answer: B**

258. A parallel plate capacitor has charge Q, potential difference V and energy U. A dielectric slab is now inserted between the two plates. Then

A) V and U both increase
B) V and U both decrease
C) V decreases, U increase
D) V increases, U decrease

Explanation: On introducing dielectric, C increases Q remains constant V decreases and U decreases. **Answer: B**

259. A capacitor of capacitance 10μF has been charged to 100V. It is then connected to another uncharged capacitor in parallel. If the common potential becomes 40V. The capacitance of the other capacitor is

A) 15μF B) 5μF C) 10μF D) 17μF

Solution: Charge is conserved, $Q_1 = Q_2 \Rightarrow C_1V_1 = (C_1 + C_2)V_2$ **Answer: A**

$(10)(100) = (10 + C_2)40 \Rightarrow C_2 = 15\mu F$

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260. A capacitor having capacitance C is charged to a voltage V. It is then removed and connected in parallel with another capcitor which is uncharged. The new charge on each capacitor is now.

A) CV B) $\frac{CV}{2}$ C) 2CV D) $\frac{CV}{4}$

Solution: Charge is conserved, $Q_1 = Q_2$

$$\Rightarrow CV_1 = 2CV_2 \Rightarrow V_2 = {}^{V_1}\!/_{2} \Rightarrow Q_2 = C_2V_2 = \frac{CV}{2}$$

Answer: B

261. The capacity of a capacitor in the presence of dielectric

A) increases C) remain constant CV

B) decreases D) may increase or decrease

Explanation: $C = \in r \in_o {}^{A}\!/_{d}$ **Answer: A**

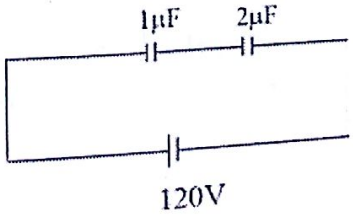
262. Two capacitors of capacitance 1μF and 2μF are connected in sereis and the system is charged to 120V. The p.d on 1μF capacitor will be

A) 40V B) 60V C) 80V D) 120V

Solution: $\frac{1}{C} = \frac{1}{1} + \frac{1}{2} \Rightarrow C = \frac{2}{3}\mu F$

$$Q = CV = \left(\frac{2}{3}\right)(120) = 80\mu C$$

$$V_1 = \frac{Q}{C_1} = \frac{80}{1} = 80V$$



Answer: C

263. A capaictor of 3μF capacitance is charged from a battery of 6V with the switch connected to terminal P. If the switch is connected to terminal Q, what will be the p.d of the system?

A) 1V B) 2V C) 4V D) 6V

Solution: When the switch is at P,

$Q = CV = (3)(6) = 18\mu C$

When the switch is at Q,

From conservation of charge,

$Q = Q_1 + Q_2 = C_1V + C_2V$

$\Rightarrow 18 = (3 + 6)V \Rightarrow V = 2V$

Answer: B

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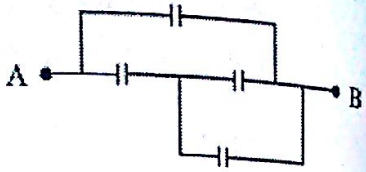
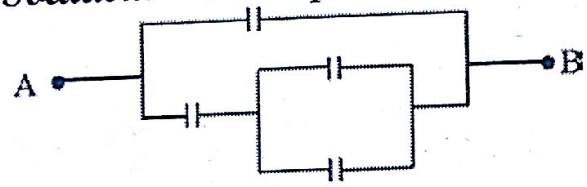
264. What is the effective capacitance between A and B if the capacitance of each capacitor is 3μF?



A) 4μF C) 6μF

B) 3μF D) 5μF

Solution: The equivalent circuit is



Answer: D

265. Two metal spheres of capacitance C1 and C2 carry some charges. If they are put in contact and then separated, then, which condition will be satisfied?

A) $\frac{Q_1}{Q_2} < \frac{C_1}{C_2}$ C) $\frac{Q_1}{Q_2} > \frac{C_1}{C_2}$

B) $Q_1 / Q_2 = \frac{C_1}{C_2}$ D) $\frac{Q_1}{Q_2} = \frac{C_2}{C_1}$

Explanation: After contact, the spheres acquire same potential

$$\Rightarrow V1 = V2 = Q_1 / Q_2 = \frac{C_1}{C_2}$$

Answer: B

266. n equal capacitors are first connected in series and then in parallel. The ratio of maximum to minimum capacitance is

A) n^2 B) n C) n^4 D) n^{-2}

Explanation: $\frac{1}{C_S} = \frac{1}{C} + \frac{1}{C} + \ldots\ldots \frac{1}{C} = \frac{n}{C}$

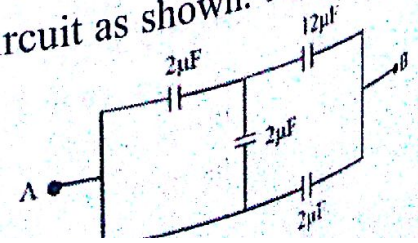
$$C_S = \frac{n}{C}$$

$$C_P = C + C + \ldots\ldots + C = nC$$

$$\frac{C_P}{C_S} = \frac{nC}{\frac{n}{C}} = n^2 \quad n^2$$

Answer: A

267. Four capacitors are connected in a circuit as shown. The effective capacitance between points A and B will be



A) 9μF C) 8μF

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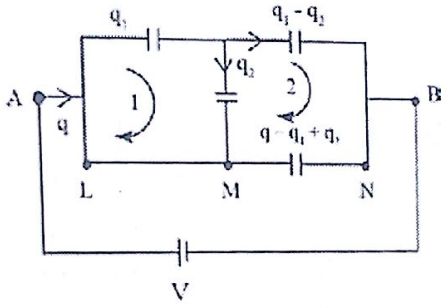
Solution: Kirchoff's law gives,

loop (1) $\frac{-q_1}{2} - \frac{q_2}{2} = 0 \Rightarrow q_1 = -q_2$

loop (2) $\frac{-q_1 - q_2}{12} + \frac{q - q_1 - q_2}{2} + \frac{q_2}{2} = 0$

$\Rightarrow -2q_1 + 2q_2 + q = 0$

$q_1 = -q_2 \Rightarrow q_1 = \frac{q}{4}$ .....(*)



Loop ALMNB

$$\frac{-q - q_1 + q_2}{2} + V = 0$$

$$V = \frac{q}{4} \quad .....(**)$$

Hence, from (*) and (**), $C_{AB} = 4\mu F$

Answer: B

Unit 12: Conceptual Questions

1. If a body is charged postively, what will happen to its mass? Explain?
2. An electron is released from rest in a region of space in an electric field. What will happen to its potential?
3. A proton is released from rest in a region of space in an electric field. What will happen to its potential?
4. How much work is required to move a charge along an equipotential line from one point to the other?
5. A parallel plate capacitor is disconnected from a battery and the plates are pulled a small distance further apart. What will happen to the charge stored, to the electric field, to the potential difference between the plates.
6. By what factor will the cpacitance of a parallel plate capacitor change if all of its dimensions are doubled?

Unit 12: Problems

Coulomb's law and electric field strength.

7. Two charges separated by a distance of 100cm causes the electrostaic force betweren the charge to increase by a factor of 4. What was their initial separation?

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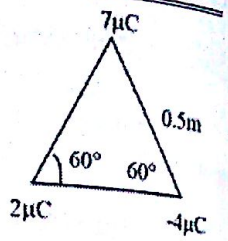
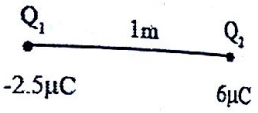
8. Calculate the net electric force on the 7μC charge? The charges are on the corners of an equilateral triangle?

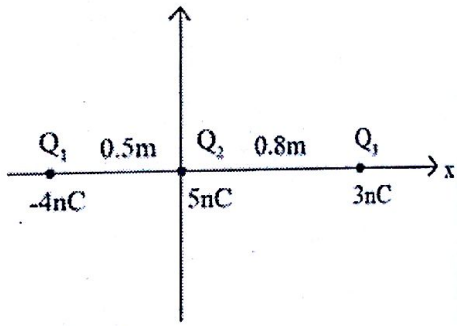


9. Determine the point at which the total electric field is zero.



10. An object with a net charge of 24μC is placed in a uniform electric field of 610N/C, directed vertically. What is the mass of the object if it "floats" in the electric field?

11. Three point charges are aligned along the x-axis.
Find the electric field at x = 2m, y = 0.



Electric potential and motion of charged particles

12. A proton accelerates from rest in a uniform electric field of 640N/C. At some later time its speed is 1.2×10^6m/s.
    a) Find the acceleration of the proton
    b) How long does it take the proton to reach this velocity?
    c) How far has it, moved in this time?
    d) What is its kinetic energy at this time?

13. The electrons in a particle beam each have a kinetic energy of 1.6×10^{-17}J. What is the electric field strength that will stop these electrons in a distance of 10cm?

14. A proton is released from rest in a uniform electric field of 6×10^5N/C directed along the positive y-axis. The proton under goes a displacement of 30cm in the direction of the electric field. Find
    a) The change in the electric potential between the plates
    b) The change in potential energy of the proton
    c) The speed of the proton after displacement

15. A proton moves in a region of a uniform electric field. The proton experiences an increase in kinetic energy of 6×10^{-14}J after being displaced 4cm in the field direction. What is the

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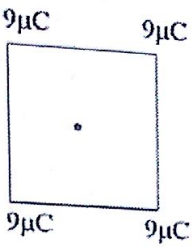
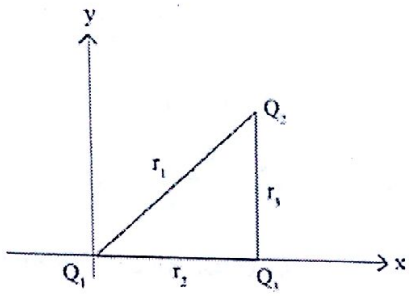
a) magnitude of the filed?
b) speed of the proton after displaced 20cm from rest
c) potential difference

16. For the charge arrangement at the corners of a square of side 2cm. Find
a) The electric field
b) The electric potential at the center of the square



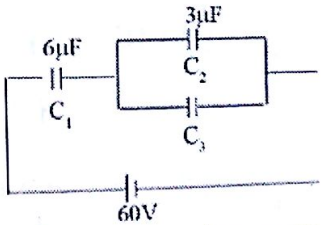
Find the electric potential energy if charges $Q_1 = 4\mu C$ is located at (0.0)m, $Q_2 = 2\mu C$ is located at (3, 4)m and charge $Q_3 = -3\mu C$ is located at (3, 0)m.



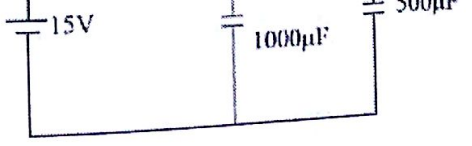
Parallel plate capacitors

17. A parallel plate capacitor is made from two aluminum foil sheet each of 4cm wide and 5cm long. Between the sheet is a Teflon ($\epsilon r = 2.1$) of 0.025mm thick. What is the capacitance?

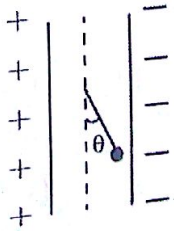
18. If the equivalent capacitance of the network shown is 4μF. Find
a) Capacitance C_3
b) charge stored on each capacitor
c) potential drop across each capacitor
d) energy stored on C_2



19. A 1000μF capacitor is charged from a 15V dc supply through a two-way switch. The switch is thrown to connect it to an uncharged 500μF capacitor as shown. Calculate
a) the initial and final charge on the 1000μF
b) the change in p. d across the 500μF capacitor after the switch is thrown
c) the initial energy stored in the 1000μF capacitor



20. A point charge of mass 0.07kg and charge 6μC is suspended by a thread between a vertical parallel plate capacitance. The plate separation is 2mm and



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the thread makes an angle of 30° with the vertical. What is the p.d between the plates?

Unit 12: Explanation to Questions

1. Postive charge is due to loss of electrons, each of which has a small mass. Hence mass decreases.
2. The electrons moves in a direction opposite to the electric field. Hence, the electron expereinces an increasing potential.
3. Its potential decreases
4. No work is done since there is no difference in potential.
5. Charge is conserved, no change. No change in the field as well but the p.d increases.
6. $C = \frac{\epsilon_o A}{d}$, area increase by 4 and separation by 2 thus, the capacitance become doubled.

Unit 12: Solution to problems

7. $F = K\frac{Q_1Q_2}{r^2} \Rightarrow F \sim \frac{1}{r^2} \Rightarrow \frac{F_1}{F_2} = \frac{r_2^2}{r_1^2} \Rightarrow \frac{1}{4} = \frac{(1)^2}{r_1^2} \Rightarrow r = 2m$

8. $F = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}$,

$$F_1 = \frac{(9\times10^9)(2\times10^{-6})(7\times10^{-6})}{(0.5)^2} = 0.5$$

$$F_2 = \frac{(9\times10^9)(4\times10^{-6})(7\times10^{-6})}{(0.5)^2} = 1 \Rightarrow F = 1.32N$$

9. $E = E_1 + E_2 = 0 \Rightarrow F = K\frac{Q_1}{(r)^2} + K\frac{Q_1}{(1+r)^2} = 0$

$$\frac{2.5}{(r)^2} = \frac{6}{(1+r)^2} \Rightarrow r = 1.8m \text{ to the left of the } -2.5\mu C.$$

10. $EQ = mg \Rightarrow m = \frac{EQ}{g} = \frac{(610)(24\times10^{-6})}{10} = 1.5\times10^{-3}Kg$

11. $E = E_1 + E_2 + E_3 = K\frac{Q_1}{r_1^2} + K\frac{Q_1}{r_2^2} + K\frac{Q_3}{r_3^2}$

$r_1 = 2.5m, r_2 = 2m \;\; r_3 = 1.2m$

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$$E_2 = 9\times10^9\left[\frac{4\times10^{-9}}{(2.5)^2}-\frac{5\times10^{-9}}{(2)^2}-\frac{3\times10^{-9}}{(1.2)^2}\right]=24\text{N/C}$$

12. a) $E=\frac{F}{q}=\frac{ma}{q}\Rightarrow a=\frac{Eq}{m}=\frac{(640)(1.6\times10^{-19})}{1.67\times10^{-27}}=6.4\times10^{10}\,\text{m/s}^2$

b) $V=0+at\Rightarrow t=\frac{V}{a}=\frac{1.2\times10^6}{6.14\times10^{10}}=1.95\times10^{-6}\,\text{sec}$

c) $S=\frac{1}{2}at^2=\frac{1}{2}(6.14\times10^{10})(19.5\times10^{-6})^2=11.7\text{m}$

d) $K_E=Eqd\Rightarrow(640)(1.6\times10^{-19})(11.7)=1.2\times10^{-15}\,\text{J}$

13. $K_E=U\Rightarrow Eqd\Rightarrow E=\frac{K_E}{qd}=\frac{1.6\times10^{-17}}{(1.6\times10^{-19})(0.1)}=10^3\text{N/C}$

14. a) $\Delta V=-Eq=-(6\times10^5)(0.3)=-1.8\times10^5\,\text{V}$

b) $\Delta U=q\Delta u=(1.6\times10^{-19})(-1.8\times10^5)=-2.88\times10^{-14}\,\text{J}$

c) $\frac{1}{2}mv^2=Eqd=\Delta V\Rightarrow V=\sqrt{\frac{2\Delta u}{m}}=\left(\frac{2\times2.88\times10^{-14}}{1.67\times10^{-27}}\right)^{1/2}$

$v=5.9\times10^6\,\text{m/s}$

15. a) $K_E=Eqd\Rightarrow E=\frac{K_E}{qd}=\frac{6\times10^{-14}}{(1.6\times10^{-19})(0.04)}=9.4\times10^5\,\text{N/C}$

b) $V=\sqrt{\frac{2qEd}{m}}=6\times10^6\,\text{m/s}$

c) $\Delta V=Ed=(9.4\times10^5)(0.2)=1.9\times10^5\,\text{V}$

16. a) $E=0$ b) $v=v_1+v_2+v_3+v_4=4\left[\frac{KQ}{r}\right]=2.3\times10^7\,\text{v}$

17. $u=u_1+u_2+u_3=K\frac{Q_1Q_2}{r_1}+K\frac{Q_1Q_3}{r_2}+K\frac{Q_2Q_3}{r_3}$,

$r_1=5, r_2=3, r_3=4.$

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$$U = 9\times10^{9}\left[\frac{(4)(2)}{5}+\frac{(4)(-3)}{3}+\frac{(2)(-3)}{5}\right]10^{-12} = -0.035J$$

18. $$C = \frac{\in r \in_o A}{d} = \frac{(2.1)(8.85\times10^{-12})(0.04)(0.05)}{0.025\times10^{-3}} = 0.15\mu F$$

19. a) $$\frac{1}{C} = \frac{1}{C_1}+\frac{1}{C_2+C_3} \Rightarrow \frac{1}{4} = \frac{1}{6}+\frac{1}{3+C_3} \Rightarrow C_3 = 9\mu F$$

b) $Q_1 = 240\mu C,\ Q_2 = 60\mu C,\ Q_3 = 180\mu C$

c) $v_1 = 40v,\ v_2 = 20v,\ v_3 = 20v$

d) $$U_2 = \frac{1}{2}Q_2V_2 = \frac{(60\times10^{-6})(20)}{2} = 600\mu J$$

20. a) $Q_i = VC = (15)(1000)\times10^{-6} = 0.015C = \text{constant} \Rightarrow Q_1+Q_2 = Q$

$$\frac{Q_1}{1000} = \frac{Q_2}{500} \Rightarrow \frac{Q_1}{1000} = \frac{Q-Q_1}{500} = \frac{0.15-Q_1}{500} \Rightarrow Q_1 - Q_f = 0.01C$$

b) $$V = \frac{Q_2}{C} = \frac{0.005}{500\times10^{-6}} = 10V$$

c) $$U = \frac{1}{2}CV^2 = \frac{1}{2}\times1000\times10^{-6}\times(15)^2 = 0.1125J$$

21. $F = mg\tan\theta \Rightarrow Eq = mg\tan\theta$

$$\Rightarrow E = \frac{mg\tan\theta}{q} \text{ and } V = Ed = \left(\frac{mg\tan\theta}{q}\right)(d)$$

$$V = \frac{(0.07)(10)(\tan30°)(2\times10^{-3})}{6\times10^{-6}} = 132V$$

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Unit 13

Steady Current and Circuit Properties

Look around you. Starting from the cell phone, Tv, radio, stove, electric light, etc to computers, washing machines, refrigerator. You may witness so much more simple and complex electrical machines on which the modern life depends on. What is more, the nervous systems of animals and humans are specialized electric circuits that carry vital signals from one part of the body to another.

13.1 Electric Current

In much the same sense as the flow of water molecules through a pipe, charged particles can flow through a wire and hence create what is called *electric current*.

Electric Current: Is the flow of electric charges from one place to another.

Suppose a charge ΔQ flows past a given point in a wire in a time Δt. In such a case, we say that the electric current I, in the wire is

$$I = \frac{\Delta Q}{\Delta t}$$

The SI unit of current is Ampere (A), named for the French physicist Andre Marie Ampere. So we have, $1A = \frac{1C}{sec}$. That is,

- A current of say 3A, for example, is a flow rate of 3 coulombs of charge every second $\left(3\frac{C}{S}\right)$.

Illustrative Examples

1. How many electrons pass a point in 5 sec if a current of 8A flows in a conductor.

   Solution: $Q = It = (8)(5) = 40c$

   $$Q = ne \Rightarrow n = {}^{Q}\!/_{e} = {}^{40}\!/_{1.6\times10^{-19}} = 2.5\times10^{20} \text{ electrons}$$

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13.2 Drift Velocity and current Density

In electrostatic conditions, the electric field is zero inside a conductor, and thus there is no current. However this doesn't mean that all charges within the conductor are at rest. In metals, some electrons are free and move randomly in all direction. Due to their randomness, there is no net flow of charge and hence no current. But, when a uniform electric field is established inside the conductor by connecting its ends with a battery, the free electrons are driven by a force $F = Eq$ causing them to move from one end to the other end of the metal with a drift velocity and hence, there is current now.

Drift Velocity: Is the average velocity that an electron reaches when an electric field is applied across a conductor.

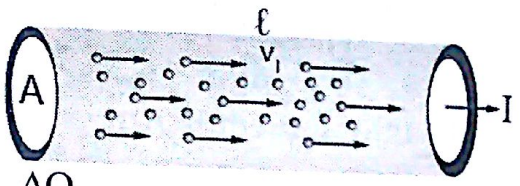
Suppose there are n free electrons per unit volume in a conductor of cross-sectional area A. We call n the *number density* of the free electrons. Assume that each particles carry a charge q and moves with a drift velocity V_d.

During the time interval Δt, all the electrons in the volume pass through the area element. The number of particles in this volume is $nAV_d\Delta t$ and the total free charge is

Total charge = (number of electrons) (charge per electron)

$$\Delta Q = (nAV_d\Delta t)\,(q)$$

The drift velocity is then, given by



$$V_d = \frac{\Delta Q}{nA\Delta tq} = \frac{\Delta Q}{\Delta t}\left(\frac{1}{Aqn}\right), \text{ but } \frac{\Delta Q}{\Delta t} = I$$

$$\Rightarrow \boxed{V_d = \frac{I}{Aqn}}$$

Current Density: Is current per cross-sectional area $J = \frac{I}{A}$

- It is a vector quantity and its unit is A/m^2.

The drift velocity interms of the currently density is given by,

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$$V_d = \frac{I}{Aqn} = \frac{I}{A}\left(\frac{1}{qn}\right), \text{ but } \frac{I}{A} = J$$

$$\boxed{V_d = \frac{J}{qn}}$$

Conceptual Examples

2. The current density of wire A and B is the same. Wire A has twice the number density of wire B. The drift velocity of electrons in wire A is

Explanation: From the relation $V_d = \frac{J}{qn}$

$$V_A = \frac{J_A}{qn_A} \text{ and } V_B = \frac{J_B}{qn_B}. \text{ But } J_A = J_B = J$$

$$n_A = 2n_B$$

Substituting this, and taking ratio $\frac{V_A}{V_B} = \frac{\frac{J_A}{q2n_B}}{\frac{J_A}{qn_B}} = \frac{J}{q2n_B} \cdot \frac{qn_B}{J}$

$$\frac{V_A}{V_B} = \frac{1}{2} \Rightarrow V_A = \frac{1}{2}V_B$$

Illustrative Examples

3. A wire 4mm in diameter transfers a charge of 960c in 80min. If there are 5.8×10^{28} free electrons per cubic meter. Find
   a) The current
   b) The drift velocity

Solution: a) $I = \frac{Q}{t} = \frac{(960)}{(80)(60)} = 0.2A$

b) $V_d = \frac{I}{nqA} = \frac{0.2}{\left(5.8\times10^{28}\right)\left(1.6\times10^{-19}\right)\left(\pi\left(2\times10^{-3}\right)^2\right)}$

$= 1.7\times10^{-6} m/s$

Ohm's Law

The current that a battery can push through a wire is the same as the water flow that a pump can push through a pipe. Greater pump pressures

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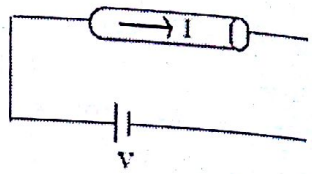
lead to large water flow rates, and, similarly, greater battery voltages lead to large electric currents. And, this is stated by ohm's law.

Ohm's Law: State that "The current produced in a given conductor is directly proportional to the difference of potential between its ends".

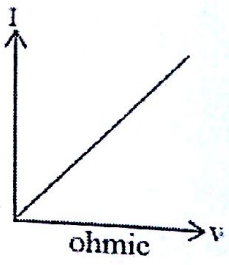
That is, when a voltage is applied across the ends of a conductor, the current in the conductor is found to be proportional to the applied voltage.



Mathematically, $\boxed{V = IR}$ ← ohm's law

Where, R (resistance) is the opposition to the flow of electric charge. Although most metals are good conductors of electricity, all offer some opposition to the surge of electric charge through them.

Remember: The effect of resistance in limiting the flow of charge was first studied by the German physicist Georg Simon Ohm in 1826, and we call the unit of resistance ohm (Ω) in his honor.



Materials that have constant resistance over a wide range of voltage are said to be ohmic and the current-voltage graph is a straight line, the slope of which is resistance, $R = \frac{V}{I}$.

Illustrative Examples

4. The difference of potential between the terminals of an electric heater is 80v when there is a current of 6A in the heater. What will the current be if the voltage is increased to 120v?

Solution: Applying ohm's law.

$$R = \frac{V_1}{I_1} = \frac{80}{6} = 13.3\Omega$$

$$\text{Then, } I_2 = \frac{V_2}{R} = \frac{120}{13.3} = 9A$$

13.3 Factors Affecting Resistance

Ohm, the German physicist reported that the resistance of a conductor at a given temperature is directly proportional to its length, inversely proportional to its cross-sectional area and dependent up on the material from which it is made.

For most metals, the resistance depends on four factors. Which are:

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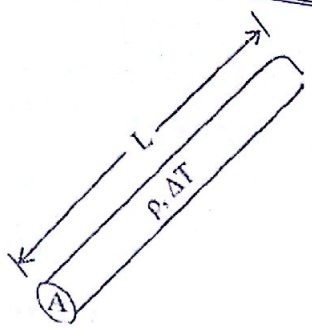
- Length (L).
- Corss–sectional Area (A)
- Kind of Material (ρ)
- Temperature (ΔT)



For a conductor of length L and cross-sectional area A at a given temperature the resistance is

$$\boxed{R=\frac{\rho\ell}{A}} \leftarrow \text{Resistance},$$

When ρ(rho) is called *resistivity* and it depends on the material the wire is made from.

Resistivity is the reciprocal of conductivity. Good conductors have low resistivity.

Conceptual Examples

5. The resistance of an ohmic conductor increases with length. Why?

   Explanation: This is because, electrons undergo more collisions with other particles as they move longer.

6. What will happen to the resistance of a wire at a given temeprature if both, the length and the diameter of the wire are doubled? Explain

   Explanation: $R=\frac{\rho A}{\ell}$, $A=\pi\frac{D^2}{4}$

   $$\Rightarrow R_1=\frac{4\rho L_1}{\pi D_1^{\,2}} \text{ and } R_2=\frac{4\rho L_2}{\pi D_2^{\,2}}$$

   taking ratio: $\frac{R_2}{R_1}=\left(\frac{L_2}{L_1}\right)\left(\frac{D_1^{\,2}}{D_2^{\,2}}\right)$, $L_2=2L_1$ and $D_2=2D_1$

   $$\frac{R_2}{R_1}=\left(\frac{2L_1}{L_1}\right)\left(\frac{D_1^{\,2}}{4D_1^{\,2}}\right)=\frac{1}{2}$$

   $$\Rightarrow R_2=\frac{1}{2}R_1, \text{ the resistance halfed}$$

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Illustrative Examples

7. A current of 5A exists in a copper wire 1.5m long and 2mm in diameter. What is the voltage ($\rho = 1.7\times10^{-8}\Omega m$)

Solution: $R = \frac{\rho\ell}{A} = \left(1.7\times10^{-8}\right)\left(\frac{1.5}{\left(\pi\times10^{-6}\right)}\right) = 0.008\Omega$

$V = IR = 0.04v$

8. A current of 0.75A flows through a copper wire 0.45mm in diameter when it is connected to a potential difference of 15v. How long is the wire? ($\rho = 1.7\times10^{-8}\Omega m$)

Solution: $R = \frac{V}{I} = \frac{\rho L}{A} \Rightarrow \left(1.7\times10^{-8}\right)\left(\frac{(L)(4)}{\left(0.45\times10^{-3}\right)^2}\right) = \frac{15}{0.75}$

$L = 180m$

9. A certain wire has a diameter of 3mm and a length of 150m. It has a resistance of 3Ω at 20°C. What is its resistivity?

Solution: $R = \frac{\rho\ell}{A} \Rightarrow \rho = \frac{RA}{\ell} = \frac{R\pi D^2}{4\ell}$

$$\rho = \frac{(3)(3.4)\left(3\times10^{-3}\right)^2}{4(150)} = 1.4\times10^{-7}\,\Omega m$$

Resistance and Temperature

For most metallic conductors, the resistance tends to increase as the temperature increases. This is because of the increase in collision among moving particles which gained higher kinetic energy at higher temperature. Experiments have shown that, the increase in resistance ΔR is proportional to the initial resistance R_o and the change in temperature ΔT. So we can have:

$$\Delta R \sim R_o \Delta T$$

$$\boxed{\Delta R = \alpha R_o \Delta T}$$

Where α is called the *temperature coefficient of resistance*. Its value is positive for metals and negative for semiconductors.

The final resistance of the conductor after the change in temperature is given by $R = R_o + \Delta R$

$$R = R_o + \alpha R_o \Delta T$$

$$\boxed{R = R_o\left[1 + a\,\Delta T\right]}$$

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Conceptual Examples

10. The resistance of any material increaes as its temepratrure increases since the molecules with in the material gain higher kinetic energy and move randomly and on their way, collide frequently with moving and non moving other molecules. True or False? Explain

Explanation: It is true for most metals but it is false for semiconductors. The resistance of semiconductors decreases as the temperature increases. Semiconductors have negative temperature coefficient of resistance which means their resistance decreases as temperature increase.

Illustrative Examples

11. An iron wire has a resistance of 200Ω at 20°C. What will its resistance be at 80°C. ($\alpha = 0.006/°C$)

Solution: $\Delta R = \alpha R_o \Delta T = (0.006)\ (200)\ (80 - 20) = 72\Omega$

Therefore, the resistance at 80°C is

$$R = R_o + \Delta R = 200 + 72$$
$$R = 272\Omega$$

12. If the resistance of a conductor is 100Ω at 20°C and 116Ω at 60°C, what is its temperature cofficiet?

Solution: $R = R_o\ [1 + \alpha\Delta T]$

$$116 = 100[1 + \alpha(60 - 20)]$$
$$1.16 = 1 + 40\alpha$$
$$\alpha = \frac{0.16}{40} = 0.004 / C°$$

13. At 20°C, a semiconductor is connected to a 5v battery and its resistance is found to be 200Ω. What is the current in the circuit when the temperature increases to 80°C. ($\alpha = -0.5 \times 10^{-3}/ C°$)

Solution: $R = R_o\ [1 + \alpha\Delta T]$

$$R = 200[1 + -0.5 \times 10^{-3}(80 - 20)]$$
$$R = 194\Omega \leftarrow \text{decreases}$$

Now, applying ohm's law,

$$I = \frac{V}{R} = \frac{5}{194}$$
$$I = 0.026A = 26mA$$

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14. A copper wire is stretched so that its length increases and its diameter decreases as a result (UEE: 2001)
A) Resistance decreases and its resistivity remains the same
B) Resistivity decreases and its resistance remains the same
C) Resistance increases and its resistivity remains the same
D) Resistivity increases and its resistance remains the same

Explanation: observe the relation, $R = \rho_o \frac{\ell}{A}$ **Answer: C**

15. The resistance of an ohmic conductor is independent of (UEE, 2003)
A) the size of the material B. the nature of the material
C. the temperature D. the voltage applied

Explanation: $R = \rho\frac{\ell}{A} = R_o\left(1 + \alpha\Delta T\right).$ **Answer: D**

16. When does a light bulb carry more current? (UEE, 2003)
A) Just when it is turned on during which the filament has low temperature and high resistance
B) Long after it is turned on as the filament warms and has high resistance
C) Just when it is turned on during which the filament has low resistance
D) Long after it is turned on as the filament warms and has low resistance

Explanation: High current is observed at low resistance

Answer: C

17. What is the resistance of a copper cable if it has a cross-sectional area of $1cm^2$ and a length of 2km? ($\rho = 2\times10^{-8}\Omega m$) (UEE, 2006)
A) 0.004Ω B) 0.004Ω C) 0.04Ω D) 0.4Ω

Solution: $R = \frac{\rho\ell}{A} = \frac{\left(2\times10^{-8}\right)(2000)}{0.0001} = 0.4\Omega$

Answer: D

18. A small radio receiver uses a battery that delivers a constant current of 25mA for 3 hours. Calculate the total charge delivered by the battery. (UEE: 2006)
A) 1080000C B) 270000C C) 540C D) 270C

Solution: $I = \frac{\Delta Q}{\Delta t} \Rightarrow \Delta Q = I\Delta t = \left(25\times10^{-3}\right)(3\times60\times60)$

$\Delta Q = 270C$

Answer: D

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19. If aluminum and copper wires with circular cross-sections have equal lengths and resistances, then the ratio of the radius of alumnimum to that of copper should be the same as that of the _____. (UEE: 2007)

A) ratio of the resistivity of aluminum to copper
B) ratio of the resistivity of copper to aluminum
C) square root of the ratio of the resistivity of aluminum to copper.
D) square root of the ratio of the reisstivity of copper to aluminum.

Explanation: $R_{Al} = \frac{\rho_{Al}\ell_{Al}}{A_{Al}} = \frac{\delta_{Al}\ell}{\pi r_{Al}^2}$ .....(*)

$R_{CU} = \frac{\rho_{CU}\ell_{CU}}{A_{CU}} = \frac{\delta_{CU}\ell}{\pi r_{CU}^2}$ .....(**)

but, $R_{Al} = R_{CU}$, taking ratio,

$\frac{\rho_{Al}}{r_{Al}^2} = \frac{\rho_{CU}}{r_{CU}^2} \Rightarrow \frac{r_{Al}}{r_{CU}} = \sqrt{\frac{\rho_{Al}}{\rho_{CU}}}$

Answer: C

20. A certain conductor of length ℓ is stretched to 2ℓ without affecting its mass density. What is the ratio of the resistance of the stretched to the unstreched conductor?

A) 4 B) 5 C) 3 D) 2

Explanation: As the lenght doubles, the area become halfed.

$R_1 = \frac{\rho\ell_1}{A_1}$ and

$R_2 = \frac{\rho\ell_2}{A_2}$, but $\ell_2 = 2\ell_1$ and $A_2 = A_1/2$

talking ratio, $\frac{R_1}{R_2} = \frac{\frac{\ell_1}{A_1}}{\ell_2/A_2} = \left(\frac{\ell_1}{A_1}\right)\left(\frac{A_2}{\ell_2}\right) = \left(\frac{\ell_1}{A_1}\right)\left(\frac{\frac{A_1}{2}}{2\ell_1}\right) = 4$

Answer: A

21. Wire B has twice the length and twice the radius of wire A. If both wires are made from the same material and if wire A has a resistance R, what is the resistance of wire B?

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A) 4R B) 2R C) $R/2$ D) $R/4$

Explanation: For wire A: $R_A = \frac{\rho \ell_A}{A_A}$ ..... (*)

For wire B: $R_B = \frac{\rho \ell_B}{A_B}$ .....(**)

talking ratio, $\frac{R_B}{R_A} = \frac{\frac{\rho \ell_B}{A_B}}{\frac{\rho \ell_A}{A_A}} = \left(\frac{\ell_B}{A_B}\right)\left(\frac{A_A}{\ell_A}\right)$,

but $\ell_B = 2\ell_A$ $r_B = 2r_A$

$$\frac{R_B}{R_A} = \left(\frac{\ell_B}{\pi r_B^2}\right)\left(\frac{\pi r_A^2}{\ell_A}\right) = \left(\frac{2\ell_A}{\pi (2r_A)^2}\right)\left(\frac{\pi r_A^2}{\ell_A}\right)$$

$$R_B = \frac{R_A}{2} = R/2 \quad R/2$$

Answer: C

13.4 Electromotive Force and Internal Resistance

When a battery is connected across a piece of metal, it drives the free electrons so that they flow through the metal and produce an electric current. The push that the battery gives to the electrons and make them move in the circuit is called *electromotive force (emf).*

emf, ε: Is the potential difference between the terminals of a battery under ideal condition, when the circuit is open.

Batteries, generators, solar cells etc are all sources of emf. They convert mechanical, chemical, thermal. etc. energy into electrical energy.

Source of emf have resistance inside them and we call it *internal resistance.*

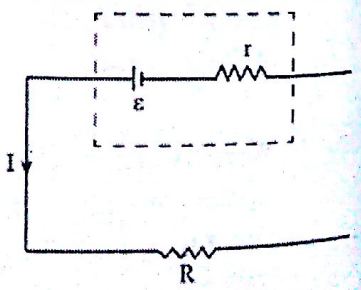
Internal resistance (r): Is the resistance inside the source inside the (battery) itself.

Now, consider a battery of emf, ε and internal resistance (r) is connected to a load or external resistance (R). Then, we have,

$\varepsilon = V_{int} + V_{ext}$

$\varepsilon = Ir + IR$

$\boxed{\varepsilon = I(r + R)}$



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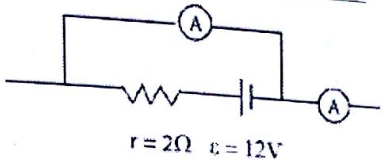
Where, $V_{int.}$ Is the voltage drop across the internal resistance.
$V_{ext.}$ Is the voltage drop across the external resistance.

Important points to remember

- The external potential drop sometimes also called *terminal potential.*
- The long vertical bar in the symbol of emf ⊣⊢ shows the positive terminal and the shorter bar the negative terminal of the battery and electrons in a circuit move from the negative to the positive terminal.

Illustrative Examples

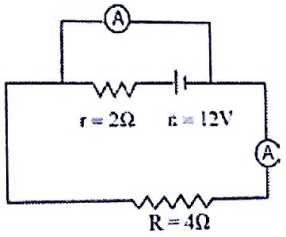
22. An open circuit is shown below. What are the readings of the voltmeter and the ammeter.



Solution: There is no current because there is no complete circuit. The ammeter reads zero. Because there is no current through the battery, there is no potential difference across the internal resistance.

- The potential difference across its terminals is equal to the emf. $\varepsilon = 12v$ and $v_{ext} = 12v$

23. A closed circuit is shown below. What are the readings of the voltmeter and the ammeter.



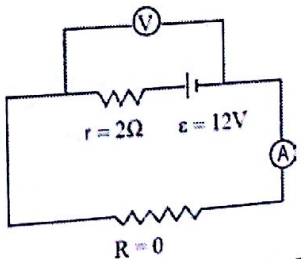
Solution: In this case, we have,

$$\varepsilon = I(r+R) \Rightarrow I = \frac{\varepsilon}{r+R} = \frac{12}{2+4} = 2A$$

And, the terminal voltage is,

$$v_{ext} = IR = (2)(4) = 8v$$

24. A short circuit is shown below. What are the voltmeter and Ammeter readings now.



Solution: Now, the external resistance is zero. Hence, the terminal voltage is zero. But the ammeter reads,

$$I = \frac{\varepsilon}{r+R} = \frac{12}{2} = 6A$$

25. When a dry cell of emf = 1.5v is connected to a bulb of 3.5Ω resistance, the current is found to be 0.4A. What is,

a) The internal resistance

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b) The internal potential drop
c) The terminal voltage.

Solution: a) $\varepsilon = I(r+R) \Rightarrow I = \frac{\varepsilon}{r+R} = \frac{1.5}{r+3.5} = 0.4$

Solving for r, $r = 0.25\Omega$

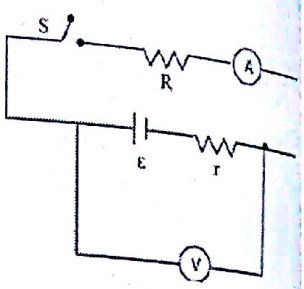
b) $v_{int} = Ir = (0.25)(0.4) = 0.1v$

c) $v_{ext} = IR = (0.4)(3.5) = 1.4v$

26. When the switch s is open, the voltmeter (V) reads 2v. When the switch is closed, the reading of the voltmeter drops to 1.75v, and the ammeter (A) reads 1.8A. Calculate.

a) The emf
b) The internal resistance

Solution: a. When the switch is open, the reading of the voltmeter is equal to the emf. Hence $\varepsilon = 2v$.



b. $\varepsilon = v_{int} + v_{ext}$

$v_{int} = \varepsilon - v_{ext} = 2 - 1.75 = 0.25v$

But, $v_{int} = Ir$

$\Rightarrow r = \frac{v_{int}}{I} = \frac{0.25}{1.8} = 0.14\Omega$

27. The terminal voltage is 11.4v when a 20Ω resistor is connected and 11.16v when a 4Ω resistor is connected. Calculate

a) The emf b) The internal resistance

Solution: a) We know, $V_{ext} = IR$

$$I_1 = \frac{V_{ext\,1}}{R_1} = \frac{11.4}{20} = 0.57A$$

$$I_2 = \frac{V_{ext\,2}}{R_2} = \frac{11.6}{4} = 2.79A$$

$\varepsilon = V_{int} + V_{ext}$

$\varepsilon = IR + Ir$

$\Rightarrow \varepsilon = 0.57(20) + 0.57r$ ..... (*), and

$\varepsilon = 2.79(4) + 2.79r$ ..... (**)

Solving at a time.

$r = 0.11\Omega$ and $\varepsilon = 11.16v$

a) $\varepsilon = 11.46v$ b) $r = 0.11\Omega$

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28. A solar cell generates a p.d of 0.1v when a 500Ω resistor is connected across it and a p.d of 0.15v when a 1000Ω resistor is substitued. What are the emf and the internal resistance of the cell? (UEE: 2001)

A) 0.3v, 1500Ω
B) 0.23v, 1000Ω
C) 0.3v, 1000Ω
D) 0.35v, 1300Ω

Explanation: $\varepsilon = IR + Ir$

$$\text{But } I = \frac{V}{R} \Rightarrow I_1 = \frac{V_1}{R_1} = \frac{0.1}{500} \text{ and } I_2 = \frac{V_2}{R_2} = \frac{0.15}{1000}$$

$$\Rightarrow \varepsilon = I_1R_1 + I_1r = \frac{0.1}{500}(500) + \frac{0.1}{500}r \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (*)$$

$$\varepsilon = I_2R_2 + I_2r = \frac{0.15}{1000}(1000) + \frac{0.15}{1000}r \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (**)$$

Solving the two equations simultaneously,
$\varepsilon = 0.3v$ and $r = 1000\Omega$

Answer: C

29. The terminal voltage of a cell supplying energy to a circuit is usually less than its emf because of the cell's. (UEE: 2002)

A) size
B) internal resitance
C) mass
D) energy

Explanation: The terminal voltage of a cell is usually less than the emf of the cell because, energy is dissipated inside the cell due to its internal resistance. **Answer: B**

30. Which of the following statement is NOT true?

A) Any device such as battery or generator can be considered as source of emf.
B) The terminal voltage can be equal to the emf of a real system.
C) Emf forces electrons move in a direction opposite to the electrostatic force
D) Emf can be defined as work done per unit charge

Explanation: In a real closed circuit, the terminal voltage is less than the emf due to the internal potential drop in the source.

Answer: B

31. A cell supplies a current of 0.9A through a 2Ω resistor and a current of 0.3A through a 7Ω resistor. The internal resistance of the cell is? (UEE: 2004)

A) 0.5Ω
B) 1.0Ω
C) 1.2Ω
D) 2.0Ω

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Solution: $\varepsilon = I(R + r)$

$\Rightarrow \varepsilon = 0.9(2 + r)$.....(*)

$\varepsilon = 0.3(7 + r)$.....(**)

Equating the two equations. $r = 0.5\Omega$

Answer: A

32. A storage battery has an emf of 25v and internal resistance of 0.2Ω. Compute its terminal voltage
    a) When it is delivering 8A
    b) When it is being charged with 8A

Solution: a) when it is discharging

$$v = \varepsilon - Ir$$
$$= 25 - 0.2(8) = 23.4v$$

b) when it is being charged,

$$v = \varepsilon + Ir$$
$$= 25 - 0.8(0.2) = 26.6v$$

33. Three resistors of 4Ω, 6Ω and 12Ω are connected in parallel, and the combination is connected in series with a 1.5v battery of 1Ω internal resistance. What is the current in the circuit.

Solution: The equivalent resistance of the three resistors is

$$\frac{1}{R} = \frac{1}{4} + \frac{1}{6} + \frac{1}{12} \Rightarrow R = 2\Omega$$

Then, R and r are in series.

$$I = \frac{\varepsilon}{R + r} = \frac{1.5}{2 + 1} = \frac{1.5}{3} = 0.5A$$

13.5 Combination of Resistors

Depending on the desired amount of current or resistance in a circuit resistors can be connected in series, in parallel or in both.

Resistors in series

When resistors are connected in series, there is only single path for the current flowing in the circuit.

- For resistors connected in series
  - The current in each resistor and in the network as a whole is the same. $I_1 = I_2 = I_n = I_T = I$
  - The total potential drop in the system is equal to the sum of the potential drops across each individual resistors

$$v = v_1 + v_2 + \ldots\ldots\ldots\ldots + v_n$$

Now, consider the series connection of three resistors shown. The total voltage is

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$v_T = v_1 + v_2 + v_3$

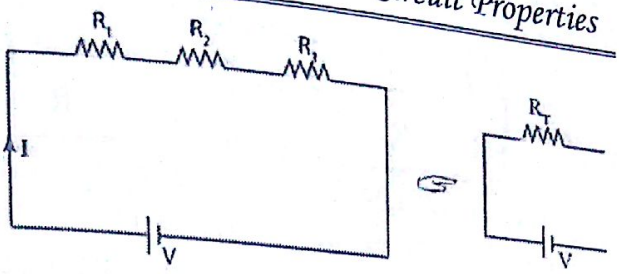
From ohm's law, $v = IR$

$I_T R_T = I_1 R_1 + I_2 R_2 + I_3 R_3$

But the current is constant

$IR_T = IR_1 + IR_2 + IR_3$

$$\boxed{R_T = R_1 + R_2 + R_3}$$



Resistors in series

In general, the equivalent resistance when n resistors are connected in series is given by

$$\boxed{R_T = R_1 + R_2 + \ldots\ldots\ldots + R_n} \leftarrow \text{equivalent resistance}$$

☞ The equivalent resistance of n resistors connected in series is equal to the sum of the individual resistance. And the equivalent resistance is greater than any of the given individual resistance.

Resistors in parallel

When resistors are connected in parallel, the current splits in different portions since it encounter several branches of paths. But, the potential drop in the network is the same due to polarity similarity.

☞ For resistors connected in parallel.

- The potential drop in the system is the same as the potential drops across each resistor. $v_1 = v_2 = v_n = v_T = v$.
- The total current in the system is equal to the sum of currents flowing in each individual resistors

$$I_T = I_1 + I_2 + \ldots\ldots\ldots\ldots\ldots + I_n$$

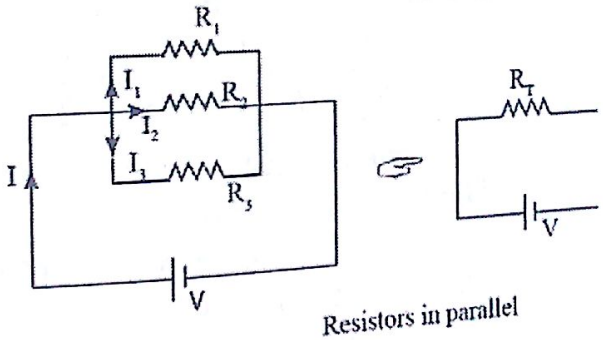
Now, consider the parallel connection of three resistors shown.

The total current is,

$$I_T = I_1 + I_2 + I_3$$

From ohm's law, $I = \dfrac{V}{R}$

$$\frac{V_T}{R_T} = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}$$



Resistors in parallel

But, voltage is constant

$$\boxed{\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$

In general, the equivalent resistance of n resistors connected in parallel is given by.

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$$\boxed{\frac{1}{R_T}=\frac{1}{R_1}+\frac{1}{R_2}+\ldots\ldots\ldots\ldots+\frac{1}{R_n}}$$ ← equivalent resistance

☞ The reciprocal equivalent resistance of n resistors connected in parallel is equal to the sum of the reciprocal resistances of each individual resistance. And the equivalent resistance is always less than any of the given individual resistance.

Conceptual Examples

34. Two identical light bulbs are connected to a battery first in series and then in parallel, in which case will the bulbs become brighter? Explain

Explanation: In series, $I=\frac{V}{2R}$, and

In parallel, $I=\frac{2V}{R}$

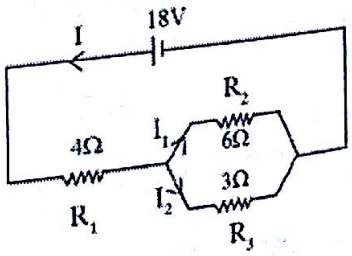
Hence, the bulbs become brighter when connected inparallel.

35. How do you connect resistors to increase the total resistance in a network?

Explanation: To increase resistance in the system, the resistors should be connected in series.

Illustrative Examples

36. Compute the equivalent resistance and the currents in each resistor.



Solution: R_2 and R_3 are in parallel.

$$\frac{1}{R_{23}}=\frac{1}{6}+\frac{1}{3}\Rightarrow\frac{1+2}{6}=\frac{1}{2}$$

$$\Rightarrow R_{23}=2\Omega$$

Then, R_1 and R_{23} are in series.

$$R=R_1+R_{23}=4+2=6\Omega$$

The current in the circuit is,

$$I=\frac{V}{R}=\frac{18}{6}=3A.$$

Moreover, the p.d across R_1 is, $V_1=R_1I=(3)(4)=12V.$

Which means, the p.d across R_2 and R_3 is $V-V_1=18-12=6V.$

Hence, $I_1=\frac{V_2}{R_2}=\frac{6}{6}=1A$ and $I_2=\frac{V_3}{R_3}=\frac{6}{3}=2A.$

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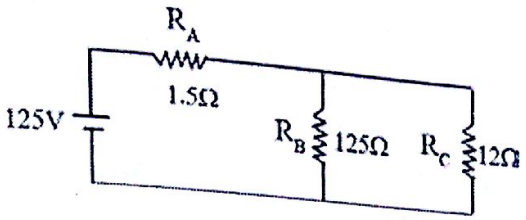
37. Find the potential drop across R_A.

Solution: First find the current in the circuit. Since R_B and R_C are in parallel,



$$\frac{1}{R}=\frac{1}{R_B}+\frac{1}{R_C}=\frac{1}{125}+\frac{1}{12}\Rightarrow R=11\Omega$$

and, R is in series with R_A.

$\Rightarrow R_T = R + R_A = 11 + 1.5 = 12.5\Omega$

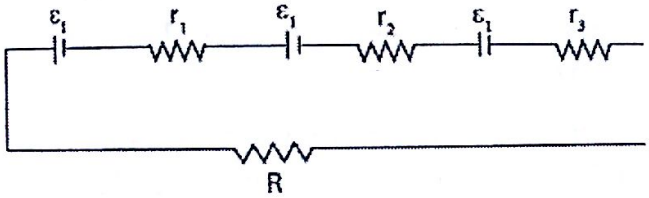
the current is then, $I = V/R = 125/12.5 = 10A$

$\Rightarrow V_A = IR_A = (10)(1.5) = 15V.$

13.6 Combination of Cells

Cells can be connected in series, in parallel or in both.

Cell in series. In this combination the positive terminal of one cell is connected to the negative terminal of the other



The total emf is, $\boxed{\varepsilon = \varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \ldots\ldots + \varepsilon_n}$

The total internal resistance is, $\boxed{r = r_1 + r_2 + \ldots\ldots + r_n}$

The current in the circuit is then, $\boxed{I=\frac{\varepsilon}{r+R}}$

Cells in parallel

In this combination, the positive terminals of all cells are connected to one point and the negative terminals to the other point.

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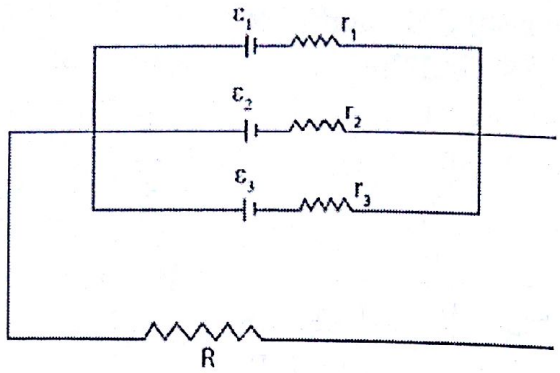

In this case, if the cells do have equal potentials, then
$\varepsilon_1 = \varepsilon_2 = \varepsilon_n = \varepsilon$
The total internal resistance is,

$$\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} + \ldots + \frac{1}{r_n}$$

Conceptual Examples

38. If you connect the positive terminal of a cell to its negative terminal directly with a wire, it will run out why?
Explanation: Because it is short circuited and the current be max.

Illustrative Examples

39. Two resistors connected in series have an equivalent resistance of 7Ω. When they are connected in parallel, their equivalent resistance is $\frac{10}{7}\Omega$. What is the resistance of each resistor (UEE: 2002).

A. 5.5Ω, 1.5Ω B. 6Ω, 1Ω
C. 5Ω, 2Ω D. 4Ω, 3Ω

Solution: $R_1 + R_2 = 7$ ....in series

$$\frac{R_1R_2}{R_1+R_2} = \frac{10}{7} \quad \text{..... .....in parallel}$$

$$\Rightarrow \frac{R_1R_2}{7} = \frac{10}{7}, \text{ but } R_1 = 7 - R_2$$

$$(7 - R_2)R_2 = 10 \Rightarrow 7R_2 - R_2^2 = 10$$

$$R_2^2 - 7R_2 + 10 = 0 \Rightarrow R_2 = 5 \text{ or } R_2 = 2$$

If $R_2 = 5\Omega$ then $R_1 = 2\Omega$ and if $R_2 = 2\Omega$ then $R_1 = 5\Omega$

Answer: C

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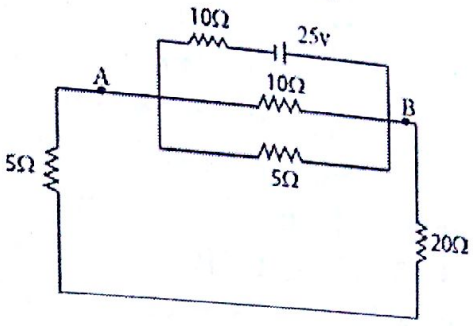
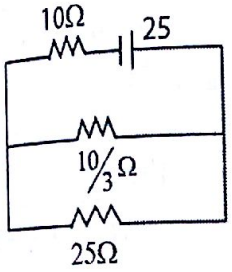
40. Consider the circuit shown below. What are the current in 20Ω resistor and the p.d between points A and B? (UEE: 2003)

A. 0.227A, 5.68v
B. 0.251A, 4.36v
C. 0.215A, 4.36v
D. 0.227A, 4.36v



Solution: The circuit can be modified to



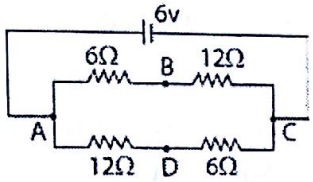
Answer: A

total current $\frac{V}{R} = \frac{25}{220/17} = 1.93A$

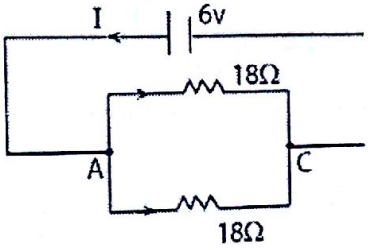
the p.d across the upper 10Ω is 19.32V
the p.d across A and B is 25–19.32 = 5.68v
the current in the 20Ω is 15.68/25 = 0.227A

41. In the circuit shown, what is the p.d between points B and D? (UEE: 2005)

A. 1v
B. 2v
C. 3v
D. 4v



Solution: The circuit reduces to



The current is $\frac{2}{3}$ A. which is then equally divided.

$V_{BA} = \left(\frac{1}{3}\right)(6) = 2v$ and $V_{DA} = \frac{1}{3}(12) = 4v$

$V_{DB} = V_{DA} - V_{BA} = 4 - 2 = 2v$

Answer: B

42. If all the resistor shown have equal resistance and the current flowing in resistor B is 4 A, then what is the current flowing into resistor G? (UEE: 2005)

A. 16 A B. 8A C. 2A D. 1A

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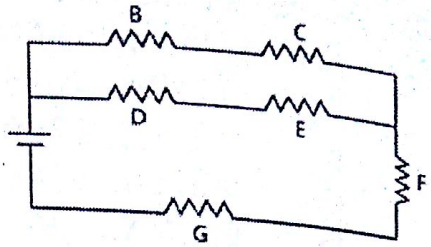
Explanation: The currents in B, C, D and E and equal since they have the same resistances and connected in series and then in parallel with the battery. The resistors in F and G are in series with the battery. Hence, the current is 8A.



Answer: B

13.7 ELECTRIC POWER

If a battery is used to establish an electric current in a conductor, chemical energy stored in the battery is continuously transformed into kinetic energy of the charge carriers. This kinetic energy is lost as a result of collisions between the charge carriers and fixed atoms in the conductor, causing an increase in the temperature of the conductor. In this way, the energy is transformed from one form to the other.

Electric power: Is the rate at which energy changes from one form to the other

When a charge ΔQ moves across a potential difference V, its electrical potential energy changes by.

$$\Delta u = \Delta QV$$

The rate at which energy changes, or power is then,

$$P = \frac{\Delta u}{\Delta t} = \frac{\Delta QV}{\Delta t}, \text{ but } \frac{\Delta Q}{\Delta t} = I$$

$$\Rightarrow \boxed{P = IV}$$

Again, from Ohm's law, $V = IR$, so we have;

$$\boxed{P = I^2R} \quad \text{or} \quad \boxed{P = \frac{V^2}{R}}$$

Conceptual Examples

43. If the current in an electric bulb drops by 2%, by what percent will the power drops?

Explanation: $P = I^2R \qquad P_1 = I_1{}^2R$ and

$P_2 = I_2{}^2 R$, but $I_2 = I_1 - \frac{2}{100} I_1 = 0.98I_1$

$\Rightarrow P_2 = (0.98I_1)^2R = 0.96I_1{}^2R = 0.96\, P_1$

Hence, P_2 is reduced by $0.04 = 4\%$

44. Two resistors, A and B are connected in series with battery. The resistance of A is twice that of B. Which resistor dissipates more power?

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Explanation: The current is the same in series connection $P = I^2R$. Thus, the power dissipated in wire A is twice that of power dissipated in wire B.

45. If electricity is transmitted over long distances through cables, some power is lost due to the increase resistance. How could you minimize the power loss? Explain

Explanation: The power loss can be minimized by using high voltage since a high voltage reduces the current. For example, increasing the voltage by a factor of 10 reduces the current by a factor of 10 which inturn reduce the power lose by a factor of 100, since $P = I^2R$.

Illustrative Examples

46. A 220v 100w bulb is connected to a 110v source. Calculate the power consumed by the bulb

Solution:
$$I = \frac{P}{V} = \frac{100}{220} = \frac{5}{11}A$$
$$R = \frac{V}{I} = \frac{220}{5/11} = 484\Omega$$
$$P = \frac{V^2}{R} = \frac{(110)^2}{484} = 25W$$

47. An electric bulb rates for 500w at 100v is used in a circuit having a 200v supply. Calculate the resistance that one must put in series with the bulb so that the bulb delivers 500W.

Solution: $R_o = \frac{V_o^2}{P} = \frac{(100)^2}{500} = 20\Omega$
$$I = \frac{V_o}{R_o} = \frac{100}{20} = 5A$$
The resistance for 200v is
$$R = \frac{V}{I} = \frac{200}{5} = 40\Omega$$
$$R_1 = R - R_o = 40 - 20 = 20\Omega$$

48. Household light bulbs are normally connected in parallel to a power supply. Suppose a 40w and a 60w light bulbs are instead connected in series. Which bulb is brighter? (UEE: 2002)

A. The 60w bulbs
B. The 40w bulb

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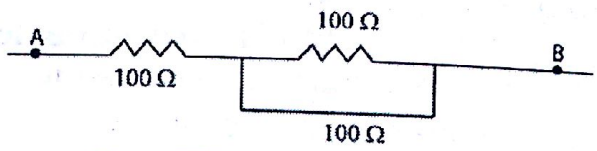
C. The bulbs are equally bright
D. The bulbs will not give light in series connection

Explanation: In series combination, the current is the same. So both bulbs have the same current. But the 60w bulb needs more energy to work at maximum capacity. Thus, bulb 40w is brighter

Answer: B

49. Three 100Ω resistors are connected as shown in the figure below. The maximum power that can be delivered to any one resistor is 25w. What is the maximum voltage that can be applied between the terminals A and B? (UEE: 2003)

A. 50v
B. 75v
C. 100v
D. 125v



Solution: $P = I^2R \Rightarrow I = \sqrt{\frac{P}{R}} = \sqrt{\frac{25}{100}} = 0.5A$

From $V = IR_T = (0.5)\,(150) = 75v$ **Answer: B**

50. Suppose that a voltage source produces 240v for a moment. By what percentage does the power output of 220v light bulb increase: (UEE: 2005)

A. 38% B. 36% C. 19% D. 18%

Explanation: $P = \frac{V_2}{R} \Rightarrow P_1 = \frac{(240)^2}{R}$ and $P_2 = \frac{(220)^2}{R}$

Taking ratio, $\frac{P_1}{P_2} = \left(\frac{240}{220}\right)^2 = 1.19 \Rightarrow P_1 = P_2(1.19) = P_2 + 0.19P_2$

Power increase by 19%

Answer: C

51. If the terminals of a battery with zero internal resistance are connected across two identical resistors in series, the total power delivered by the battery is 8w. If the same battery is connected across the same resistors in parallel, what is the total power delivered by the battery? (UEE: 2006)

A. 2w B. 4w C. 16w D. 32w

Solution: $P = \frac{v^2}{R} \Rightarrow P_1 = \frac{V^2}{2R}$, in series

$8 = \frac{V^2}{2R} \Rightarrow \frac{v^2}{R} = 16$

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$$P_2 = \frac{2v^2}{R}, \text{ in parallel}$$

$$P_2 = 2(16) = 32 \text{ w}$$

Answer: D

13.8 Kirchhoff's Rules

Ohm's law is insufficient to solve the problems of complicated circuits. In 1882, The German physicist Gustav Kirchhoff facilitated two simple rules to analyze the current and voltage in a more complex circuits. Kirchhoff's rules are named as *junction* and *loop rules*

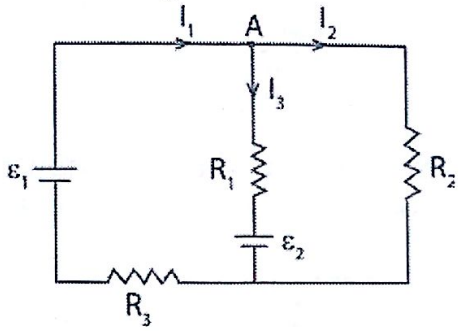
Kirchhoff's junction rule

This rule is based on the conservation of charge and is stated as:

Point (Junction) rule: "The sum of currents entering a point must be equal to the sum of currents leaving that point."

$\Rightarrow (\Sigma I \text{ entering} = \Sigma I \text{ leaving}) \longleftarrow$ point rule

As an example, consider the simple circuit shown



Now, if we consider point A, or junction A

The current that enteres in to point A is I_1 while, the currents I_2 and I_3 leaves this point, thus, according to point rule, we have.

$$I_1 = I_2 + I_3$$

Kirchhoff's Loop rule

This rule is based on the law of conservation of energy

Loop rule: "The sum of the voltages supplied to a closed loop must be the same as the sum of potential drops in that closed loop."

$\Rightarrow (\Sigma \text{ supplied} = \Sigma I R) \longleftarrow$ loop rule

$\Rightarrow$ potential rise = potential drop

As an example, consider the simple circuit shown above

For loop 1

$$\varepsilon_1 + \varepsilon_2 = I_3 R_1 + I_1 R_3$$

For loop 2

$$\varepsilon_2 = I_3 R_1 + I_2 R_2$$

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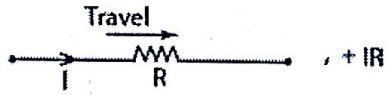
Important points to remember

- **Junction:** Is any point in a circuit where three or more wires meet
- **Loop:** Is any closed path in a given circuit
- **Junction rule:** The algebraic sum of all currents meeting at any junction in a circuit must be zero.
- **Loop rule:** The algebraic sum of all potential differences around any closed loop in a circuit is zero.

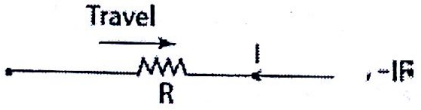
Sign convensions while applying the loop rule

When applying the loop rule, you must choose a direction for traversing the loop, and be consistent in going either clockwise or counter clockwise. As you traverse the loop, record voltage drops and rises according to the following rules.

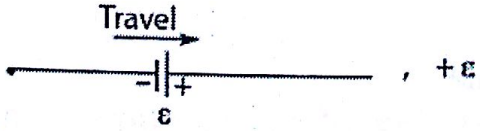
- If a resistor is traversed in the direction of the current the change in electric potential across the resistor is + IR.



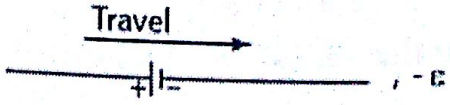
- If a resistor is traversed in the direction opposite to the current, the change in electric potential across the resistor is – IR



- If a source of emf is traversed in the direction of emf (from – to +) terminal, the change in electric potential is + ε



- If a source of emf is traversed in the direction opposite to the emf (from + to –) terminal, the change in electric potential is, –ε



Illustrative Examples

52. Find the currents I_1, I_2 and I_3

Solution: Applying point rule at Junction A

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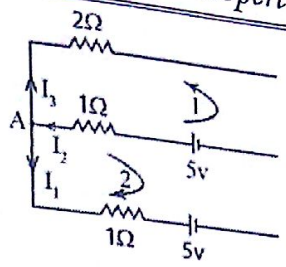
ΣI in $= \Sigma I$ out

$\Rightarrow I_2 = I_1 + I_3$ ..... (*)



Applying loop rule

loop–1, $\Sigma\varepsilon = \Sigma I R$

$-5 = -2I_3 - I_2$

$\Rightarrow I_2 + 2I_3 = 5$ .....(**)

loop – 2, $\Sigma\varepsilon = \Sigma I R$

$-5 + 5 = -I_1 - I_2$

$\Rightarrow I_1 = -I_2$ .....(***)

Now, from, equations (*) and equations (***)

$I_2 = I_1 + I_3$

$\Rightarrow 2I_2 = I_3$

and from equation (**)

$I_2 + 2I_3 = 5$

$I_2 + 2(2I_2) = 5$

$\Rightarrow I_2 = 1A$, $I_1 = -1A$ and $I_3 = 2A$

- The minus sign of I_1 is due to the wrong direction chosen

53. In the circuit shown, $I_1 = 3A$ and ε and R are unknown. What are the currents I_2 and I_3

Solution: point rule (point a)

ΣI in $= \Sigma I$ out

$I_3 = I_1 + I_2$ ..... (*)

Loop rule

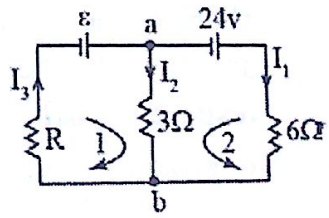
→ loop – 2, assume counter clock movement

$\Sigma\varepsilon = \Sigma I R$

$-24 = -6I_1 + 3I_2$, but $I_1 = 3$

$-24 = -6(3) + 3I_2$

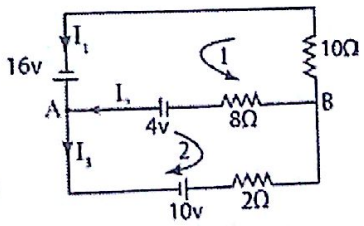
$I_2 = -2A$



and from equation (*),

$I_3 = I_1 + I_2 = 3 - 2 = 1A$

54. Find the currents I_1, I_2 and I_3



Solution: point rule, point A

ΣI in $= \Sigma I$ out

$I_1 + I_2 = I$ .....(*)

Loop rule

Loop – 1. $\Sigma\varepsilon = \varepsilon IR$

$16 - 4 = 10I_1 - 8I_2$

$12 = 10I_1 - 8I_2$ ..... (**)

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Loop – 2. $\Sigma \varepsilon = \varepsilon IR$

$-10 - 4 = -8\, I_2 - 2I_3$

$-14 = -8I_2 - 2I_3$ ....(***)

From the above equations,

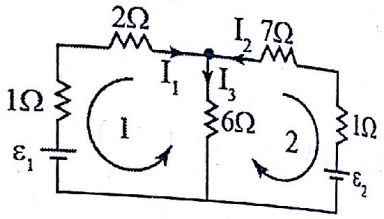
$I_1 = 2A$, $I_2 = 1A$ and $I_3 = 3A$

55. For the circuit shown below, determine ε_1 and ε_2 if $I_1 = \frac{1}{3}A$ and $I_2 = \frac{1}{2}A$

Solution: Point Rule. Point B, Σ I in = Σ I out

$I_1 + I_2 = I_3$

$\Rightarrow I_3 = \frac{1}{3} + \frac{1}{2} = \frac{5}{6}A$



Loop rule

Loop – 1, $\Sigma\varepsilon = \Sigma IR$

$-\varepsilon_1 = -6I_3 - 2\, I_1 - I_1$

$-\varepsilon_1 = -6\left(\frac{5}{6}\right) - 2\left(\frac{1}{3}\right) - \frac{1}{3}$

$\Rightarrow \varepsilon_1 = 6v$

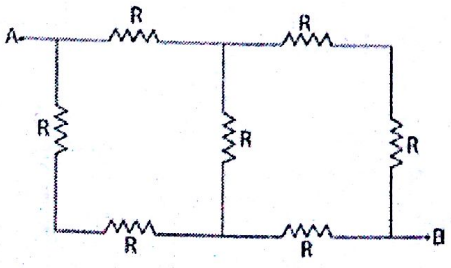
Loop – 2, $\Sigma\varepsilon = \Sigma\, I\, R$

$-\varepsilon_2 = -6I_3 - 7I_2 - I_2$

$-\varepsilon_2 = -6\left(\frac{5}{6}\right) - 7\left(\frac{1}{2}\right) - \frac{1}{2}$

$\Rightarrow \varepsilon_2 = 9v$

56. What is the equivalent resistance between the terminal points A and B in the network shown. Assume that the resistance of each resistor is 10Ω



Solution: Let a battery of emf ε with no internal resistance be connected across A and B

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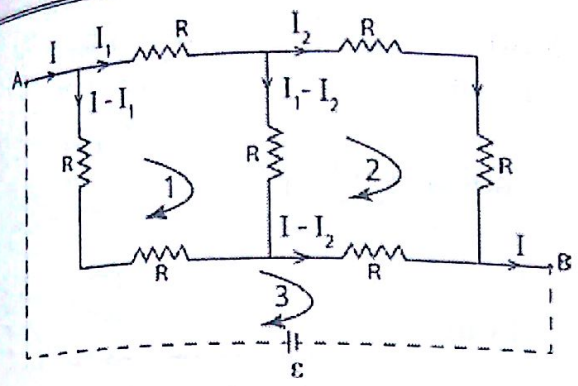


Now apply Kirchhoff loop rule

Loop – 1, $\Sigma\varepsilon = \Sigma I R$

$$O = -RI_1 - R(I_1 - I_2) + 2R(I - I_1)$$

$$\Rightarrow 4I_1 - I_2 = 2I \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (*)$$

Loop – 2, $\Sigma\varepsilon = \Sigma IR$

$$O = -2RI_2 + R(I - I_2) + R(I_1 - I_2)$$

$$\Rightarrow -I_1 + 4I_2 = I \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (**)$$

From equation (*) and (**)

$$I_1 = \frac{3}{5}I \text{ and } I_2 = \frac{2}{5}I$$

Loop – 3, $\Sigma\varepsilon = \Sigma IR$

$$\varepsilon = R(I - I_1) + R(I - I_1) + R(I - I_2)$$

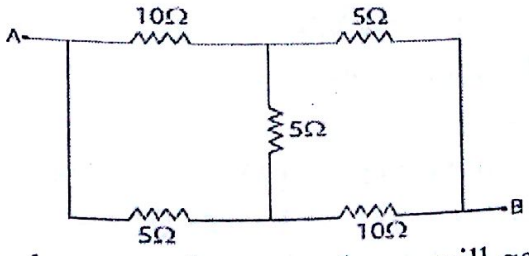
$$\Rightarrow \varepsilon = \frac{7}{5}IR, \text{ after substituting for } I_1 \text{ and } I_2$$

Now, from ohm's law

$$\varepsilon = IR_{AB} \Rightarrow IR_{AB} = \frac{7}{5}IR$$

$$R_{AB} = \frac{7}{5}R = \frac{7}{5}(10) = 14\Omega$$

57. Find the equivalent resistance between points A and B of the circuit shown.



Hint: use the above producer, and you will get $R_{AB} = 7\Omega$.

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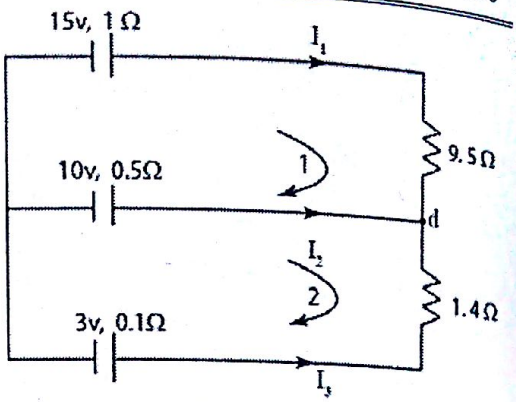
58. For the circuit shown. Find

a. The currents I_1, I_2 and I_3

b. The p.d between point b and a



Solution: point rule

⟶ For point d, ΣI in = Σ I out

$I_1 + I_2 + I_3 = 0$ ..... (*)

Loop rule

→ For loop – 1, $\Sigma\varepsilon = \Sigma IR$

$15 - 10 = I_1 + 9.5\, I_1 - 0.5\, I_2$

$5 = 8.5I_1 - 0.5\, I_2$ ..... (**)

→ For loop – 2, $\Sigma\varepsilon = \Sigma IR$

$10 - 3 = 0.5\, I_2 - 1.4I_3 - 0.1\, I_3$

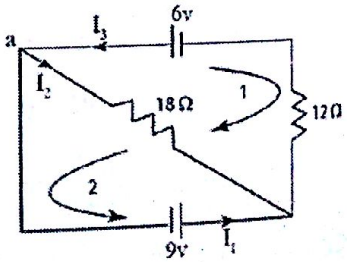
$7 = 0.5\, I_2 - 1.5I_3$ ..... (***)

Solving the above equations, we get

a. $I_1 = 2A$, $I_2 = -8A$ and $I_3 = 6A$

b. $V_{ba} = v_a - v_b = -15 + (1)(2) = -13v$

59. Find the currents in all the wires of the circuit shown



Solution: point rule

→ For point – a, Σ I in = Σ I out

$I_3 = I_1 + I_2$ ..... (*)

Loop rule

→ Loop – 1, $\Sigma\varepsilon = \Sigma IR$

$-6 = 18\, I_2 - 12I_3$ ..... (**)

→ Loop – 2, $\Sigma\varepsilon = \Sigma IR$

$9 = -18\, I_2$

$\Rightarrow I_2 = -0.5$ A

Substituting I_2 in to equation (**), we get I_3

$\Rightarrow I_3 = 1.25$ A

Finally, into equation (*)

$I_1 = I_3 - I_2 = 1.25 - (-0.5) = 1.75$ A

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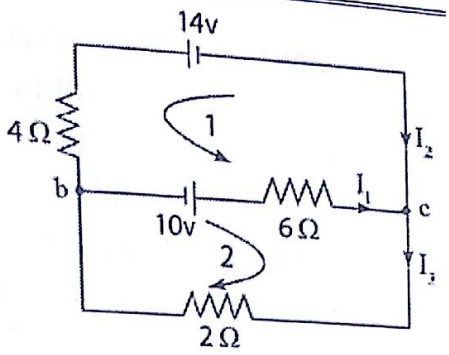
60. For the circuit shown



a. Find the currents, I_1, I_2 and I_3

b. The p.d between points b and c

Hint: use the above procedures and you will have

a. $I_1 = 2A$, $I_2 = -3A$ and $I_3 = -1A$

b. $V_c - V_b = 10 - 6I_1 = 10 - 6(2) = -2V$

61. Find the readings of the ideal ammeter and voltmeter

Solution: point rule

→ At point c, ΣI in $= \Sigma I$ out

$$I_1 + I_3 = I_2 \ldots.(*)$$

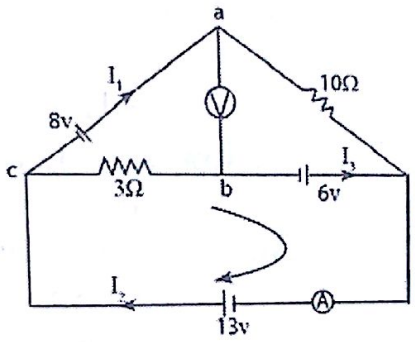
Loop rule

→ For the rectangle, potential

Rise = potential drop

$\Sigma\varepsilon = \Sigma I R$

$13 + 6 = 3I_3, \Rightarrow I_3 = 6.3A$



→ For the larger triangle, potential rise = potential drop

$$\Sigma\varepsilon = \Sigma IR$$

$8 - 6 = 10I_1 - 3I_3 \Rightarrow I_1 = 2.1A$

Now, substituting the values of I_1 and I_3 into equation (*)

$I_2 = 2.1 + 6.3 = 8.4$ A ← The ammeter reading

To obtain the voltmeter reading, we calculate.

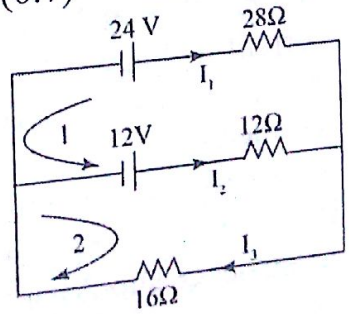
$V_a - V_b = 8 + 3\ (6.3) = 27$ v ← The voltmeter reading

62. Find the potential difference between points A and B

Solution: $V_B - V_A = -6 - (3)\ (2) + 9 - (3)\ (0.7) = -5.1$v

The minus sign indicates that point A is at higher potential.

63. The current in the 16Ω resistor is? (UEE: 2006)



A. 0.64 A

B. 0.15 A

C. 0.49 A

D. 1.28 A

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Solution: Point rule

Take point at the right end

$I_1 + I_2 = I_3$ ..... (*)

Loop rule

→ loop – 1, assume a counter clock wise traverse

$\Sigma\varepsilon = \Sigma IR$

$12 - 24 = 12\ I_2 - 28I_1$

$-12 = 12\ I_2 - 28I_1$ ..... (**)

→ Loop – 2, assume a clockwise traverse

$\Sigma\varepsilon = \Sigma IR$

$12 = 12I_2 + 16I_3$ ..... (***)

Substitute for I_1, $I_1 = I_3 - I_2$ into equation (**) We have

$-12 = 12\ I_2 - 28I_1 \Rightarrow -12 = 12\ I_2 - 28\ (I_3 - I_2)$

$-12 = -16\ I_2 - 28\ I_3$ ..... (****)

Now, solving equation, (***) and (****) at a time,

$48 = 48\ I_2 + 64\ I_3$

$-36 = -48I_2 - 56\ I_3$

$\Rightarrow\ I_3 = 1.5A$

No Answer

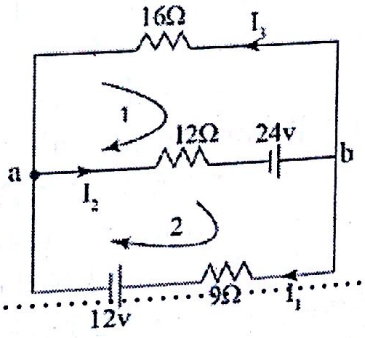
64. In the circuit shown, what is the p.d between point b and a?

A. 19.2 v

B. 24 v

C. 13 v

D. 12 v



Solution: point rule

Take point a, Σ in = Σ I out

$I_1 + I_3 = I_2$ ..... (*)

Loop rule

Loop – 1, $-24 = -12I_2 - 16I_3$ and loop – 2, $24 - 12 = 12\ I_2 + 9I_1$

Solving, we get $I_2 = 0.91A$

$V_b - V_a = 24 - 12\ (0.91) = 13V$

Answer: C

13.9 Electrical Measuring Instruments

Current, voltage and resistance in a circuit are measured using Ammeter, voltmeter and Ohmmeter respectively.

Ammeter

A current – measuring instrument is called an *Ammeter*. It is connected in series with the resistor and should have a very small resistance so that its presence should not alter the current in the circuit.

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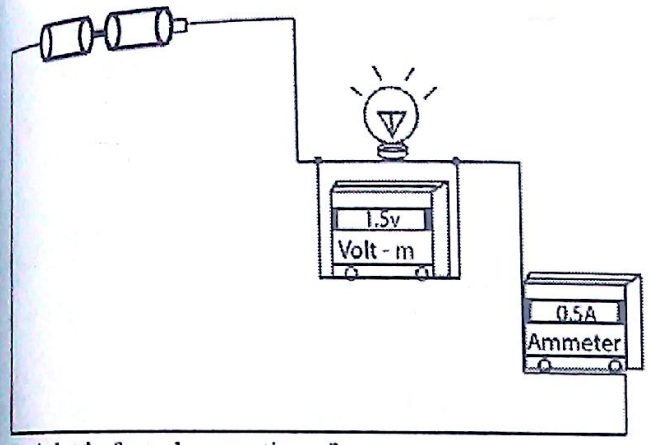
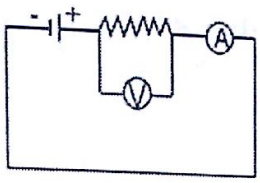
Voltmeter

A voltage measuring instrument is called a *voltmeter*. It is connected in parallel with the resistor and should have a very large resistance so that the passage of current through it should not destruct its function.

As to the connection, consider the following.



A sketch of actual connection of Ammeter and Voltmeter in circuit



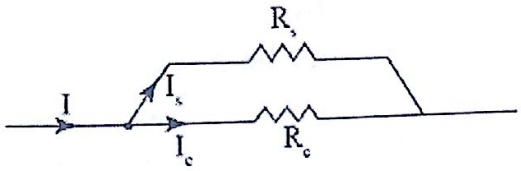
≡ A schematic diagram of the circuit shiwn before.

Modifying a Galvanometer into an Ammeter and a Voltmeter.

Galvanometer as a Ammeter

To modify a galvanometer as an ammeter, we should connect a resistor called *shunt resistor* whose resistance is very small in parallel with the coil resistance of the galvanometer.

That is,



Where: I is the current to be measured

I_C is the coil full scale deflection current

R_C is the coil resistance

R_S is the required shunt resistance

Now, since R_S and R_C are in parallel,

$$V_S = V_C$$

$$I_S R_S = I_C R_C$$

$$R_s = \frac{I_C R_C}{I_s}, \text{ but } I = I_C + I_S$$

$$\Rightarrow I_S = I - I_C$$

$$R_s = \frac{I_C R_C}{I - I_C}$$ ← The required shunt resistance

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Galvanometer as a Voltmeter

To change a galvanometer in to a voltmeter, we need to connect a resistor called *multiplier resistor* whose resistance is very large in series with the coil resistance of the galvanometer.

In this case, I

So we can have,

$$I_m = I_c$$

$$\frac{V_m}{R_m} = \frac{V_c}{R_c}$$

$$R_m = \frac{V_m R_c}{V_c}, \text{ but } v = V_c + V_m \Rightarrow V_m = V - V_c$$

$$Rm = \frac{(V - V_c)R_c}{V_c}, \text{ since } V_c = I_c R_c$$

$$\boxed{R_m = \frac{V - I_c R_c}{I_c}} \leftarrow \text{The required multiplier resistance}$$

Illustrative Examples

65. A galvanometer of full scale deflection 5mA is to be converted into a 10A ammeter. If its coil has a resistance of 50Ω, what value shunt must be fitted?

Solution: $R_s = \frac{I_c R_c}{I - I_c} = \frac{(0.005)(50)}{(10 - 0.005)} = 0.025\Omega$

66. A galvanometer of resistance 50Ω and full – scale deflection 5mA is to be made into a 10 v voltmeter. How can this be done?

Solution: $R_m = \frac{V - I_c R_c}{I_c} = \frac{10 - (0.005)(50)}{0.005} = 1950\Omega$

67. Before the ammeter was added, the current in the circuit was 0.03A. When the ammeter is added, it reads 0.02A. Calculate the resistance of the ammeter. Assume the voltage is 12v

Solution: Before $V = IR \Rightarrow R = V/I = \frac{12}{0.03} = 400\Omega$

After, $R_A + R = \frac{V}{I}$, ammeter is in series.

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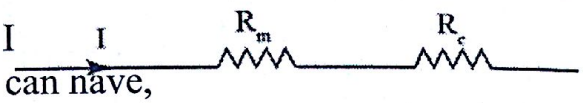
Galvanometer as a Voltmeter

To change a galvanometer in to a voltmeter, we need to connect a resistor called *multiplier resistor* whose resistance is very large in series with the coil resistance of the galvanometer.

In this case, 

So we can have,

$$I_m = I_c$$

$$\frac{V_m}{R_m} = \frac{V_c}{R_c}$$

$$R_m = \frac{V_m R_c}{V_c}, \text{ but } v = V_c + V_m \Rightarrow V_m = V - V_c$$

$$Rm = \frac{(V - V_c)R_c}{V_c}, \text{ since } V_c = I_c R_c$$

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Solution: Before $V = IR \Rightarrow R = V/I = \frac{12}{0.03} = 400\Omega$

After, $R_A + R = \frac{V}{I}$, ammeter is in series.

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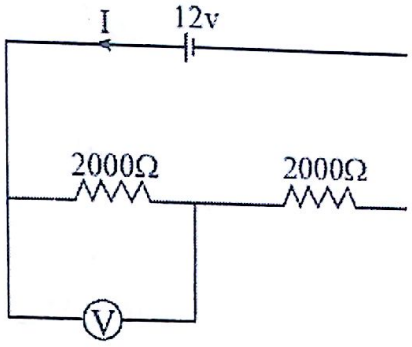
$$R_A = \frac{V}{I} - R = \frac{12}{0.02} - 400$$

$$R_A = 200\Omega$$

68. The voltmeter reads 4v when it is in the circuit. Calculate its resistance.

Solution: The equivalent resistance of the 2000Ω and the resistance of the voltmeter is.

$$\frac{2000\,R_v}{2000 + R_v}$$



and, the current in the circuit is

$$I = \frac{8}{2000} = 4\times10^{-3}A$$

Now, from Ohm's law, $R = \frac{V}{I}$

$$\frac{2000R_v}{2000 + R_v} = \frac{4}{4 \times 10^{-3}}$$

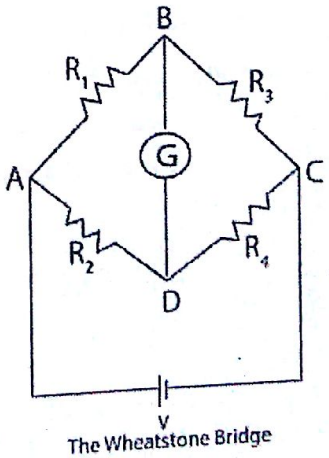
Solving for R_v, $R_v = 2000\Omega$

The Wheatstone Bridge

The circuit we now know as the *Wheatstone bridge* was first invented by Samuel Hunter Christie Wheatstone in 1933. The Wheatstone bridge has many practical applications. One very common application in industry today is to monitor sensor devices such as strain gauge which changes its resistance as the temperature or pressure changes.

Now, consider the basic Wheatstone bridge circuit.

The galvanometer G and the four resistors are connected as shown. The Wheatstone is said to be balanced when no current flows in the galvanometer. That is when potential of B = potential of D



The Wheatstone Bridge

Condition of balance, $\boxed{\frac{R_1}{R_2} = \frac{R_3}{R_4}}$

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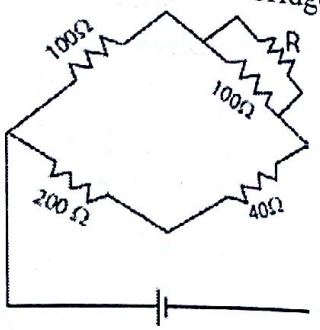
If one of the resistors changes even a little bit in value, the bridge will become unbalanced and current will flow through the galvanometer. Thus, the galvanometer pointer will deflect.

Illustrative Examples

69. The circuit shown represents a balanced Wheatstone bridge. Calculate the value of resistance R.



Solution: For balanced bridge

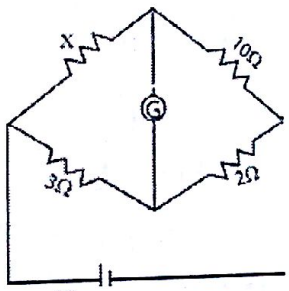
$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

$$\frac{100}{200} = \frac{R_3}{40} \Rightarrow R_3 = 20$$

$$\frac{1}{R_3} = \frac{1}{100} + \frac{1}{R} \Rightarrow \frac{1}{20} = \frac{1}{100} + \frac{1}{R}$$

Thus, $R = 25\Omega$

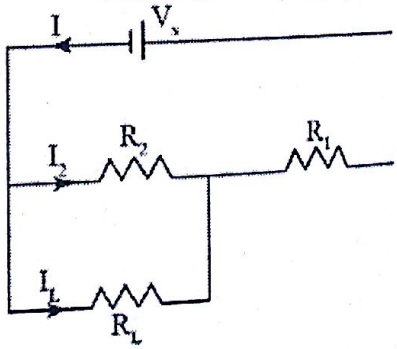
70. For the balanced Wheatstone bridge, find x



Solution: $\frac{x}{3} = \frac{10}{2} \Rightarrow x = 15\Omega$

The Potentiometer

The potentiometer is an instrument that can be used to measure the emf of a source without drawing any current from source. It has a sliding contact and acts as an adjustable potential divider.



By adjusting the position of the sliding contact, it is possible to adjust value of the potential difference v_1 and v_2

When the load resistance R_L is in parallel with R_2, then the potential difference V_L across R_L is found as,

$I = I_2 + I_L$

$\frac{V_s}{R} = \frac{V_2}{R_2} + \frac{V_L}{R_L}$, but $V_2 = V_L \rightarrow$ parallel to each other

$$\frac{V_s}{\frac{R_2 R_L}{R_2 + R_L} + R_1} = V_L\left(\frac{R_2 + R_L}{R_L R_2}\right), R = \frac{R_2 R_L}{R_2 + R_L} + R_1$$

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$$\frac{V_s(R_2+R_L)}{R_2R_L+R_1R_2+R_1R_L}=V_L\left(\frac{R_2+R_L}{R_LR_2}\right)$$

$$\Rightarrow V_L=\frac{R_LR_2}{R_2R_L+R_1R_2+R_1R_L}V_s,\ \text{but } R_L>>R_1 \text{ and } R_2, \text{ usually } R_1R_2\approx 0$$

$$\boxed{V_L=\frac{R_2}{R_1+R_2}V_s}$$

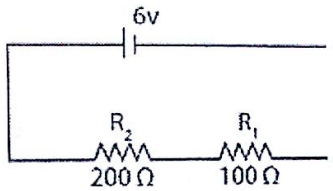
Similarly, if R_L is in parallel with R_1, then V_L is

$$\boxed{V_L=\frac{R_1}{R_1+R_2}V_s}$$

As can be seen, the supply voltage V_s is divided by the circuit in proportion to the value of R_1 and R_2

Illustrative Example

71. What is the output voltage across
    a. The 200Ω resistor
    b. The 100Ω resistor



Solution:

a.

$$V_L=\frac{R_2}{R_1+R_2}V_S=\frac{200}{100+200}\times 6=4v$$

b. $$V_L=\frac{R_1}{R_1+R_2}V_s=\frac{100}{100+200}\times 6=2V$$

Additional Examples

72. A wire of resistance R is cut into n equal parts. These parts are then connected in parallel. The equivalent resistance of the combination is

A. nR B. R/n C. n/R D. R/n^2

Explanation: Resistance of each part is, $\frac{R}{n}$

$$\frac{1}{R_P}=\frac{1}{R_1}+\frac{1}{R_2}+.....+\frac{1}{R_n}$$

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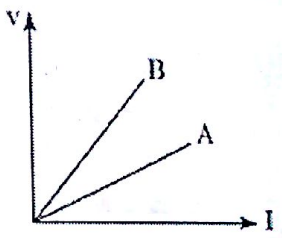
$$\frac{1}{R_P}=\frac{n}{R/n}\Rightarrow\frac{1}{R_P}=\frac{n^2}{R}, \text{ thus, } R_P=\frac{R}{n^2}$$ **Answer: D**

73. A piece of copper and a piece of germanium are cooled from the room temperature down to 80k
   A. The resistance of each of them increases
   B. The resistance of each of them decreases
   C. The resistance of copper increases and that of germanium decreases
   D. The resistance of copper decreases and that of germanium increases.

   Explanation: Resistance of a metal decreases with decrease of temperature. Resistance of a semiconductor increases with decrease of temperature. $\propto$ is + ve for metals and –ve for semiconductor **Answer: D**

74. The V–I graph for parallel and series combination of two metallic resistors is shown. Which graph represents parallel combination?



   A. A B. B
   C. both D. neither A nor B

   Explanation: parallel combination reduces resistance.

   Answer: A

75. A steady current flow in a metallic conductor of non-uniform cross- section. The quantity remain constant along the length of the conductor is
   A. Drift speed only
   B. Current, electric field and drift speed
   C. Current and drift speed
   D. Current only

   Explanation: $V_d=\frac{I}{nqA}$, A varies and so does V_d **Answer: D**

76. Potential difference of a cell in an open circuit is 6v but it falls to 4V when a current of 2A is drawn from it. The internal resistance of the cell is
   A. 1Ω B. 2Ω C. 3Ω D. 4Ω

   Solution: $\varepsilon = v_{in} + v_{ex} \Rightarrow Ir + 4 = 6$

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$$Ir = 2r = 6 - 4 \Rightarrow r = \frac{2}{2} = 1\Omega$$

Answer: A

77. A student has 10 identical resistors each of resistance 'r' The minimum resistance made by him from the given resistors is

A. 10r B. $r/10$ C. $\frac{r}{100}$ D. $r/5$

Explanation: $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots\ldots + \frac{1}{R_{10}} = \frac{10}{r} \Rightarrow R = r/10$

Answer: B

78. A 50v battery is connected across a 10Ω resistor. The current is 4.5A. The internal resistance of the battery is

A. Zero B. 0.5Ω C. 1.1Ω D. 5Ω

Solution: $\varepsilon = I(r + R) \Rightarrow r + R = \frac{\varepsilon}{I} = \frac{50}{4.5} = 11.1$

$$r = 11.1 - R = 11.1 - 10 = 1.1\Omega$$

Answer: C

79. When both the length and area of a wire are doubled, then its resistance will

A. Become half B. Remain the same
C. Be double D. Become four timers

Explanation: $R_1 = \rho\frac{\ell}{A}$, now when ℓ and A are double,

$$R_2 = \rho\frac{(2l)}{(2A)} = \rho\frac{\ell}{A}$$

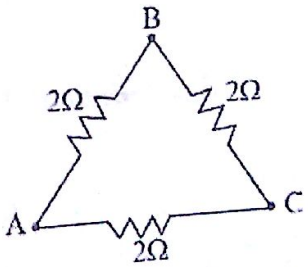
Answer: B

80. Three resistors, each of 2Ω, are connected together in a triangular shape. The resistance between any two vertices will be

A. $\frac{4}{3}\Omega$ B. $\frac{3}{4}\Omega$ C. 3Ω D. 6Ω

Solution: Say between A and B, we have 2 +2 = 4Ω now, the remaining 2Ω and the 4Ω calculated are in parallel.

$$\frac{1}{R} = \frac{1}{2} + \frac{1}{4} = \frac{3}{4} \Rightarrow R = \frac{4}{3}\Omega$$



Answer: A

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81. A piece of wire is cut into four equal parts and the piece are bundled together side by side to form a thick wire compared with that of the original wire, the resistance of the bundle is

A. the same B. $\frac{1}{4}$ C. $\frac{1}{8}$ D. $\frac{1}{16}$

Explanation: $R^1 = \frac{1}{4^2}R = \frac{1}{16}R$ **Answer: D**

82. What is the number of equal parts into which a conductor having a resistance $R_o = 100\Omega$ should be cut to obtain the resistance $R = 1\Omega$ if the parts are connected in parallel?

A. 5 B. 10 C. 20 D. 2

Explanation: $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + ... \frac{1}{R_n} = \frac{n}{\frac{R}{n}}$

$$\frac{1}{1} = \frac{n^2}{R} \Rightarrow n = \sqrt{R_o} = 10$$ **Answer: B**

83. The resistivity of iron is 1×10^{-7} Ωm. The resistance of an iron wire of particular length and thickness is 1Ω. If both the length and the diameter of wire are doubled, then the resistivity in Ωm will be

A. 1×10^{-7} B. 2×10^{-7}
C. 4×10^{-7} D. 8×10^{-7}

Explanation: Resistivity is a property of the material, it does not depend up on the geometrical dimentions. **Answer: A**

84. The resistance of a wire of uniform diameter d and length ℓ is R. The resistance of another wire of the same material but diameter 2d and length 4ℓ will be

A. 2R B. R C. $\frac{R}{2}$ D. $\frac{R}{4}$

Explanation: $R = \frac{\rho\ell}{A} = \frac{4\rho\ell}{\pi d^2}$

$$R^1 = \frac{4\rho(4\ell)}{4\pi d^2} = \frac{4\rho\ell}{\pi d^2} = R$$ **Answer: B**

85. The resistance of a metallic conductor increases with temperature due to

A. Change in carrier density

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B. Change in the dimensions of the conductor
C. Increase in the number of collisions among the carriers
D. Increase in the rate of collisions between the carriers and the vibrating atoms of the conductor.

Answer: D

86. A conductor of length ℓ and uniform cross – sectional area A is made of material of resistivity ρ. What is its conductance?

A. $\frac{A}{\rho\ell}$ B. $\frac{A}{\rho}$ C. $\frac{\rho A}{\ell}$ D. $\frac{\ell}{\rho}$

Explanation: conductance is the inverse of resistance

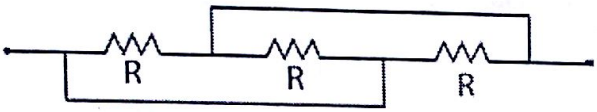
Since, $R = \rho\frac{\ell}{A} \Rightarrow$ conductance $= \frac{A}{\rho\ell}$

Answer: A

87. What is the equivalent resistance for the given network

A. 3R B. $\frac{R}{3}$ C. $\frac{3R}{2}$ D. $2R/3$

Explanation:



It is the combination of three resistors in parallel

$$\frac{1}{R_T} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} = \frac{3}{R} \Rightarrow R_T = R/3$$

Answer: B

88. A conductor of resistance 12.8Ω is cut into a number of parts. All these parts are connected in parallel. The resistance of the combination so obtained is $\frac{1}{5}\Omega$. The number of parts is

A. 4 B. 6 C. 8 D. 12

Explanation: $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots\ldots + \frac{1}{R_n}$, $R_o = 12.8\Omega$

$$\frac{1}{R_T} = \frac{1}{R_o/n} + \frac{1}{R_o/n} + \ldots\ldots + \frac{1}{R_o/n}, \quad R_1 = R_2 = R_n = \frac{R_o}{n}$$

$$\frac{1}{R_T} = \frac{n}{R_o/n} = \frac{n^2}{R_o}$$

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Loop – 2. $\Sigma \varepsilon = \varepsilon IR$

$-10 - 4 = -8\, I_2 - 2I_3$

$-14 = -8I_2 - 2I_3$ ....(***)

From the above equations,

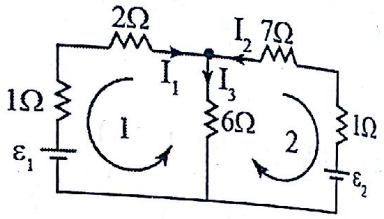
$I_1 = 2A$, $I_2 = 1A$ and $I_3 = 3A$

55. For the circuit shown below, determine ε_1 and ε_2 if $I_1 = \frac{1}{3}A$ and $I_2 = \frac{1}{2}A$

Solution: Point Rule. Point B, Σ I in = Σ I out

$I_1 + I_2 = I_3$

$\Rightarrow I_3 = \frac{1}{3} + \frac{1}{2} = \frac{5}{6}A$



Loop rule

Loop – 1, $\Sigma\varepsilon = \Sigma IR$

$-\varepsilon_1 = -6I_3 - 2\, I_1 - I_1$

$-\varepsilon_1 = -6\left(\frac{5}{6}\right) - 2\left(\frac{1}{3}\right) - \frac{1}{3}$

$\Rightarrow \varepsilon_1 = 6v$

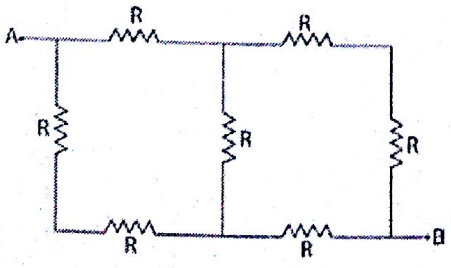
Loop – 2, $\Sigma\varepsilon = \Sigma\, I\, R$

$-\varepsilon_2 = -6I_3 - 7I_2 - I_2$

$-\varepsilon_2 = -6\left(\frac{5}{6}\right) - 7\left(\frac{1}{2}\right) - \frac{1}{2}$

$\Rightarrow \varepsilon_2 = 9v$

56. What is the equivalent resistance between the terminal points A and B in the network shown. Assume that the resistance of each resistor is 10Ω



Solution: Let a battery of emf ε with no internal resistance be connected across A and B

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A. 1.5A B. 2A C. 2.5A D. 9A

Solution: Total resistance $= (4)\left(\frac{1}{4}\right) + 1 = 2\Omega$

Total emf $= (1.5)\ (2) = 3v$

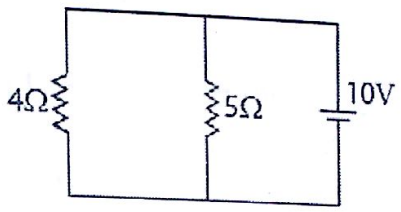
current. $I = \frac{3}{2} = 1.5A$

Answer: A

93. What is the current flowing through the 5Ω resistor

A. 10A B. 5A C. 4A D. 2A

Solution: The 4Ω and 5Ω are parallel hence, the p.d across them are the same, 10v.



$I = \frac{V}{R} = \frac{10}{5} = 2A$

Answer: D

94. A wire of resistance R is divided in 10 equal parts. These parts are connected in parallel. The equivalent resistance of such connection will be

A. 0.01R B. 0.1R C. 10R D. 100R

Explanation: $\frac{1}{R_T} = \frac{1}{R/10} + \frac{1}{R/10} + \ldots\ldots + \frac{1}{R/10} = \frac{10}{R/10} = \frac{100}{R}$

$R_T = \frac{R}{100} = 0.01R$

Answer: A

95. Two wires of same metal have the same length but their cross – sections are in the ratio 3:1. They are joined in series. The resistance of the thicker wire is 10Ω. The total resistance of the combination will be

A. 40Ω B. $\frac{40}{3}\Omega$ C. $\frac{5}{2}\Omega$ D. 100Ω

Explanation: cross- sectional area is reduced by a factor of 3. Resistance is increased by a factor of 3. So, the given arrangement is a series combination of 10Ω and 30Ω.

$\Rightarrow R = 10 + 30 = 40\Omega$

Answer: A

96. How many times will the total resistance of 'n' identical resistors be increased if the parallel arrangement is changed to series arrangement?

A. n times B. n^2 times

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C. $\sqrt{n}$ times D. n^3 times

Explanation: $\frac{1}{R_P} = \frac{1}{R} + \frac{1}{R} + ... + \frac{1}{R} = \frac{n}{R}$

$$R_P = \frac{R}{n}$$

$$R_s = R + R + ... + R = nR$$

$$\frac{R_S}{R_P} = \frac{nR}{\frac{R}{n}} \Rightarrow R_S = n^2 R_P$$

Answer: B

97. A battery has an emf of 12v and an internal resistance of 0.5Ω. When an external 5.5Ω resistor is connected across the terminals of the battery, the potential difference between the terminals will be

A. 12V B. 11.5 V C. 11 V D. 2V

Solution: $I = \frac{\varepsilon}{r+R} = \frac{12}{0.5+5.5} = \frac{12}{6} = 2A$

$V_{ext} = IR = (2)(5.5) = 11$ v

Answer: C

98. A wire has resistance of 12Ω. It is bent in the form of a circle. The effective resistance between the two points on any diameter of the circle is

A. 3Ω B. 6Ω C. 12Ω D. 24Ω

Explanation: Parallel combination of two resistances, each of 6Ω. Hence, the result is 3Ω

Answer: A

99. A metal block has length 2m, width 1m and thickness o.5m. If the resistance across the length is R, then the resistance across the thickness is

A. $R/16$ B. $R/8$ C. $R/4$ D. 4R

Explanation: Case I. $R = \rho \frac{2}{(1)(0.5)} = 4\rho$

Case II. $R^1 = \rho \frac{0.5}{(1)(2)} = \rho/4$

taking ratio, $R^1 = R/16$

Answer: A

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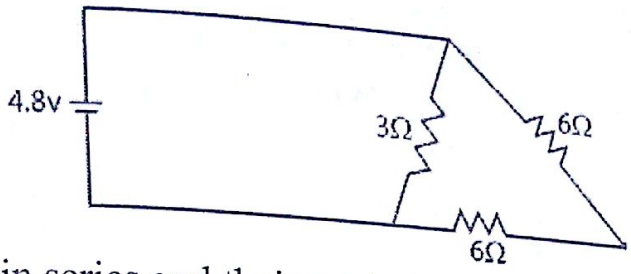
100. The current in the circuit shown is?

A. 8.31 A
B. 6.82 A
C. 4.92 A
D. 2A



Solution: The two 6Ω are in series and their equivalent is in parallel to the 3Ω

$$\frac{1}{R} = \frac{1}{3} + \frac{1}{12} = \frac{4+1}{12} = \frac{5}{12}$$

$$R = \frac{12}{5} \text{ and } I = \frac{v}{R} = \frac{4.8}{\left(12/5\right)} = 2A$$

Answer: D

101. There is a current of 40A in a wire of 10^{-6} m^2 cross section. If the number of free electrons per m^3 is 10^{29}, then the drift velocity is

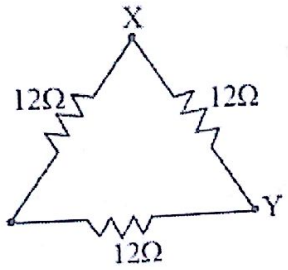
A. 250 mm/s
B. 25 mm/s
C. 2.5 mm/s
D. 125 mm/s

Solution: $V_d = \frac{I}{nAq} = \frac{40}{(10^{29})(10^{-6})(1.6\times10^{-19})} = 2.5$ mm/s

Answer: C

102. A wire of resistance 36Ω is bent in the form of an equilateral triangle. The resistance between two vertices is

A. 12 Ω
B. 24 Ω
C. 8Ω
D. 36Ω



Explanation: $\frac{36}{3} = 12\Omega$

Say the resistance b/n x and y is obtained by adding the two 12Ω resistances in series and then in parallel with the third one.

$$R_s = 12 + 12 = 24\Omega$$

$$\frac{1}{R} = \frac{1}{R_s} + \frac{1}{12} = \frac{1}{24} + \frac{1}{12} = \frac{3}{24}$$

$$\Rightarrow R = 8\Omega$$

Answer: C

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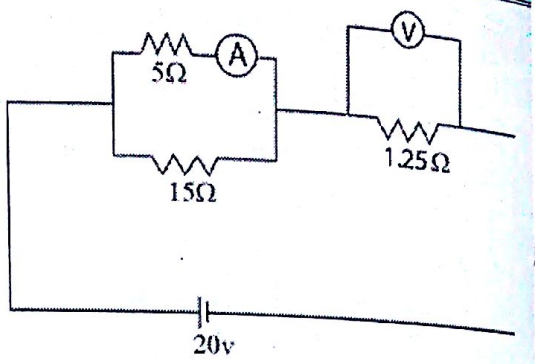
103. An ideal ammeter and ideal voltmeter are connected as shown. What is their reading



A. 6.25A, 3.75V
B. 3A, 5V
C. 3A,3.75V
D. 6A,625V

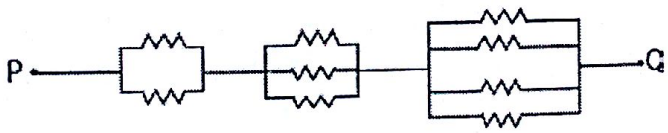
Explanation: Calculate the current through the 5Ω and the p.d across the 1.25Ω **Answer: B**

104. Two resistances are joined in parallel. The combined resistance is $\frac{6}{5}\ \Omega$. One of the resistances wires is broken and the effective resistance become 2Ω. Then, the resistance of the wire that got broken was

A. $\frac{3}{5}\ \Omega$ B. 2Ω C. $\frac{6}{5}\ \Omega$ D. 3Ω

Explanation: $\frac{6}{5} = \frac{2R}{R+2} \Rightarrow R = 3\Omega$ **Answer: D**

105. In the circuit shown, every resistance has 24 Ω value. The resistance between terminals P and Q is



A. 26 Ω
B. 32 Ω
C. 36 Ω
D. 42 Ω

Answer: A

106. A battery of four cells in series, each having an emf of 1.4v and an internal resistance of 2Ω, is to be used to charge a small 2v capacitor what is the charging current.

A. 0.1 A B. 0.2A C. 0.3 A D. 0.45A

Solution: $I = \frac{V}{R} = \frac{4(1.4)-2}{4(2)} = 0.45A$ **Answer: D**

107. A wire has a resistance of 10Ω. What will be its resistance if it is stretched by one tenth of its original length?

A. 12.1Ω B. 14.2Ω C. 10 Ω D. 10.1Ω

Explanation: $m_1 = m_2 \Rightarrow V_1 = V_2$

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$$A_1\left(\ell + \frac{\ell}{10}\right) = A\ell \Rightarrow A_1 = \frac{10}{11}A$$

$$R = \rho\frac{\ell}{A} \text{ and } R_1 = \rho\frac{\ell_1}{A_1}$$

Taking ratio, $\frac{R_1}{R} = \frac{\ell_1}{A_1} = \frac{A}{\ell} = \left(\frac{11}{10}\right)\left(\frac{11}{10}\right) = 1.21$

$R_1 = R(1.21) = 12.1\Omega$ **Answer: A**

108. Two resistors of 15Ω and 30Ω are connected in parallel. What should be the value of R to be connected in series with the other two so that the net resistance will be 20Ω?

A. 5Ω B. 10Ω C. 15Ω D. 120Ω

Solution: $\frac{1}{R_1} = \frac{1}{15} + \frac{1}{30} = \frac{3}{30} \Rightarrow R_1 = 10\Omega$

$R_T = R + 10 \Rightarrow 20 \Rightarrow R = 10$ **Answer: B**

109. The current in a circuit having an external resistance of 3.75Ω is 0.5A. When a resistance of 1Ω is introduced into the circuit, the current becomes 0.4A. The emf of the source is

A. 4v B. 3v C. 2v D. 1v

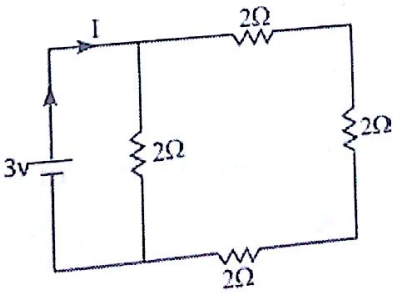
Explanation: $I_1 = \frac{\varepsilon}{r+R} \Rightarrow \frac{\varepsilon}{r+3.75} = 0.5$ ..... (*)

$I_2 = \frac{\varepsilon}{r+3.75+1} \Rightarrow \frac{\varepsilon}{r+4.75} = 0.40$ ..... (**)

From the two equations $\varepsilon = 2v$ **Answer: C**

110. What is current I in the circuit shown

A. 1A B. 0.5A C. 1.2A D. 2A



Solution: The three 2Ω are in series

$R_1 = 2 + 2 + 2 = 6\Omega$

The 6Ω and the fourth 2Ω are in parallel.

$$\frac{1}{R} = \frac{1}{6} + \frac{1}{2} = \frac{1+3}{6} = \frac{4}{6}$$

$$R = \frac{6}{4} = 3/2$$

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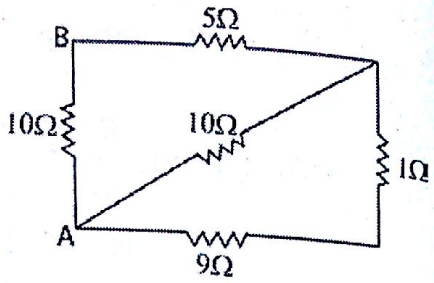
$$I = V/R = 3/(3/2) = 2A$$

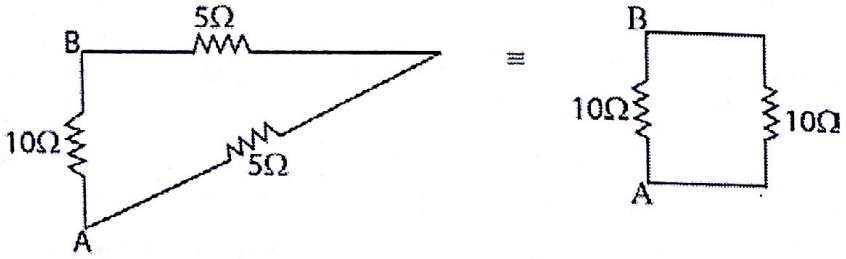
Answer: D

111. The equivalent resistance between the points A and B is

A. 5Ω
B. 10Ω
C. 15Ω
D. 1.5Ω



Solution: The 1Ω and 9Ω are in series and their result is in parallel with the 10Ω Then, the circuit become



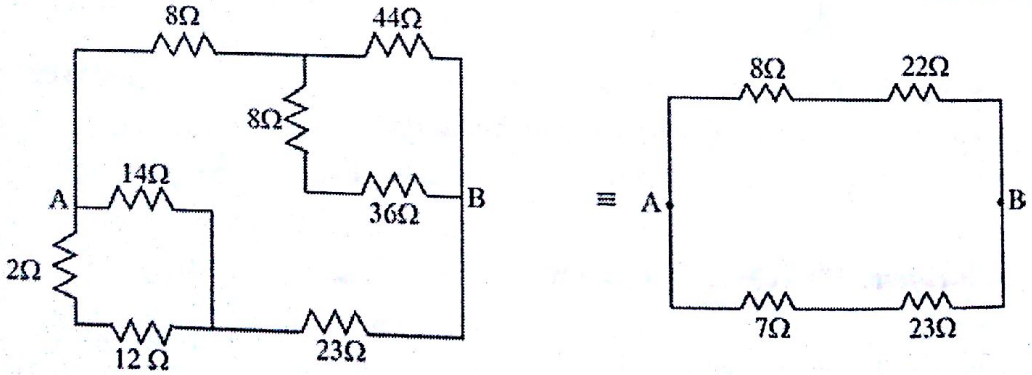
Answer: A

112. The equivalent resistance between A and B is

A. 60Ω B. 30Ω C. 15Ω D. 90Ω

Solution: The circuit modified to

Answer: C



113. The value of current I in the circuit shown is

A. 0.1A B. 0.5A C. $\frac{1}{3}$A D. $\frac{1}{6}$A

Solution: $\frac{1}{R} = \frac{1}{60} + \frac{1}{20} + \frac{1}{10} = \frac{1}{6}$

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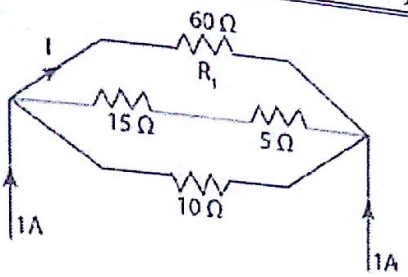
$R = 6\Omega$

$V = IR = (1)(60) = 6v$

$I = \frac{V}{R_1} = \frac{6}{60} = 0.1A$



Answer: A

114. Three resistors of values 2Ω, 3Ω and 6Ω are to be connected to produce an effective resistance of 4Ω. This can be done by connecting

A. 6Ω resistance in series with the parallel combination of 2Ω and 3Ω

B. 3Ω resistance in series with the parallel combination of 2Ω and 6Ω

C. 2 Ω resistance in series with the parallel combination of 3 Ω and 6 Ω

D. 2 Ω resistance in parallel with the parallel combination of 3Ω and 6 Ω

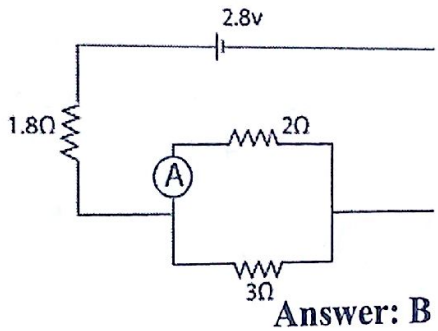
Answer: C

115. If the ammeter reads 0.48A. The internal resistance of the battery is

A. 1 Ω B. 0.5 Ω C. 1.5 Ω D. 2 Ω

Solution: $I = \frac{\varepsilon}{r+R} = \frac{2.8}{r+3}$

$0.48 = \frac{2.8}{r+3} \Rightarrow r = 0.5\Omega$

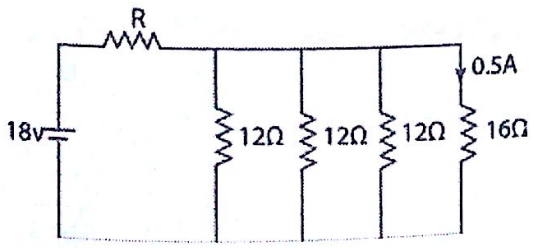


Answer: B

116. The value of R is

A. 8 Ω B. 16 Ω C. 4 Ω D. 20 Ω

Solution:



Answer: C

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117. The resistance of a conductor is 5Ω at 50°c and 6Ω at 100°c. Its resistance at O°C is

A. 3Ω B. 4Ω C. 1Ω D. 2Ω

Solution: $R = R_o (1 + \alpha \Delta T)$

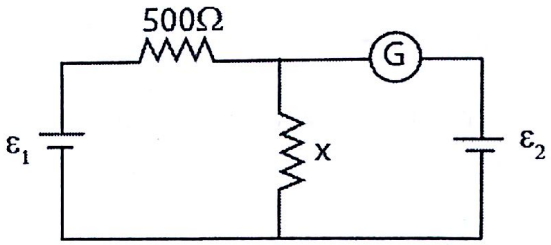
$$\left.\begin{aligned} 5 &= R_o (1 + 50\alpha) \\ 6 &= R_o (1 + 100\alpha) \end{aligned}\right\} \text{solving at a time}$$

$\Rightarrow R_o = 4\Omega$ **Answer: B**

118. If battery ε_1 has an emf of 12v and zero internal resistance while battery ε_2 has an emf of 2v. If the galvanometer G reads zero, the value of resistance x is.

A. 10Ω B. 100Ω C. 500Ω D. 200Ω

Solution:



$$\varepsilon_2 = Ix,\ I = \frac{\varepsilon_1}{R} = \frac{12}{500 + x}$$

$$2 = \left(\frac{12}{500 + x}\right)x \Rightarrow x = 100\Omega$$ **Answer: B**

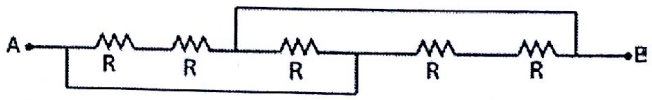
119. What is the effective resistance between points A and B

A. $\frac{4}{3}R$

B. R

C. $\frac{2}{3}R$

D. $\frac{1}{2}R$



Answer: D

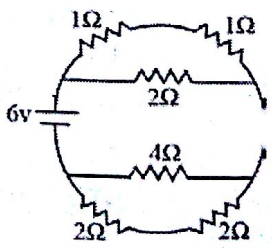
120. What is the current in the circuit

A. 1A

B. 2A

C. 1.5A

D. 3A



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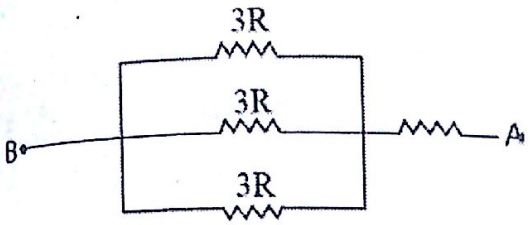
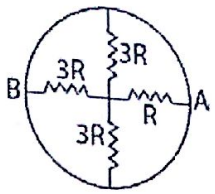
Solution: $I = \frac{V}{R} = \frac{6}{3} = 2A$

Answer: B

121. The equivalent resistance between point A and B is
A. 2R B. 4R C. 7R D. 10R

Solution: The circuit is equivalent to



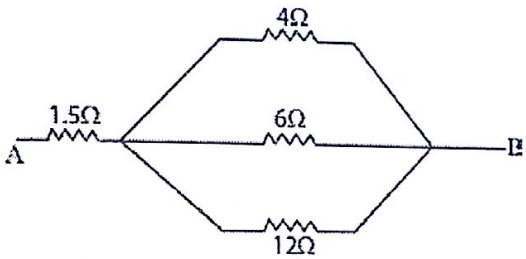


Answer: A

122. If the voltage between point A and B is 18v, the current in the 6Ω resistor is
A. 1.7 A B. 2.5A C. 0.8A D. 5A

Solution: $\frac{1}{R_1} = \frac{1}{4} + \frac{1}{6} + \frac{1}{12}$

$\frac{1}{R_1} = \frac{1}{2} \Rightarrow R_1 = 2\Omega$



The p.d across the parallel combination is $\left(\frac{2}{2+1.5}\right)(18) = \frac{36}{3.5}$

The current through the 6Ω is $I = V/R = \frac{36}{3.5/6} = 1.7A$

Answer: A

123. The resistance of 20m long wire is 5Ω. The wire is stretched to 40m length. The resistance will now be
A. 4Ω B. 10Ω C. 20Ω D. 200Ω

Explanation: Length is double, resistance will increase by a factor of four

Answer: C

124. Resistance of a conductor is doubled keeping the voltage constant. The rate of generation of heat will
A. become one fourth B. be halved
C. be doubled D. become four times

Explanation: Power $= IV = V^2/R$

Answer: B

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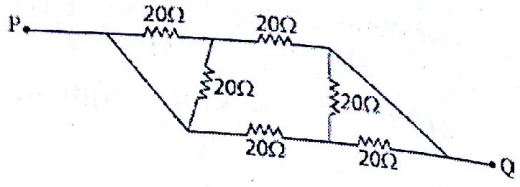
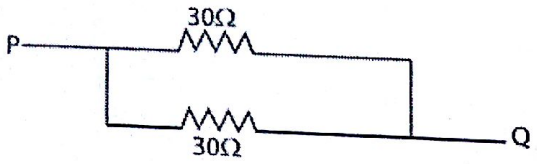
125. The effective resistance between point P and Q is
A. 15Ω B. 40Ω C. 60Ω D. 80Ω

Solution: The circuit reduced to





$\Rightarrow R = 15\Omega$

Answer: A

126. When the external resistance of a circuit is 10Ω, the current in it is 10A. When the resistance becomes 20Ω, the current drops to 8A. The external resistance corresponding to a 4A current is
A. 70Ω B. 62Ω C. 58Ω D. 96Ω

Solution:
$$\left.\begin{aligned} I = \frac{\varepsilon}{r+R} \Rightarrow 10 = \frac{\varepsilon}{r+10} \\ 8 = \frac{\varepsilon}{r+20} \end{aligned}\right\} r = 30\Omega,\ \varepsilon = 400v$$

$$4 = \frac{400}{R+30} \Rightarrow R = 70\Omega$$

Answer: A

127. The resistance of a wire is 40Ω. If it is stretched to double its length, its new resistance will be
A. 20Ω B. 160Ω C. 80Ω D. 320Ω

Explanation: $R = (2)^2 R_o = (4)(40) = 160\Omega$ **Answer: B**

128. A battery of internal resistance r having no load resistance has emf E. what is the observed p.d cross the terminal of the battery when a load resistance R = r is connected to its terminals
A. 2E B. E C. E/2 D. E/4

Explanation: $I = \frac{E}{R+r} = \frac{E}{R+R} = \frac{E}{2R}$

$$V = E - Ir = E - \left(\frac{E}{2R}\right)(R) = E/2$$ **Answer: C**

129. Four wires of equal length and of resistance 10Ω each are connected in the from of a square. The equivalent resistance between two opposite corners of the square is
A. 10Ω B. 40Ω C. 20Ω D. $5/2\Omega$

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Explanation: For opposite corners of the square, the 20Ω and 20Ω are in parallel.

Answer: A

130. Voltmeter V_1 and V_2 are connected in series across a DC line V_1 reads 80 V and has per volt 200Ω resistance. V_2 has 32kΩ resistance. The line voltage is

A. 120 v B. 160v C. 220 v D. 24o v

Solution: $I = \frac{V}{R} = \frac{1}{200}$ A

$$V_2 = IR = \left(32\times10^{3}\right)\left(\frac{1}{200}\right) = 160v$$

$$V = V_1 + V_2 = 80 + 160 = 240V$$

Answer: D

131. Two wires of the same material having radii in the ratio 1: 2 carry currents in the ratio 4: 1. The ratio of drift velocities of electrons in them is

A. 1:16 B. 16: 1 C. 1: 4 D. 4:1

Explanation: $V_2 = \frac{I}{nAg} \sim \frac{I}{r_2}$

$$\Rightarrow \frac{V_1}{V_2} = \left(\frac{I_1}{I_2}\right)\left(\frac{r_2}{r_1}\right)^2 = \left(\frac{4}{1}\right)\left(\frac{2}{1}\right)^2 = \frac{16}{1}$$

Answer: B

132. The equivalent resistance across A and B in the given circuit is

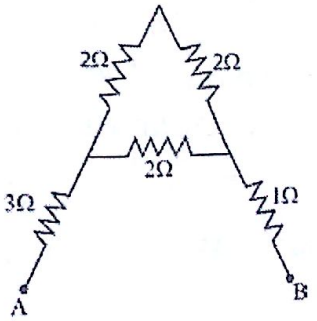
A. 16Ω B. 12Ω C. 5.3Ω D. 13Ω

Solution: The top two 2Ω and the bottom 2Ω are in

$$\frac{1}{R_1} = \frac{1}{4} + \frac{1}{2}$$

$$R_1 = 4/3$$

$$R_T = 4/3 + 3 + 1 = 5.3\Omega$$



Answer: C

133. Two uniform wires A and B are of the same metal and have equal masses. The radius of wire A is twice that of wire B. The total resistance when they are connected in parallel is

A. 4Ω when the resistance of wire A is 4.25Ω
B. 5Ω when the resistance of wire A is 4Ω
C. 4Ω when the resistance of wire B is 4.25 Ω
D. 5Ω when the resistance of wire B is 4Ω

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Explanation: $R \sim \frac{1}{r^4}$

$$\therefore \frac{R_A}{R_B} = \left(\frac{r_B}{r_A}\right)^4 = \left(\frac{r_B}{2r_B}\right)^4 = \frac{1}{16}, \text{ since } r_A = 2r_B$$

$\Rightarrow R_B = 16\,R_A$

If $R_A = 4.25\Omega$, $R_B = 16 \times 4.25 = 68\Omega$

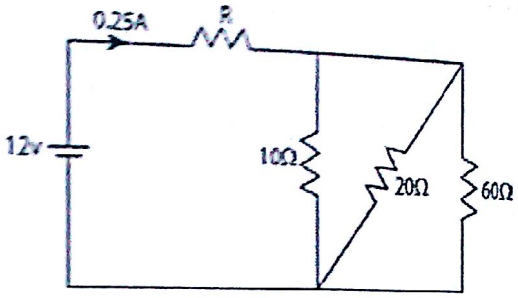
∴ when connected in parallel

$$R = \frac{(4.25)(68)}{4.25 + 68} = 4\Omega$$

Answer: A

134. What is the value of resistance R in the circuit shown

A. 42Ω
B. 62Ω
C. 72 Ω
D. 82 Ω



Explanation:

$$\frac{1}{R_1} = \frac{1}{10} + \frac{1}{20} + \frac{1}{60} = \frac{10}{60} \quad R_1 = 6\Omega$$

$$R_T = R + R_1 = 6 + R$$

$$I = \frac{V}{R_T} \Rightarrow \frac{12}{6+R} = 0.25 \Rightarrow R = 42\Omega$$

Answer: A

135. A bulb rated 220v, 100w is connected across 110v line. The power consumed will be

A. 25w B. 40w C. 75w D. 50w

Explanation: power, $P = \frac{V^2}{R} \sim V^2$

$$\Rightarrow \frac{P^1}{P} = \left(\frac{V^1}{V}\right)^2 \Rightarrow P^1 \left(\frac{V^1}{V}\right)^2 P = \left(\frac{110}{220}\right)^2 \times 100 = 25w$$

Answer: A

136. What is the current supplied to the circuit

A. 1A B. 2A C. 4A D. 6A

Solution: The 2Ω and 6Ω are in parallel

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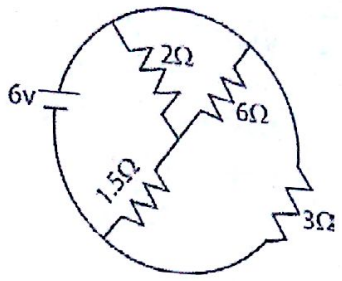
$\frac{1}{R_1} = \frac{1}{2} + \frac{1}{6} = \frac{4}{6} \Rightarrow R_1 = 1.5\Omega$

The 1.5 Ω and R_1 are in series

$R_2 = 1.5 + R_1 = 1.5 + 1.5 = 3\Omega$

The 3Ω and R_2 are in parallel

$\Rightarrow R = \frac{(3)(3)}{3+3} = \frac{9}{6} = \frac{3}{2} = 1.5\Omega$



$I = \frac{V}{R} = \frac{6}{1.5} = 4A$

Answer: C

137. A wire of radius r has resistance R. It is stretched to wire of radius $\frac{r}{2}$. then the resistance of the wire becomes

A. 2R B. 4R C. 16R D. zero

Explanation: $R = \rho\frac{\ell}{A}$

Answer: C

Kirchhoff's rule and Wheatstone's Bridge

138. A current of 2A flows in a system of conductors shown below. The potential difference $V_A - V_B$ will be

A. 2v B. 1v C. −1v D. −2v

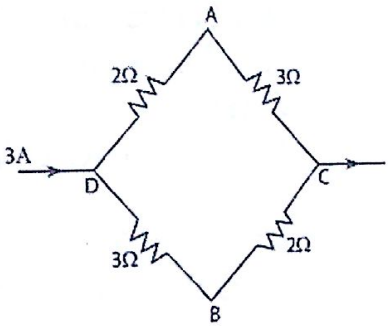
Explanation: The resistance of DAC is equal to the resistance of DBC = 5Ω

∴ The 2A current is divided equally

Hence, $V_D - V_A = (1)(2) = 2v$

$V_D - V_B = (1)(3) = 3V$

$V_A - V_B = (V_D - V_B) - (V_D - V_A) = 3 - 2 = 1v$



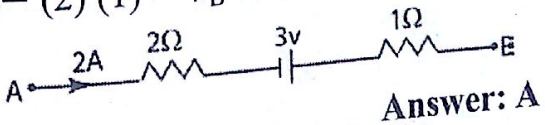
Answer: B

139. The given circuit is part of a closed circuit. The potential difference between points A and B is.

A. 9v B. −9v C. 3v D. 6v

Explanation: $V_A - (2)(2) - 3 - (2)(1) = V_B$

$V_A - V_B = 9v$



Answer: A

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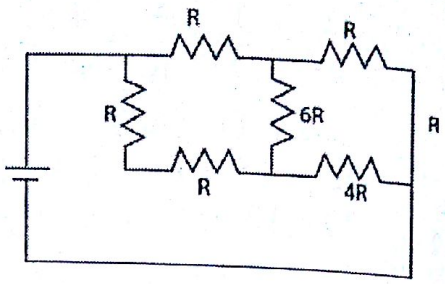
140. A battery of internal resistance 4Ω is connected as shown. In order that maximum power can be delivered to the network, the value of R is



A. $\frac{4}{9}\Omega$ C. $\frac{8}{3}\Omega$

B. 2Ω D. 18Ω

Answer: A

Explanation: The circuit is a balanced Wheatstone's bride. Thus, there is no p.d across the 6R, it can be omitted.

$R = \frac{(3R)(6R)}{3R+6R} = 2R$. For P_{max}, $R_{ext} = R_{int}$

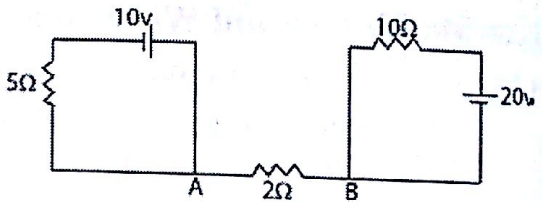
$2R = 4 \Rightarrow R = 2\Omega$

Answer: B

141. What is the current through the 2Ω resistor

A. 0A B. 2A C. 0.3A D. 0.4A

Explanation: The p.d across the 2Ω is zero
Hence, the current is zero.



Answer: A

142. What is the current through the 10Ω resistor

A. 0.1A B. 0.2A C. 0.3A D. 0.4A

Solution:

Loop Rule (AOBA ↻)

$\Sigma\varepsilon = \Sigma I R$

$4 = 10I_1 + 20 I_2$ ....(1)

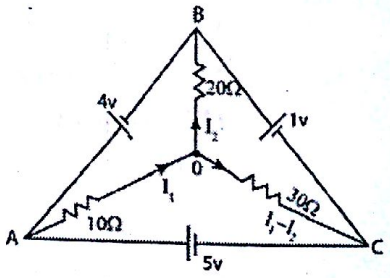
Loop Rule (AOCA ↻

$\Sigma\varepsilon = \Sigma IR$

$-5 = -10 I_1 - 30 (I_1 - I_2)$

$-5 = -10I_1 - 30I_1 - 30I_2$

$-5 = -40I_1 - 30I_2$ ..... (2)

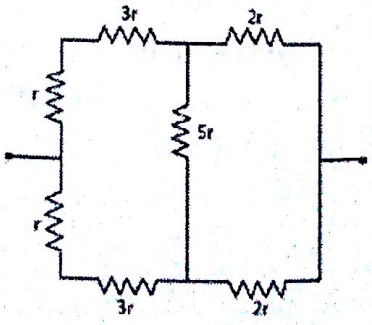


From equation (1) and (2), $I_1 = 0.2A$

Answer: B

143. The resistance of the circuit shown is

A. 18r
B. 9r
C. 6r
D. 3r



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Explanation: The circuit is a balanced wheat stone bridge with two 6r resistors in parallel.

Answer: D

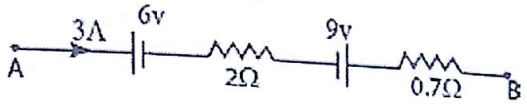
144. The potential difference between points A and B
A. 3v B. 15v C. −5.1v D. 5.1v

Explanation: Using Kirchhoff's rule

$\Sigma\varepsilon = IR \Rightarrow V_A - V_B - 6 + 9 = (3)(2) + (3)(0.7)$

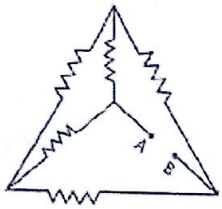
$V_A - V_B = 5.1v$ or

$V_B - V_A = -5.1v$



Answer: C

145. If each resistance is 2Ω. The total resistance between point A and B is
A. 1Ω C. 3Ω
B. 4Ω D. 2Ω



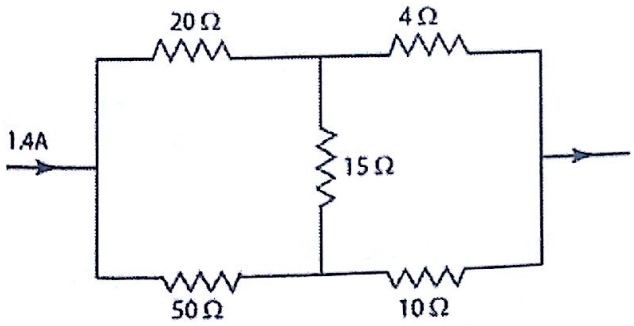
Explanation: The arrangement is a balanced Wheatstone bridge

Answer: D

146. The current through the 4Ω resistor is
A. 1.4A B. 0.4A C. 1.0A D. 0.7A

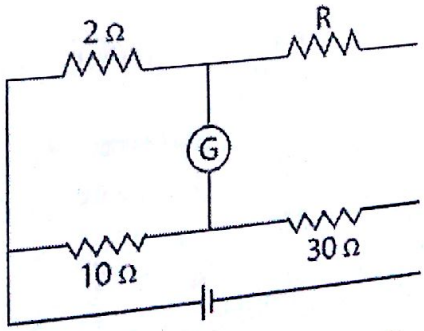
Explanation: The arrangement is a Wheatstone bridge

$$\frac{I_1}{I} = \frac{R_2}{R} \Rightarrow I_1 = \left(\frac{R_2}{R}\right)(I) = \frac{60}{84}(1.4) = 1A$$



Answer: C

147. If there is no deflection then R is equal to
A. 2Ω
B. 30Ω
C. 6Ω
D. $\frac{2}{3}\Omega$



Answer: C

Explanation: $\frac{2}{10} = \frac{R}{30} \Rightarrow R = 6\Omega$

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148. The current flowing through a coil of resistance 800Ω is to be reduced by 80%. What value of shunt should be connected across the coil?

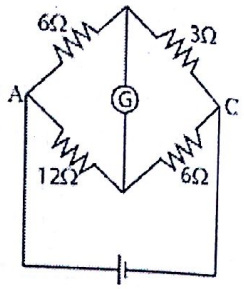
A. 200Ω B. 80Ω C. 8Ω D. 10Ω

Solution: Current ratio, 20%: 80% ⇒ 1: 4 Resistance ratio, 4: 1

⇒ 800: 200

Answer: A

149. If the galvanometer is of 10Ω resistance and potential difference of 2v is maintained across AC, the current through the galvanometer is

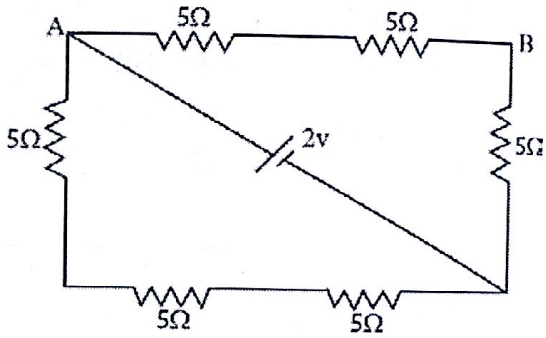


A. 8.7mA
B. 7.8 mA
C. 8.7A
D. zero

Explanation: It is a case of balanced Wheatstone bridge

Answer: D

150. The potential difference between points A and B is



A. $\frac{2}{3}$ v
B. $\frac{8}{9}$ v
C. $\frac{4}{3}$ v
D. 2v

Explanation:

$$I = \frac{V}{R} = \frac{2}{7.5} = \frac{20}{75} \text{ A}$$

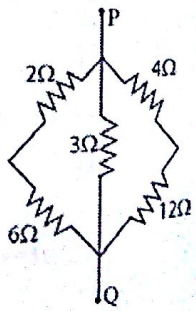
Current in the upper branch is

$$I = \frac{1}{2}\left(\frac{20}{75}\right) = \frac{10}{75} \text{ A}$$

$$V_{AB} = (10)\left(\frac{10}{75}\right) = \frac{4}{3} \text{ v}$$

Answer: C

151. The p.d between points P and Q in the circuit shown is



A. 4.5 v B. 12v C. 24v D. 8v

Explanation: From a balanced Wheatstone Bridge, the 3Ω resistance is not considered

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$$R = \frac{(8)(16)}{8+16} = \frac{16}{3}\Omega$$

$$V = IR = \left(\frac{3}{2}\right)\left(\frac{16}{3}\right) = 8v$$

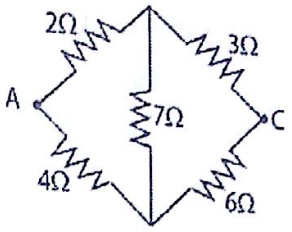
Answer: D

152. The equivalent resistance between points A and C is

A. $\frac{10}{3}\Omega$ B. 22Ω C. 15Ω D. 10.6Ω

Explanation: The network is a balanced Wheatstone bridge.

$$R = \frac{(5)(10)}{5+10} = \frac{10}{3}\Omega$$

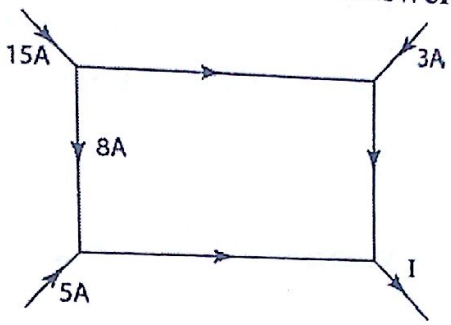


Answer: A

153. The value of current I in the circuit is

A. 3A
B. 13A
C. 23A
D. −3A

Explanation: $I = 10 + 3 = 13A$



Answer: B

154. A cell supplies a current of 0.9A through a 2Ω resistor and a current of 0.3A through a 7Ω resistor. The internal resistance of the cell is

A. 0.5Ω B. 1Ω C. 1.2Ω D. 2Ω

Solution: $\varepsilon = I(R + r)$, so we have

$\varepsilon = I_1(R_1 + r)$ $\varepsilon = I_2(R_2 + r)$

$\Rightarrow I_1(R_1 + r) = I_2(R_2 + r)$

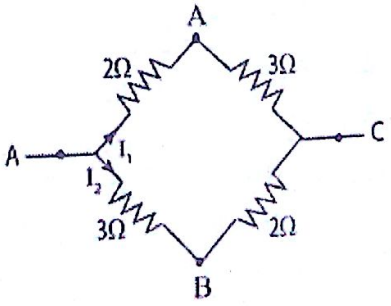
$0.9(2 + r) = 0.3(7 + r) \Rightarrow 1.8 + 0.9r = 2.1 + 0.3r$

$0.6r = 0.3 \Rightarrow r = 0.5\Omega$

Answer: A

155. There is a current of 2A in the system shown. What is the potential difference between point A and B

A. −2v
B. −1v
C. 1v
D. 2v



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Solution:

Using Kirchhoff's loop rule. Considering a clock wise traverse,

$\Sigma\varepsilon = \Sigma IR$

$O = 2I_1 + 3I_1 - 2I_2 - 3I_2 \Rightarrow I_1 = I_2$

But, $I_1 + I_2 = 2$, thus, $I_1 = I_2 = 1A$

Now, $V_A = 2I_1 = 2(1) = 2v$ and $V_B = 3I_2 = 3(1) = 3v$

$V_A - V_B = 2 - 3 = -1V$

156. If a 2A current flows through a to b, what is the potential difference between a and b? **Answer: B**

A. 12v B. 16v C. 20v D. 24v

Solution: $V_A + 2(2) - 16 - V_B = 0$

$\Rightarrow V_A - V_B = 12v$

157. The effective resistance of two resistance when connected in series is 32Ω, and it is 6Ω when they are in parallel. What are the values of the two resistors? **Answer: A**

A. 16Ω, 16Ω B. 20Ω, 12Ω

C. 5Ω, 27Ω D. 8Ω, 24Ω

Solution: $\frac{R_1R_2}{R_1+R_2} = 6$ in parallel

$R_1 + R_2 = 32$ in series

$\Rightarrow R_2 = 32 - R_1$, hence $\frac{R_1(32-R_1)}{32} = 6$

Solving this, $R_1 = 8\Omega$ and $R_2 = 24\ \Omega$

158. The equivalent resistance of two resistors R and 2R when they are connected in parallel is 8Ω. What is the equivalent resistance of the two resistors when they are connected in series? **Answer: D**

A. 36 Ω B. 24Ω C. 18Ω D. 12Ω

Solution: $\frac{R_1R_2}{R_1+R_2} = \frac{(R)(2R)}{R+2R} = \frac{2R^2}{3R} = 8 \Rightarrow R = 12\Omega$

Now, when connected in series

$R + 2R = 12 + 2(12) = 36\Omega$

159. The potential difference applied across a series combination of 4Ω and 12Ω resistors is 240v. What is the potential difference across the 4Ω resistor? **Answer: A**

A. 180V B. 150V C. 120 V D. 60V

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Solution: The current in the circuit is $I = \frac{V}{R} = \frac{240}{4+12} = 15A$

Now, using ohm's law, $V = IR = (15)(4) = 60V$

Answer: D

160. A wire of 100m length and a 2mm diameter has $\pi \times 10^{-8}$ Ωm resistivity. What is the resistance of the wire?

A. 1Ω B. 0.1Ω C. 0.25Ω D. 2.5Ω

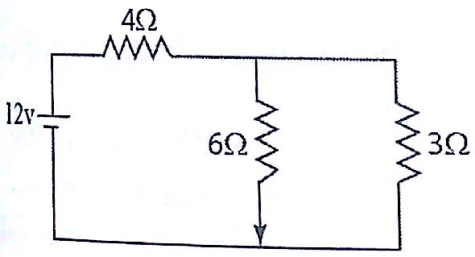
Solution: $R = \rho \ell / A = \frac{(\pi \times 10^{-8})(100)}{\pi \times (1 \times 10^{-3})^2} = 1\Omega$

Answer: A

161. A 3Ω and a 6Ω resistors are connected in parallel. This combination is then connected with a 4Ω resistor and a 12v battery in series. What is the potential difference across the 4Ω resistor?

A. AV B. 8V C. 6V D. 10V

Solution: First of all, let's find the current in the circuit. To do so, we need to find the effective resistance



$R_1 = \frac{(3)(6)}{(3+6)} = 2\Omega$

$R_T = 4 + 2 = 6\Omega$

$\Rightarrow I = \frac{V}{R} = \frac{12}{6} = 12A$. The p.d across the 4Ω is then,

$V = IR = (2)(4) = 8V$

Answer: B

162. A 4Ω speaker is connected across the output terminal of a power amplifier that delivers 64w to it. The current flowing to the speaker is?

A. 8A B. 4A C. 2A D. 16A

Solution: $P = IV = I^2R \Rightarrow I = \sqrt{P/R} = \sqrt{\frac{64}{4}} = 4A$

Answer: B

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163. For the circuit shown below, what is the current I in terms of $\varepsilon_1, \varepsilon_2$ and R

A. $I = \frac{\varepsilon_1 + \varepsilon_2}{R}$ B. $I = \frac{\varepsilon_1 - \varepsilon_2}{R}$

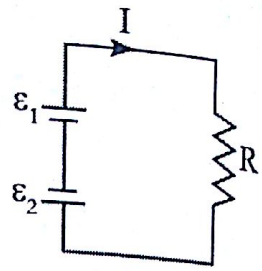
C. $I = \frac{\varepsilon_2}{R} + \varepsilon_1$ D. $I = \frac{\varepsilon_1}{R} + \varepsilon_2$

Solution: Using Kirchhoff's rule (loop rule)

$$\Sigma\varepsilon = \Sigma IR$$

Traversing in the clockwise direction

$$\varepsilon_1 - \varepsilon_2 = IR \Rightarrow I = \frac{\varepsilon_1 - \varepsilon_2}{R}$$



Answer: B

Unit 13: Conceptual Questions

1. Can the terminal voltage of a batter every be greater than its emf? explain
2. Electrical devices are often rated with a voltage and a current as 120v, 5A. Batteries, however, are rated only with a voltage, example 1.5v. Why?
3. What happens to the current in the other two lamps if one lamp in a three lamp series circuit burn out?
4. An engineer needs a 10Ω resistor and a 90Ω resistor. But, there are only 30Ω resistors in the market. What should he do?
5. Two lamps have different resistance one larger than the other. If they are connected in parallel, which is brighter (dissipates more power)?
6. The elements of heater is very hot while the wires carrying current are cold. Why?
7. Why is it dangerous to replace a 15A fuse in a circuit with a fuse of 30A?
8. Explain why a car batter "goes dead" faster on a cold morning than on a normal day?

Unit 13: Problems

Electric Current and Drift Velocity

9. A small radio receiver used a battery that delivers a constant current of 50mA for 6 hours. Calculate
   a. the total charge delivered by the battery
   b. the number of electrons pumped

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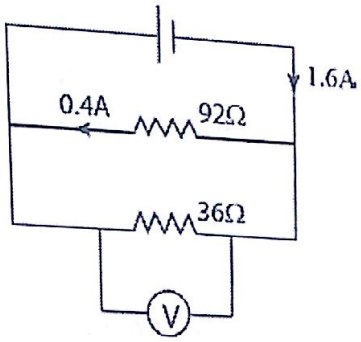
10. Copper contains 8×10^{28} free electrons per m^3. If the wire made of copper has 1.5×10^{-6} m^2 cross – sectional area and a current of 0.5A. Calculate the drift velocity of the free electrons?
11. Find the number of charge carries per unit volume in a copper wire of cross- sectional area $2mm^2$ carrying a current of 1.5A, if the drift velocity of the charge carries is 0.00028 m/s.

Resistance and Resistivity

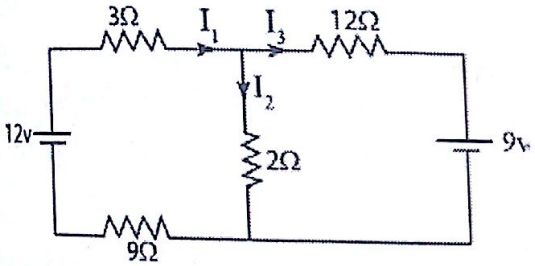
12. A student connects the ends of a pencil lead to a 6v supply and measures the current to be 8.6A. The lead is a rod of graphite of length 7.5cm and diameter 1.4mm calculate
    a. the resistance of the pencil
    b. the resistivity of graphite

Kirchhoff's Rule

13. From the circuit shown below, calculate
    a. the current in the 36Ω resistor
    b. the reading on the voltmeter

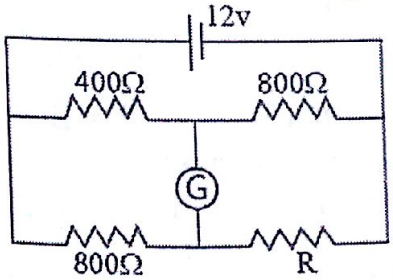


14. What are the value of I_1, I_2 and I_3 in the circuit shown



Measuring Instruments

15. A galvanometer has a resistance of 60Ω and is of 5mA full – scale deflection. How would you modify it so as to get a 10A ammeter?
16. For the balanced Wheatstone Bridge. Find the value of R



Unit 13: Explanation to Questions

1. Yes! this is so when the battery is charging
2. Electrical appliance has a given resistance. Hence a definite current for a given voltage. But, batteries may be used in different diveces of

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different resistance. Hence, the current is not constant, for this reason only voltage is specified.

3. It will run off
4. He need to connect the three 30Ω resistor in parallel to get the 10Ω and in series to get the 90Ω resistor.
5. $P = \frac{V^2}{R}$, P is large for small R. The brighter is the one with the smaller resistance.
6. Heater is made of nichrome which has high resistivity.
7. The tolerance of the electronic devices used may be less than 30A. hence, it will cause many devices to burn out.
8. On a cold morning, the chemical reaction inside the battery is very small as the system gives thermal energy to the environment

Unit 13: Solution to problems

9. a. $\Delta Q = I\Delta t = (50\times10^{-3})(6\times 60\times 60) = 1080c$

   b. $\Delta Q = ne \Rightarrow n = \frac{\Delta Q}{e} = \frac{1080}{1.6\times10^{-19}} = 6.75\times10^{21}$ electrons

10. $V_d = \frac{I}{nAq} = \frac{0.5}{(8\times10^{28})(1.5\times10^{-6})(1.6\times10^{-19})} = 2.6\times10^{-5}\ m/s$

11. $n = \frac{I}{V_d Aq} = \frac{1.5}{(0.00028)(2\times10^{-6})(1.6\times10^{-19})} = 1.67\times10^{28}$ electrons m/s^3

12. a. $R = V/I = 6/8.6 = 0.7\Omega$

    b. $R = \frac{\rho\ell}{A} \Rightarrow \rho = \frac{RA}{\ell} = \rho = \frac{(0.7)(\pi)(0.7\times10^{-3})^2}{(7.5\times10^{-2})} = 2.1\times10^{-5}\Omega m$

13. a. From point rule, I = 1.2 A b. V = IR = (1.2) (36) = 43.2v
14. $I_1 = 0.8A$, $I_2 = 1.3A$ and $I_3 = -0.5A$
    Use point and loop rules of Kirchhoff
15. $R_s = \frac{I_C R_C}{I - I_C} = \frac{(5\times10^{-3})(60)}{10 - 5\times10^{-3}} = 3\times10^{5}\Omega$
16. $\frac{400}{800} = \frac{800}{R} \Rightarrow R = 1600\Omega$

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Unit 14

Magnetism

Surprisingly, magnetism has become very important in everyday life since it is the basis of many electronic devices. For example, magnets are used in electric meters, motors, loudspeakers, bells, and magnetic tapes and disks which are used in sound and video recording equipments and for computer data storage. Magnetic fields are used in magnetic resonance imaging devices to explore the human body with better accuracy and greater safety than x-rays can do. What is more, giant super-conducting magnets are used in the cyclotrons and magnetic bottles possibly the key to future space navigation systems.

14.1 Concepts of a Magnetic Field

Magnets do have North pole and South pole. If a bar magnet is broken in half, each half will have both a north pole and a south pole, just like any other bar magnet. And,

If two magnets are brought together, they will attract or repel each other.

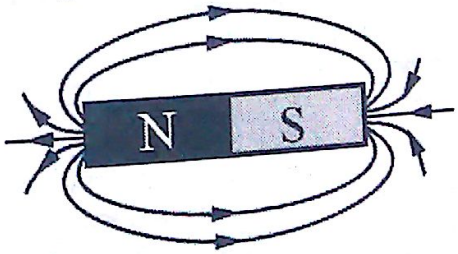
- Like poles repel each other, and unlike poles attract. Just as an electric charge creates an electric field, in its vicinity, so does a magnet.

Magnetic Fsield: Is a region where a magnet exerts a force on its surroundings.

Magnetic field is visualized by imaginary lines called *magnetic field lines*.

The magnetic field lines are closed loops that leave the North Pole of a magnet and enter the South Pole outside the magnet.

- Moreover, the direction of the magnetic field at any point in space is the direction indicated by the North Pole of a compass needle placed at that point.



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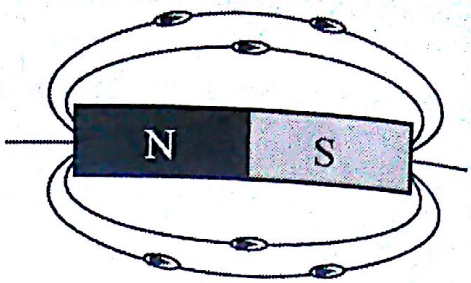
Important points to remember



- The magnetic field is stronger in regions where the field lines are close together such as near the poles and weaker where they are far apart.
  - The magnetic field lines never cross each other.
  - The source of any magnetic field is the motion of electric charges.

Magnetic Properties of Matter

Magnetism, describes how the atoms of materials respond to a magnetic field. Thus, different materials may respond to a magnetic field differently.

Diamagnetism: Is the tendency of a material to oppose an applied magnetic field. It is the property of all materials.

Paramagnetic materials: Are materials having unpaired electrons, which will tend to align themselves in the same direction as the applied magnetic field, thus reinforcing, that is, strengthening it.

Ferromagnetic materials: Are materials having unpaired electrons, which will align with the applied magnetic field and parallel to each other. They keep this alignment even when the applied field is removed. Iron, Nickel and Cobalt are common examples of ferro-magnetic material.

14.2 Earth's Magnetism

The Earth can be thought of as a huge bar magnet. The geographic north pole is the south pole of the magnet, and the geographic south pole is the north pole of the magnet. That is why a bar magnet suspended by a string swing freely in a north-south direction along the horizontal plane, since unlike poles attract each other. The same is true for a compass needle.

- The Earth's magnetism is thought to come from the circulating electric currents in the molten iron core of the Earth, as expressed in the dynamo theory of magnetism.

Conceptual Example

1. Can magnetic field lines cross one another?

   Explanation: The direction in which a compass point at any given location is the direction of the magnetic field at that point. Since a

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compass can point in only one direction, there must be only one direction for the field. If field lines were to cross, it means, there are two directions for the field at the crossing point and this is not possible. Hence, magnetic field lines never cross.

2. Explain diamagnetic, paramagnetic and ferromagnetic materials.

Explanation:

- Diamagnetic is the property of all materials. It is the tendency of material to oppose an applied field.
- Paramagnetic materials reinforce a field because they have unpaired electrons which will tend to align themselves in the same direction as the applied field.
- Ferromagnetic materials have unpaired electrons. They align parallel in the same direction as the field and remain aligned even when the field is removed.

14.3 Magnetic Force on Charged Particle

When a charged particle is placed in a magnetic field, it experience a magnetic force, which is given by,

$\vec{F} = q(\vec{v} \times \vec{B})$ ⟵ in vector form

$\Rightarrow \boxed{F = qv\,B \sin\theta}$ ⟵ in magnitude

Where, F is the magnetic force on the particle

v is the speed of the particle

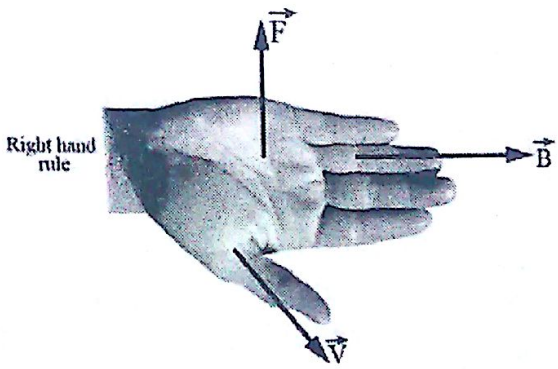
B is the magnetic field strength whose unit is Telsa (T)

θ is the angle between $\vec{v}$ and $\vec{B}$

The direction of the force on the particle is determined using the right hand rule.

Right hand rule

"Extend the right hand so the fingers point along the direction of the magnetic field and the thumb points along the velocity of the charge. The palm of the hand then faces in the direction of the magnetic force that acts on a positive charge."



Note that: If the moving charge is negative instead of positive, the direction of the magnetic force is method for finding the force direction on a moving negative charge is that, first assume the charge is positive and use RHR to find the direction of the force.

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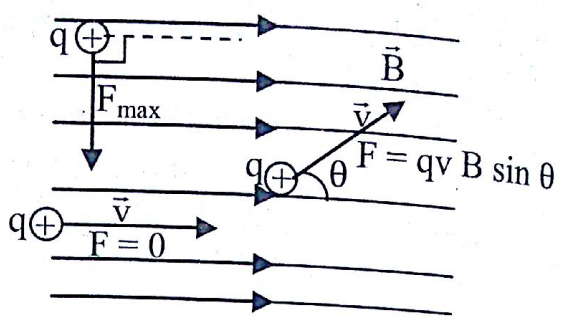
Then, reverse this direction to find the direction of the force acting on the negative charge."

☛ **Important points to remember**

- For a charge to experience a magnetic force. It must be moving. No magnetic force on a stationery charge.



The magnetic force on a moving charged particle

- If the charge moves parallel or anti parallel to the field, the charge experiences no magnetic force. If $\vec{v}//\vec{B}$ then $\vec{F} = 0$.
- If the charge moves perpendicular to the field, the charge experiences the maximum possible force. If $\vec{v} \perp \vec{B}$, then, $F = qvB$.
- If the charge moves at an angle θ with respect to the field, then the magnitude of the force on the charged particle is given by, $F = qv\,B \sin\theta$.

Conceptual Examples

3. Two charged particles move at right angles to a magnetic field and deflect in opposite directions. Can one conclude that the particles have opposite charges?
   Explanation: No. The particles may have charge of the same sign but move in opposite directions along the same line.
4. An electron moves with constant velocity through a region of space that is free of electric field. Can one conclude that the magnetic field is zero in this region? Explain
   Explanation: No. If the electron moves in the same or opposite direction to the field, the magnetic force on it will be zero. As a result, its velocity will remain constant.
5. An electron moving in the positive x direction, at right angles to a magnetic field, experiences a magnetic force in the positive y direction. What is the direction of the magnetic field?

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Explanation: Applying the right hand rule, the direction of the magnetic field is found to be in the positive z direction.

6. No work is done by a magnetic force exerted on a charged particle moving in a magnetic field. Why?

Explanation: Because, magnetic force always acts perpendicular to the motion of the charge.

Illustrative Examples

7. A charge of 4μC is moving with a velocity. $\vec{v} = (2i + 3j + 5\hat{k}) \times 10^6$ m/s in a uniform magnetic field $\vec{B} = (i + 2j + 4\hat{k})$T. Find the force on the charged particle.

Solution: $\vec{F} = q(\vec{v} \times \vec{B})$

$$\vec{F} = 4 \times 10^{-6}[(2i + 3j + 5k) \times 10^6 \times (i \times 2j + 4\hat{k})]$$

$$\vec{F} = (8i - 12j + 4\hat{k})N$$

8. A proton moving at a speed of 5×10^6 m/s enters a magnetic field of 0.4T pointing to the right. If the proton makes an angle of 30° with the field. Find

A. The magnitude and direction of the force on the proton.

B. The acceleration of the proton.

C. What would be the force and acceleration if the particle were an electron instead of a proton?

Solution: a. $F = qv\,B \sin\theta = (1.6 \times 10^{-19})(5 \times 10^6)(0.4)(\sin 30°)$

$F = 1.6 \times 10^{-13}$N directed upward using RHR

b. $F = ma \Rightarrow a = \dfrac{F}{m} = \dfrac{1.6 \times 10^{-13}}{1.67 \times 10^{-27}} = 9.6 \times 10^{13}\ m/s^2$

c. The magnitude of the magnetic force on the electron is the same as that on the proton since both have the same velocity and charge. However, the direction of the force on the electron is downward, that is opposite to that on the proton. And the acceleration of the electron is,

$$a = \frac{F}{m} = \frac{1.6 \times 10^{-13}}{9.11 \times 10^{-31}} = 1.8 \times 10^{17}\ m/s^2$$

9. A uniform magnetic field of 2T in the z direction exert a force on a proton moving at 3 m/s in the x direction. What is the magnitude of the force?

Solution: $F = qvB = (1.6 \times 10^{-19})(3)(2) = 9.6 \times 10^{-19}$N

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10. Find the magnitude of the force felt by an electron traveling at an angle of 60° to the earth's magnetic field at 1.4×10^7 m/s. The earth's magnetic field is 5×10^{-5}T (UEE 2004)

A. 0N C. 1.1×10^{-16}N
B. 9.7×10^{-17}N D. 8.6×10^{-7}N

Solution:

$F = qv B \sin\theta = (1.6 \times 10^{-19})(1.4 \times 10^7)(5 \times 10^{-5})(\sin 60°)$

$F = 9.7 \times 10^{-17}$N

Answer: B

Motion of Charged Particle in Electric and Magnetic Fields

When a charged particle q is moving with velocity $\vec{v}$ in a region where there is an electric field $\vec{E}$ and magnetic field $\vec{B}$, the total force is the vector sum of the electric force $\vec{F}_e$ and the magnetic force $\vec{F}_m$.

$$\vec{F} = \vec{F}_e + \vec{F}_m$$

$$\Rightarrow \vec{F} = \boxed{q\vec{E} + q\vec{v} \times \vec{B}} \longleftarrow \text{Lorentz force}$$

Illustrative Example

11. An electron which is moving with a velocity of $(3i + 4j)10^6$ m/s enters a region of space where there is a magnetic field, $\vec{B} = 0.3i + 0.4j + 0.2k$ T and electric field $\vec{E} = (4i + 3j)10^6$N/C. Compute the force on the electron.

Solution: $\vec{F} = \vec{F}_e + \vec{F}_m = q\vec{E} + q\vec{v} \times \vec{B}$

$\vec{F} = 1.6 \times 10^{-19}(4i + 3j)10^6 + 1.6 \times 10^{-19}[(3i + 4j)10^6 \times (0.3i + 0.4j + 0.2k)]$

$\vec{F} = (7.7i + 3.8j)10^{-13}$N

When the force on a particle as a result of the electric field is the same as the force on the particle as a result of the magnetic field

$F_e = F_m$

$qE = qv B \sin\theta$

$$\Rightarrow \boxed{v = \frac{E}{B}\sin\theta}$$

Illustrative Example

12. Find the speed of an electron travelling perpendicular to the electric field and magnetic fields of magnitude 0.3kv/m and 3.0T respectively.

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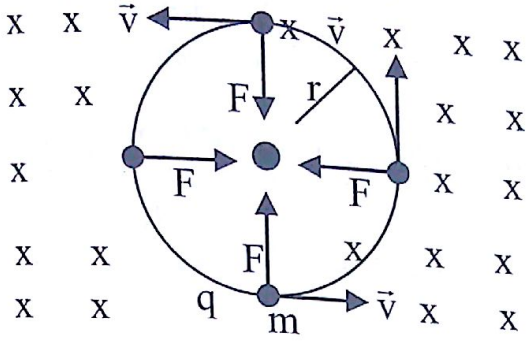
Solution: $v = \frac{E}{B}\sin\theta = \frac{0.3\times1000}{3}\sin 90° = 1000\text{m/s}$

Circular Motion of Particles in Magnetic Fields

Suppose a positively charged particle is moving with a velocity $\vec{v}$ in a magnetic field $\vec{B}$ that is directed into the page.



The magnetic force on the particle is then

$$F_m = qvB$$

This force causes the particle to follow a circular path of radius r which then can be justified using the right hand rule. Thus, the magnetic force provides the centripetal force

$$F_m = F_c$$

$$qvB = \frac{mv^2}{r}$$

$$\Rightarrow \boxed{qB = \frac{mv}{rr}} \text{ also, } \boxed{r = \frac{mv}{qB}}$$

Where: m is the mass of the particle

r is the radius of the circle

- **Keep in mind that:**
  - The faster and more massive the particle, the larger the circle.
  - The stronger the magnetic field and the greater the charge, the smaller the circle.

Period and Angular Velocity of the Particle

The time required for the particle to complete one cycle is obtained from, $t = \frac{s}{v}$.

$$\Rightarrow T = \frac{2\pi r}{v}, \text{ but } r = \frac{mv}{qB}$$

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$\Rightarrow T = \frac{2\pi mv}{vqB}$, thus

$$\boxed{T = \frac{2\pi m}{qB}} \longleftarrow \text{period}$$

The frequency is then, $f = \frac{1}{T} = \frac{qB}{2\pi m}$

And, the angular velocity of the particle is $\omega = 2\pi f$, $f = \frac{qB}{2\pi m}$

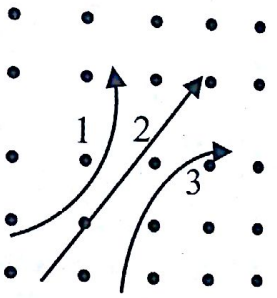
$$\Rightarrow \boxed{\omega = \frac{qB}{m}} \longleftarrow \text{Angular velocity}$$

Conceptual Examples

13. If a charged particle moves in a straight line through some region, can you say that the magnetic field in that region is zero?
Explanation: No. Another field, such as electric or gravitational may produce another force on the particle resulting in a net force of zero.

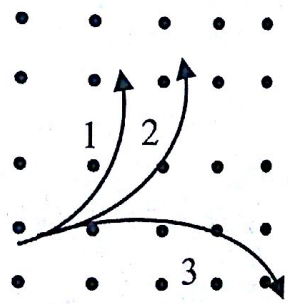
14. A charged particle moves in a circular path in the presence of a constant magnetic field. Does the particle gain energy? Explain
Explanation: No. Because, the magnetic force is always perpendicular to the displacement of the particle, thus no work is done on the particle and its speed remains constant. Hence, its kinetic energy does not change.

15. Three particles travel through a region of space where the magnetic field is out of the page. For each of the three particles, state whether the particle's charge is positive, negative, or zero.



Explanation: Using the right hand rule, particle 1 is negative, particle 2 has no charge and particle 3 is positive.

16. Consider the paths of particles 1, 2 and 3. If these particles have the same mass and speed, list them in order of increasing magnitude of charge.



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Explanation: Since the radius of curvature is, $r = \frac{mv}{qB}$, hence a large charge implies a small radius. It follows that $|q_1| > |q_2| > |q_3|$.

17. Consider two identical particles moving in a circle of small radius and large radius respectively. Which particle will take longer time to complete one cycle?

Explanation: Time to complete one cycle, is given by

$T = \frac{2\pi m}{qB}$ ⟵ independent of radius,

Thus, the particles will take the same amount of time.

Illustrative Examples

18. An electron moving perpendicular to a magnetic field of 4.6×10^{-3}T follows a circular path of radius 2.8mm. What is the electron's speed?

Solution:

$$v = \frac{rqB}{m} = \frac{(2.8\times10^{-3})(1.6\times10^{-19})(4.6\times10^{-3})}{9.11\times10^{-31}} = 2.26\times10^{6}\,m/s$$

19. A proton is moving horizontally through a vertical field of 0.04T, with a velocity of 2.4×10^6 m/s. Calculate

A. The radius of the path
B. The acceleration
C. The period of revolution

Solution: a) $r = \frac{mv}{qB} = \frac{(1.67\times10^{-27})(2.4\times10^{6})}{(1.6\times10^{-19})(4\times10^{-2})} = 62.6cm$

b) $a = \frac{v^2}{r} = \frac{(2.4\times10^{6})}{0.626} = 9.2\times10^{12}\,m/s$

c) $T = \frac{2\pi r}{v} = \frac{2\times3.14\times0.626}{2.4\times10^{6}} = 1.6\times10^{-6}\,sec$

20. A particle of mass m is fired into a magnetic field of strength B at a speed v. The particle travels in a circular path inside the field with a radius r. which of the following expressions gives the magnitude of the charge on the particle? (UEE 2005)

A. $\frac{mr}{v^2B}$ B. $\frac{mv^2}{Br}$ C. $\frac{vB}{mr}$ D. $\frac{mv}{Br}$

Explanation: $F_m = F_c \Rightarrow qvB = \frac{mv^2}{r}$

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$$q = \frac{mv}{Br}$$

Answer: D

Compression between Electric and Magnetic Forces on a Charged Particle

Electron Force	Magnetic Force
• Acts in the direction of the electric field	• Acts perpendicular to the magnetic field.
• Independent of the particle velocity $\vec{F} = \vec{E}q$	• Depends on the particle velocity $\vec{F} = q(\vec{v} \times \vec{B})$
• Does work on the particle, $W = Eqd$	• Does no work $W = 0$
• It changes the speed of the particle, thus causes the particle to accelerate.	• It does not change the particles speed but changes its direction.

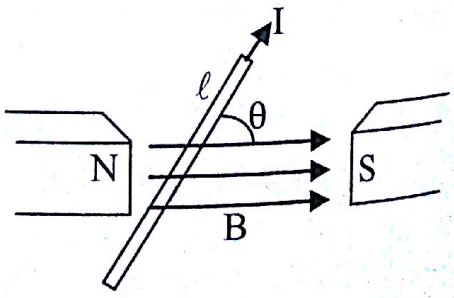
Conceptual Example

21. How can the motion of a charged particle be used to distinguish between a magnetic field and an electric field in a certain region?
Explanation: The magnetic force on a moving charged particle is always perpendicular to the direction of motion. However, the electric force on a charged particle is always parallel to the electric field.

14.4 Magnetic Force on a Current Carrying Wire

A magnetic force acts on a single charged particle moving through an external magnetic field. Thus, a current carrying wire also experiences a magnetic force when placed in an external magnetic field. This follows from the fact that electric current is resulted from a collection of many charged particles in motion; hence, the magnetic force on the wire is due to the sum of the individual magnetic force on the charged particles.
To calculate the magnetic force on a current carrying wire, consider a wire of length ℓ and cross – sectional area A, carrying a current I and placed in an external magnetic field B. Also, assume the charges in the wire move with same drift velocity v. The magnetic force on charge of moving with velocity v is:

$$\vec{F} = q(\vec{v} \times \vec{B})$$



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To find the total magnetic force on the wire, we multiplying the magnetic force on one charge by the number of charges in the wire. Because the volume of the wire is $A\ell$, the number of charges in that wire is $nA\ell$, where n is the number of charges per volume. Hence, the total magnetic force on the wire of length ℓ is

$$\vec{F} = (q\vec{v} \times \vec{B})nA\ell$$

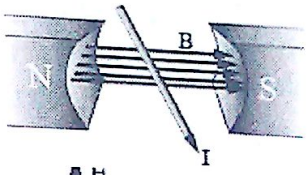
But, $I = nqvA$

Therefore, $\boxed{\vec{F} = I(\vec{\ell} \times \vec{B})}$ ⟵ in vector form

$$\Rightarrow \boxed{F = I\ell B \sin\theta}$$ ⟵ in magnitude

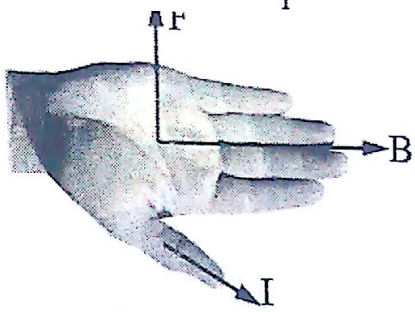
Where, θ is the angle between ℓ and B

- The direction of the magnetic force is determined using the right hand rule.



Right Hand Rule

"Point the fingers of your right hand in the direction of the magnetic field. Point your thumb in the direction of the conventional current in the wire. The palm of your hand then faces in the direction of the force acting on the wire. "



- **Important points to remember**
  - The force on the wire is maximum when the magnetic field is perpendicular to the current.
    That is, when $\vec{\ell} \perp \vec{B}$, $F = I\ell B$
  - The force on the wire is zero when the magnetic field is parallel or anti parallel to the current
    That is, when $\vec{\ell} \,//\, \vec{B}$, $F = 0$
  - The force on the wire when the magnetic field is an any angle θ with the current is, $F = I\ell B \sin\theta$ ⟵ general case

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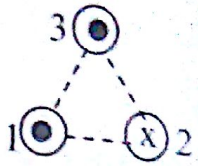
Conceptual Examples

22. A current carrying wire is placed in a region with a uniform magnetic field. Can the wire experience zero magnetic force? How?



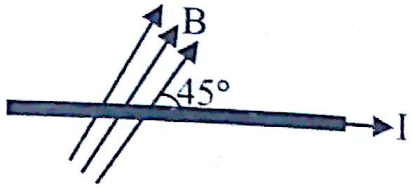
Explanation: If the wire is placed parallel or anti-parallel to the field the force on it is zero since, $F = I\ell B \sin\theta$.

23. Consider three wires each carrying equal currents. Wire 1 and 3 are out of the page, and wire 2 into the page. What is the direction of magnetic force on wire 3?

Answer: To the left

Illustrative Examples

24. A 1m wire carrying a current of 50A from west to east is placed in a magnetic field of 1.2T that runs in the north east direction. Find the force on the wire.



Solution: $F = I\ell B \sin\theta = (50)(1)(1.2)\sin 45°$

$F = 42.4N$, upward

25. A 50cm wire carrying a current of 3A in the x direction lies in a magnetic field of $(2i + 3k)T$. What is the magnetic force on the wire?

Solution: $\vec{F} = I(\vec{\ell} \times \vec{B}) = 3[(0.5i) \times (2i + 3k)] = -4.5jN$

26. A 20 g wire with a current of 1.5A is placed in a 0.08T magnetic field. If 4.5m of the wire hangs across the field at an angle of 30°. Calculate

A. The magnetic force
B. The acceleration

Solution: a) $F = I\ell B \sin\theta = (0.08)(1.5)(4.5)\sin 30° = 0.27N$

b) $a = \frac{F}{m} = \frac{0.27}{0.02} = 13.5 m/s^2$

27. A 15A current is directed along the positive x direction and perpendicular to the field. A magnetic force per unit length of 12N/m acts on the wire in the negative y direction. Calculate the magnetic field.

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Solution: $F = BI\ell \Rightarrow B = \frac{F}{I\ell} = \left(\frac{F}{\ell}\right)\left(\frac{1}{I}\right) = (12)\left(\frac{1}{15}\right)$

$\Rightarrow B = 0.8T$, along positive z direction

28. A long straight wire carrying a current of 3A flowing in a direction parallel to the unit vector $\frac{1}{\sqrt{3}}(i + j + k)$ is placed in a magnetic field of $(0.75i + 0.4k)T$. What is the magnetic force per unit length of the wire? (UEE 2001)

A. $(0.69i - 0.61j - 1.3\hat{k})N/m$ C. $(0.49i - 0.51j - 1.37\hat{k})N/m$

B. $(0.39i - 0.4j - 2.3\hat{k})N/m$ D. $(0.59i - 0.71j - 0.13\hat{k})N/m$

Solution: $F = I(l \times B) = 3\left[\frac{1}{\sqrt{3}}(i + j + k) \times (0.75j + 0.4\hat{k})\right]N$

$$\frac{F}{\ell} = (0.69i - 0.61j - 1.3\hat{k})N/m$$

Answer: A

29. A wire with mass of 0.5 g/cm carries a 2A current horizontally towards south. What are the direction and magnitude of the minimum magnetic field needed to lift this wire upward (UEE 2002)

A. 0.25T west C. 0.50T west

B. 0.25T east D. 0.50T east

Solution: To lift the wire upward, the minimum magnetic force needed should balance the weight of the wire.

$F_m = W$

$$I\ell B = mg \Rightarrow B = \frac{mg}{I\ell} = \left(\frac{m}{\ell}\right)\left(\frac{g}{I}\right)$$

$$B = (0.05)\left(\frac{10}{2}\right) = 0.25T \text{ due east}$$

Answer: B

30. A wire carries a steady current of 2.4A. A straight section of the wire is 0.75m long and lies along the x axis with in a uniform magnetic field, B = 1.6T along +z axis. If the current is in the +x direction, what is the magnetic force on the section of wire (UEE 2005).

A. 2.88jN B. 1.62jN C. -1.62jN D. -2.88jN

Solution: $\vec{F} = I(\vec{\ell} \times \vec{B}) = 2.4\ (0.75i \times 1.6k) = -2.88jN$

Answer: D

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31. A wire carries a steady current of 2.4A. A straight section of the wire is 0.75m long and lies in xy plane along the x axis within a uniform magnetic field B = 1.6kT. If the current is in the positive x direction, what is the magnetic force on the section of wire? (UEE 2006)

A. 2.88N, in the negative y direction
B. 2.88N, in the positive y direction
C. 3.84N, in the negative y direction
D. 3.84N, in the positive y direction

Solution: $\vec{F} = I(\vec{\ell} \times \vec{B}) = 2.4(0.75i \times 1.6k) = -2.88jN$ **Answer: A**

32. An electron moving at a speed of 1.5×10^7 m/s enters a region of uniform magnetic field of 2×10^{-3}T perpendicularly. The path followed by the electron is circular. What is the radius of the circular path? (UEE: 2007)

A. 0.43cm B. 4.3cm C. 43cm D. 430cm

Solution: $r = \frac{mv}{qB} = \frac{(9.11\times10^{-31})(1.5\times10^{7})}{(1.6\times10^{-19})(2\times10^{-3})} = 4.3\times10^{-2}\,m = 4.3cm$

Answer: B

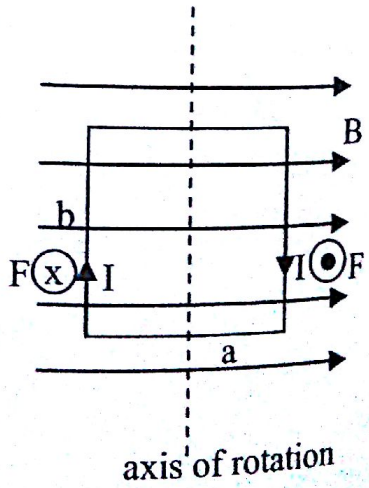
33. If a wire carrying current in the plane of this paper toward the top of the page experiences a magnetic force toward the right edge of the page, what will be the direction of the magnetic field causing this force? (UEE 2007)

A. In the plane of the page and toward the left edge.
B. In the plane of the page and toward the bottom edge.
C. Perpendicular and directed out of the plane of the page.
D. Perpendicular and directed into the plane of the page.

Explanation: Applying the right hand rule. **Answer: C**

14.5 Torque on a current carrying loop

Consider a rectangular loop of wire with sides a and b placed in a uniform magnetic field that is parallel to the plane of the loop. From this arrangement, it is clear that on the horizontal segments (sides marked 'a') the loop experiences zero force, since they are parallel to the field. The



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vertical segments (sides marked 'b') are perpendicular to the field, hence they experience forces of magnitude F = IbB. One of these forces is into the page and the other out of the page.

If we imagine an axis of rotation through the center of side a, the forces exert a torque that tends to rotate the loop clockwise. The magnitude of this torque for each vertical segment is the force, F = IbB times the moment arm, $\frac{a}{2}$. Noting that both vertical segments exert a torque in the same direction, we see that the torque is simply the sum of the torque produced by each segment.

$$\tau = IbB\left(\frac{a}{2}\right) + IbB\left(\frac{a}{2}\right)$$

$$\tau = IBba$$

Observing that the area of the rectangular loop is A = ba, therefore we can express the torque as follows:

$$\tau = IAB$$

And we can have

$$\tau = I(\vec{A} \times \vec{B})$$

$$\boxed{\tau = IAB \sin\theta}$$

Torque exerted on a loop of Area A and N turns is given by

$$\boxed{\tau = NI\,AB \sin\theta}$$

Where θ is the angel between the plane of the loop and the magnetic force exerted on each side of the loop. Equivalently, θ is the angle between the normal and the magnetic field.

- **Note that:** There is no torque when the magnetic field is perpendicular to the plane of a loop. This is because when θ = 0, the moment arm is zero, hence the torque.

Conceptual Examples

34. What is the net magnetic force on a closed current loop in a uniform magnetic field?

Explanation: Zero. A net torque exists, but the net force is zero.

35. When will the torque became maximum? When will it be zero? Explain

Explanation: For none zero force and area, the torque is maximum when the magnetic field is parallel to the plane of the loop. The torque is zero for none zero force and area when the magnetic field is perpendicular to the plane of the loop.

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Illustrative Examples

36. Find the torque on a loop of wire area $12cm^2$ at 30° to magnetic field of strength 50mT with a current of 4A flowing through it.

Solution: $\tau = IAB \sin\theta = (4)(12 \times 10^{-4})(50\times10^{-3}) \sin 30°$

$\tau = 1.2 \times 10^{-4}$ N.m

37. A rectangular coil with 200 turns is 5 cm high and 4cm wide. When the coil is placed in a magnetic field of 0.5T, its maximum torque is 0.2N.m. What is the current in the coil?

Solution: $\tau = NIAB \Rightarrow I = \dfrac{\tau}{NAB} = \dfrac{0.2}{(200)(0.05)(0.04)(0.5)}$

$I = 1A$

Note: The direction of the current will determine whether the torque on the coil is clockwise or counterclockwise.

Magnetic dipole moment

A current loop creates a magnetic dipole moment. If we consider a coil of wire consisting of N turns, carrying current I, and having area A, then the magnetic moment of such a coil is given by:

$$\boxed{\mu = NIA}$$

Magnetic moment is a vector quantity and its direction is determined using the right hand rule.

When a magnetic dipole moment is placed in a magnetic field, it experiences a torque. This torque is given by

$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

This can be written as

$$\boxed{\tau = \mu B \sin\theta}$$

Illustrative Example

38. A current of 17mA is maintained in a single circular loop with a circumference of 2m. A magnetic field of 0.8T is directed parallel to the plane of the loop.

a) Calculate the magnetic moment of the loop.

b) What is the magnitude of the torque exerted on the loop by the magnetic field?

Solution:

a) $\mu = NIA$, $c = 2\pi r \Rightarrow 2 = 2\pi r$, $\pi r = 1m$, $A = \pi r^2$ $\pi\left(\dfrac{1}{\pi}\right)^2 = \dfrac{1}{\pi}$

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$$\mu = (1)(17 \times 10^{-3})\left(\frac{1}{3.14}\right)$$

$$\mu = 5.4 \times 10^{-3}\ A.m^2$$

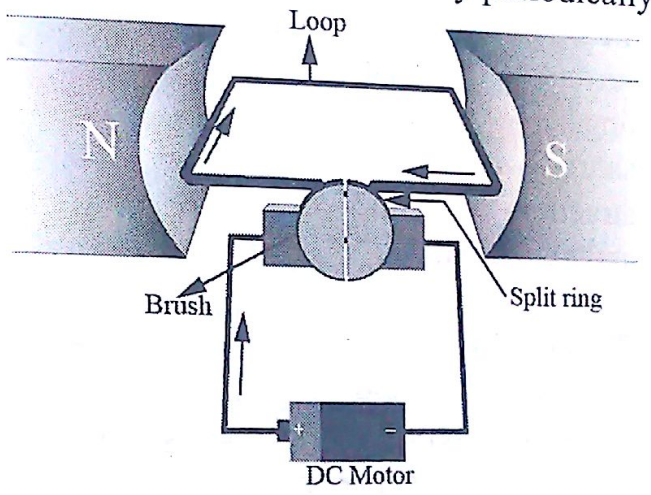
b) $\tau = \mu B = (5.4 \times 10^{-3})(0.8)$

$$\tau = 4.3 \times 10^{-3}\ N.m$$

A Direct Current Motor

It's hard to imagine life in the 21 century without electric motors. Computer disk drives, CD players, DVD players, food processors, automobiles, furnaces, washing machines, air conditioners etc all contain motors. The motor convert electrical energy into rotational kinetic energy that is mechanical energy.

A simple DC motor consist current carrying loop called *armatures*, split – ring commutator, brushes and DC source. The split – ring commutator provides continuous rotation of the loop in one direction by periodically reversing the current direction. The brushes, which are usually made from graphite are used to reduce sparking and heat.

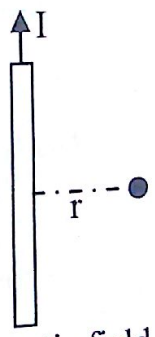


The working principle of electric motor or the galvanometer is the rotation of a current carrying coil placed in a magnetic field. Thus, these devices use the concept of torque for their operation.

14.6 Magnetic Field due to a Wire

During a lecture demonstration in 1819, the Danish scientist Hans Christain Oersted found that an electric current in a wire deflected a nearby compass needle. This momentous discovery lead Oersted in 1820 to experimentally confirm that a current – carrying conductor produces a magnetic field.

The magnitude of the magnetic field produced by the current – carrying wire is found to be directly



Magnetic field at distance r

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proportional to the current and inversely proportional to the distanc from the wire.

$$\Rightarrow B \sim \frac{I}{r}$$

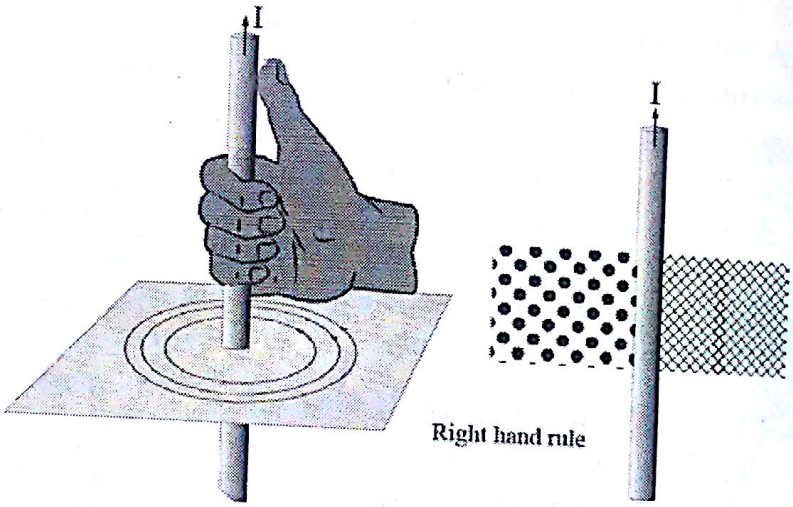
$$\boxed{B = \frac{\mu o I}{2\pi r}}$$

Where: μ_o is called *permeability of free space* and has the value: $\mu_o = 4\pi \times 10^{-7} \frac{T.m}{A}$.

The direction of the magnetic field produced by the current – carrying wire is determined by the right hand rule.

Right Hand Rule

"Point the thumb of your right hand in the direction of the current. Your fingers then naturally curl in the direction of the magnetic field."



Right hand rule

☛ **Note that:** The magnetic field lines are represented by concentric circles.

- **The dot (•)** represents the field lines are out of the page. And,
- **The cross (×)** represents, the field lines are into the page.

Conceptual Examples

39. What will be the direction of compass needles that are placed around a straight wire when
    a. There is no current in the wire.
    b. There is current in the wire.

    Explanation:

    a) The compass needles all will point in the same direction.
    b) The compass needles will defect in the direction of the magnetic field.

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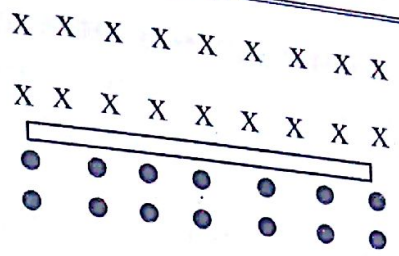
40. The magnetic field shown below is due to a horizontal current carrying wire. Does the current in the wire flow to the left or to the right?



Explanation: If you grasp the wire in such a way that your right hand fingers are into the page from above and out of the page below the wire the thumb will point to the left. Thus, the current in the wire flows to the left.

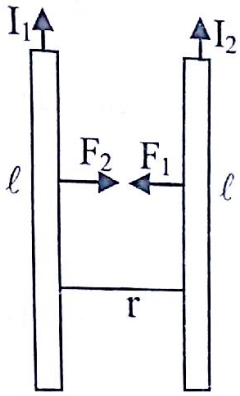
Illustrative Example

41. Find the magnitude of the magnetic field 1m from a long, straight wire carrying a current of 1A.

Solution: $B = \frac{\mu_o I}{2\pi r} = \frac{(4\pi \times 10^{-7})(1)}{(2\pi)(1)} = 2 \times 10^{-7}\,T$

14.7 Force Between Current Carrying Wire

We know that a current – carrying wire in a magnetic field experiences a force. We also know that a current – carrying wire produces a magnetic field. It follows, then, that one current – carrying wire will exert a force on another.



Force between wires

Now, consider two long, straight, parallel wires separated by distance r and carrying currents I_1 and I_2 in the same direction. The magnetic force on one wire due to a magnetic field set up by the other wire can be calculated as follows. For example, let us see the force on wire 2 due to the magnetic field of wire 1. The current I_1, sets up magnetic field, B_1 at wire 2. Which is given by,

$$B_1 = \frac{\mu_o I_1}{2\pi r}$$

And according to the relation, $F = I\ell B$,
The magnitude of the magnetic force on wire 2 due to the magnetic field of wire 1 is then $F_2 = B_1 I_2 \ell$, substitute for B_1

$F_2 = \left(\frac{\mu_o I_1}{2\pi r}\right) I_2 \ell$, $F_1 = F_2$; From Newton's 3rd law $\boxed{F = \frac{\mu_o I_1 I_2 \ell}{2\pi r}}$

The force per unit length is given by $\boxed{\frac{F}{\ell} = \frac{\mu_o I_1 I_2}{2\pi r}}$

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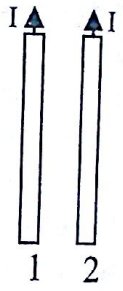
Important points to remember

- The force that wire 1 exerts on wire 2 is equal in magnitude and opposite in direction to the force that wire 2 exerts on wire 1 due to Newton's 3rd law.
- Wires carrying – currents in the same direction attract each other while wires carrying – currents in opposite direction repel each other.
- The force between two parallel wires carrying a current is used to define the SI unit of current, the ampere.
  "The ampere is that current which if maintained in two straight parallel wires of infinite length of negligible cross – section and placed one meter apart in vacuum, would produce between these two wires a force equal to 2×10^{-7} Newton per meter of length."

Conceptual Examples

42. Parallel wires exert magnetic force on each other. What about perpendicular wires? Explain



Explanation: There is no net force on the wires, but there is a torque.

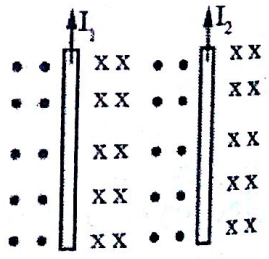
43. In the figure shown, $I_1 = 2A$ and $I_2 = 6A$. Which of the following is true?

a) $F_1 = -3F_2$ b) $F_1 = -F_2$ c) $F_1 = \frac{F_2}{3}$

Explanation: We know, $F_1 - F_2 = \frac{\mu_o I_1 I_2 \ell}{2\pi r}$ hence, $F_1 = F_2$

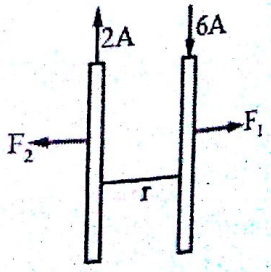
Answer: B

44. Why do two straight parallel wires carrying currents in the same direction attract each other?



Explanation: It is because, when the wires carry currents in the same direction, the magnetic field between them is opposite and hence, the force is attractive. Right hand rule can verify this fact.

45. If two parallel wires carry currents of 2A and 6A respectively, in opposite directions, which wire exert a larger force on the other? Explain



Explanation: Even if the wires carry currents

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of different magnitude, the force one exerts on the other in equal in magnitude and opposite in direction according to Newton's 3rd law.

$$F_1 = F_2 = \frac{\mu_o I_1 I_2 \ell}{2\pi r}$$

Illustrative Examples

46. Two wires separated by a distance of 12cm carry currents in the same direction. The current in one wire is 2.5A and the current in the other wire is 4.5A. Calculate:
a) The magnetic field midway between the wires.
b) The force per unit length of wire.

Solution: a) The magnetic field due to wire 1 is

$$B_1 = \frac{\mu_o I_1}{2\pi r} = \frac{(4\pi \times 10^{-7})(2.5)}{(2\pi)(6 \times 10^{-2})} = 8.3 \times 10^{-6} \text{ into the page}$$

The magnetic field due to wire 2 is,

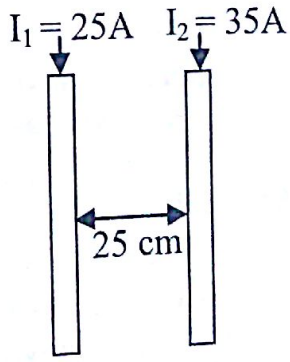
$$B_2 = \frac{\mu_o I_2}{2\pi r} = \frac{(4\pi \times 10^{-7})(4.5)}{(2\pi)(6 \times 10^{-2})} = 15 \times 10^{-6} \text{ out of the page}$$

The net field midway the wires is
$B = B_2 - B_1 = 15 \times 10^{-6} - 8.3 \times 10^{-6} = 6.7 \times 10^{-6}$T out of the page

b) $$F = \frac{\mu_o I_1 I_2 \ell}{2\pi r} \Rightarrow \frac{F}{\ell} = \frac{\mu_o I_1 I_2}{2\pi r} = \frac{(4\pi \times 10^{-7})(2.5)(4.5)}{(2\pi)(12 \times 10^{-2})}$$

$\frac{F}{\ell} = 1.9 \times 10^{-5}$ N/m attractive

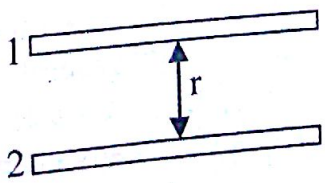
47. What is the force per unit length between the wires shown below?



Solution: $$F = \frac{\mu_o I_1 I_2 \ell}{2\pi r} \Rightarrow \frac{F}{\ell} = \frac{\mu_o I_1 I_2}{2\pi r}$$

$$\frac{F}{\ell} = \frac{(4\pi \times 10^{-7})(25)(35)}{2\pi \times 25 \times 10^{-2}} = 7 \times 0^{-4} \text{ N/m}$$

48. A wire with a weight per unit length of 0.08N/m is suspended directly above a second wire. The top wire carries a current of 60A. Find the separation between the wires.



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Solution: $F = \frac{\mu_o I_1 I_2 \ell}{2\pi r} \Rightarrow \frac{F}{\ell} = \frac{\mu_o I_1 I_2}{2\pi r}$

Since, $\frac{F}{\ell} = 0.08 \Rightarrow r = \frac{\mu_o I_1 I_2}{2\pi \times 0.08}$

$$r = \frac{(4\pi \times 10^{-7})(30)(60)}{2\pi \times 0.08} = 0.0045\text{m} \Rightarrow r = 4.5\text{mm}$$

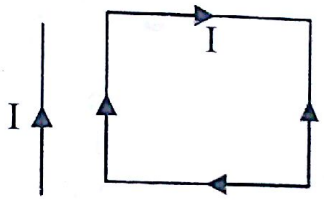
49. Two parallel copper wires are 1 cm apart. Lighting sends a 10,000A pulse of current along each conductor. What is the force per unit length on one conductor? (UEE 2002)

A. 1000N/m B. 200N/m C. 20N/m D. 2000N/m

Solution: $\frac{F}{\ell} = \frac{\mu_o I_1 I_2}{2\pi r} = \frac{(4\pi \times 10^{-7})(10,000)(10,000)}{2\pi(0.01)} = 2000\text{N/m}$

Answer: D

50. A rectangular loop carrying a current is situated near a long straight wire such that the wire is parallel to one of the sides of the loop and it is in a plane of the loop. If a steady current is established in the wire as shown, the loop will (UEE 2005)



A. Remain stationary.
B. Move toward the wire.
C. Move away from the wire.
D. Rotate about an axis parallel to the wire.

Explanation: Since the straight wire and the side of the loop near the wire carry currents in the same direction there is attraction force between the wire and the loop. **Answer: B**

51. Which one of the following statements is correct about the effect and formation of a magnetic field? (UEE 2007)

A. A magnetic field exerts a non zero force on a stationary charged object.
B. A magnetic field exerts a non zero force on a conducting wire placed in the field.
C. A uniform magnetic field can be created at all points around a long current carrying wire.
D. A magnetic field can be created in the space surrounding any charged object that is in motion. **Answer: D**

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52. The torque experienced by a current carrying loop, in an external, magnetic field is (UEE 2008)
   A. Independent of the number of turns of the loop.
   B. Independent of the current in the loop.
   C. Dependent on the orientation of the loop.
   D. Dependent on magnetic field generated according to Lenz's law.

Answer: C

53. Two long wires of the same length and cross – sectional area carry the same amount of current. Which one of the following statements is correct about the magnetic field and the magnetic forces on the wires? (UEE 2008)
   A. The magnetic field is zero at the midpoint between the wires when the currents are in the same direction.
   B. The force between the wires is attractive when they carry currents in opposite direction.
   C. The magnetic fields between the wires reinforce each other when the currents are in the same direction.
   D. The magnetic field due to one is proportional to the current through the other wire.

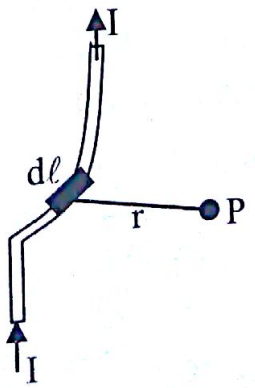
Answer: A

The Biot – Savart Law

In the 19th Century, Jean Biot and Felix Savart, investigate the magnetic field due to a very small element of current that is part of a large current distribution.

Bilot – Savart Law states that: "The magnetic field contribution dB at a point by a current element of length $d\ell$ at a point P a distance r from that element is by $\boxed{d\vec{B} = \frac{\mu_o I d\ell}{4\pi r^2} \times \vec{n}r}$"

$$\Rightarrow dB = \frac{\mu_o I d\ell \sin\theta}{4\pi r^2}$$



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31. A wire carries a steady current of 2.4A. A straight section of the wire is 0.75m long and lies in xy plane along the x axis within a uniform magnetic field B = 1.6kT. If the current is in the positive x direction, what is the magnetic force on the section of wire? (UEE 2006)

A. 2.88N, in the negative y direction
B. 2.88N, in the positive y direction
C. 3.84N, in the negative y direction
D. 3.84N, in the positive y direction

Solution: $\vec{F} = I(\vec{\ell} \times \vec{B}) = 2.4(0.75i \times 1.6k) = -2.88jN$ **Answer: A**

32. An electron moving at a speed of 1.5×10^7 m/s enters a region of uniform magnetic field of 2×10^{-3}T perpendicularly. The path followed by the electron is circular. What is the radius of the circular path? (UEE: 2007)

A. 0.43cm B. 4.3cm C. 43cm D. 430cm

Solution: $r = \frac{mv}{qB} = \frac{(9.11\times10^{-31})(1.5\times10^{7})}{(1.6\times10^{-19})(2\times10^{-3})} = 4.3\times10^{-2}\,m = 4.3cm$

Answer: B

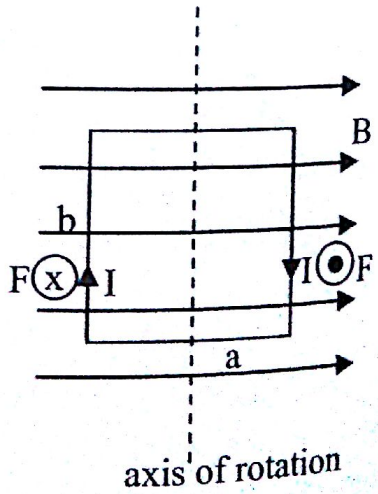
33. If a wire carrying current in the plane of this paper toward the top of the page experiences a magnetic force toward the right edge of the page, what will be the direction of the magnetic field causing this force? (UEE 2007)

A. In the plane of the page and toward the left edge.
B. In the plane of the page and toward the bottom edge.
C. Perpendicular and directed out of the plane of the page.
D. Perpendicular and directed into the plane of the page.

Explanation: Applying the right hand rule. **Answer: C**

14.5 Torque on a current carrying loop

Consider a rectangular loop of wire with sides a and b placed in a uniform magnetic field that is parallel to the plane of the loop. From this arrangement, it is clear that on the horizontal segments (sides marked 'a') the loop experiences zero force, since they are parallel to the field. The



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vertical segments (sides marked 'b') are perpendicular to the field, hence they experience forces of magnitude F = IbB. One of these forces is into the page and the other out of the page.

If we imagine an axis of rotation through the center of side a, the forces exert a torque that tends to rotate the loop clockwise. The magnitude of this torque for each vertical segment is the force, F = IbB times the moment arm, $\frac{a}{2}$. Noting that both vertical segments exert a torque in the same direction, we see that the torque is simply the sum of the torque produced by each segment.

$$\tau = IbB\left(\frac{a}{2}\right) + IbB\left(\frac{a}{2}\right)$$

$$\tau = IBba$$

Observing that the area of the rectangular loop is A = ba, therefore we can express the torque as follows:

$$\tau = IAB$$

And we can have

$$\tau = I(\vec{A} \times \vec{B})$$

$$\boxed{\tau = IAB \sin\theta}$$

Torque exerted on a loop of Area A and N turns is given by

$$\boxed{\tau = NI\,AB \sin\theta}$$

Where θ is the angel between the plane of the loop and the magnetic force exerted on each side of the loop. Equivalently, θ is the angle between the normal and the magnetic field.

- **Note that:** There is no torque when the magnetic field is perpendicular to the plane of a loop. This is because when θ = 0, the moment arm is zero, hence the torque.

Conceptual Examples

34. What is the net magnetic force on a closed current loop in a uniform magnetic field?

Explanation: Zero. A net torque exists, but the net force is zero.

35. When will the torque became maximum? When will it be zero? Explain

Explanation: For none zero force and area, the torque is maximum when the magnetic field is parallel to the plane of the loop. The torque is zero for none zero force and area when the magnetic field is perpendicular to the plane of the loop.

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